

SESSION 3

Disorders of Vascular Flow

Hyperemia & Congestion · Edema · Hemorrhage · Thrombosis · Embolism · Infarction

PART 1 **Hyperemia, Congestion and Edema**

PART 2 **Hemorrhage**

PART 3 **Thrombosis**

PART 4 **Embolism**

PART 5 **Infarction**

PART 6 **Review Questions, Master Tables and Glossary**

SYNTHESIZED FROM

Lecture slides — “Disorder of Vascular Flow”

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Robbins & Kumar, *Basic Pathology*, 11th ed.

Ch. 3 — Hemodynamic Disorders, Thromboembolism and Shock; Ch. 8 — Blood Vessels

Sattar, *Fundamentals of Pathology* (Pathoma)

Hemostasis & related disorders; Ch. 7 — Vascular Pathology

How to Use This Guide

This guide is built to be read *once, slowly* — and then revisited many times, quickly. Everything you need is inside it; you should not have to open the source textbooks again.

The organizing idea

All six topics in this session are variations on a single question: **what happens when blood is in the wrong place, in the wrong amount, or in the wrong physical state?** Blood can pool where it should have drained (congestion), its watery fraction can leak out of the vessel (edema), the whole of it can escape through a breach in the wall (hemorrhage), it can solidify inside a vessel that is still alive (thrombosis), that solid mass can travel downstream (embolism), and where it finally lodges the tissue beyond it dies (infarction). Read in that order, the five sections form one continuous story rather than six separate lists.

The four kinds of boxes

Box	What it contains and how to use it
High-Yield	Facts that are repeatedly examined. If you memorize nothing else, memorize these.
Exam Trap	A place where two concepts are easily confused, or where the intuitive answer is the wrong one. These are where marks are actually lost.
Mnemonic	Used <i>only</i> where a list is genuinely arbitrary and must be memorized — not as a substitute for understanding.
Clinical Correlation	How the mechanism shows up at the bedside: the symptom, the sign, the test, the treatment.
Mechanism chain	Green numbered boxes. Each step carries a <i>Why</i> line explaining why that step forces the next one. Follow the chain rather than memorizing it — if you can re-construct the logic, you can answer a question you have never seen.

Suggested study sequence

1. **First pass (≈3 h).** Read Parts 1–5 straight through, including the green mechanism chains, skipping the tables.
2. **Second pass (≈1.5 h).** Re-read using only the headings, the definition boxes and the comparison tables. The tables are the compressed form of the text — they only make sense after the first pass.
3. **Third pass (≈1.5 h).** Do all 28 vignettes in Part 6 *closed-book*, then read every explanation — including for the questions you got right.
4. **Night before (≈40 min).** High-Yield boxes + Exam Traps + the glossary only.

A NOTE ON THE SOURCES

Where the lecture slides, Robbins and Pathoma disagree in emphasis, this guide follows the mechanism that is best supported and flags the discrepancy. Where the slides use an older or Chinese-curriculum term (e.g. *hyaline thrombus*, *anemic infarct*), the Robbins equivalent (*fibrin microthrombus*, *white/pale infarct*) is given alongside it, because your examiner may use either.

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PART ONE

01 Hyperemia, Congestion and Edema

Too much blood inside the vessels, and too much water outside them. These two problems are physically different but mechanically linked: the same raised venous pressure that engorges a capillary bed also pushes fluid out of it. Understanding that link is the single most useful thing in this Part.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Distinguish active hyperemia from passive congestion by mechanism, colour, temperature and clinical setting.
- List and explain at least three morphological features each of chronic pulmonary congestion and nutmeg liver.
- Explain, from first principles, why hemosiderin-laden macrophages appear in the alveoli of a patient in left heart failure.
- Derive all four mechanisms of non-inflammatory edema from the Starling equation.
- Separate a transudate from an exudate on protein content, specific gravity and cellularity, and say what each implies.

1.1 The Normal Microcirculation

Before you can call something abnormal you need the baseline. A capillary bed sits between an **arteriole** (inflow, high pressure, controlled by smooth-muscle tone) and a **venule** (outflow, low pressure, essentially passive). The volume of blood standing in the bed at any moment is simply the balance of those two: *volume = what comes in – what goes out*.



Fig 1.1 Normal capillary bed. Arteriolar inflow (red) and venular outflow

There are therefore only two ways to increase the local blood volume, and they are mirror images of one another: (blue) are balanced; the bed contains a modest, steady volume of blood.

- **Open the tap wider** — arteriolar dilation increases inflow. This is **hyperemia** (active).
- **Block the drain** — impaired venous outflow traps blood in the bed. This is **congestion** (passive).

Both are grouped under the umbrella term **hyperemia** in the classical literature — "a local increase in the volume of blood caused by dilatation of the small vessels" — but in modern usage *hyperemia* alone means the active arterial form, and the passive form is called *congestion*. Your examiner will expect the modern usage.

HIGH-YIELD

Hyperemia and congestion both increase *intravascular* volume. Edema increases *extravascular* (interstitial) volume. Congestion very often causes edema, because the raised venous pressure that produces the congestion is also the force that drives fluid across the capillary wall — but they are not the same thing and are not interchangeable terms.

1.2 Hyperemia — the Active, Arterial Process

DEFINITION

Hyperemia (active hyperemia) is an **active** increase in local blood volume produced by **arteriolar dilation**, which augments arterial inflow into a tissue.

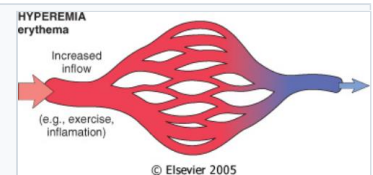


Fig 1.2 Hyperemia. Increased arteriolar inflow engorges the bed with oxygenated blood — hence erythema (redness) and local warmth.

The word *active* is doing real work here. The vessel is not passively filling; a signal has actively relaxed arteriolar smooth muscle. The commonest signals are neurogenic (sympathetic withdrawal, as in blushing), metabolic (adenosine, CO_2 , H^+ , K^+ accumulating in exercising muscle), and inflammatory (histamine, bradykinin, prostaglandins, nitric oxide at a site of injury).

○ MECHANISM — WHY HYPEREMIC TISSUE IS RED AND WARM

- 1 A vasodilator signal (metabolic, neurogenic or inflammatory) relaxes arteriolar smooth muscle.
WHY: Arteriolar tone is the main resistance point of the circulation; relaxing it drops downstream resistance more than any other single change.
- 2 Resistance falls, so flow through the capillary bed rises sharply.
WHY: $\text{Flow} \propto \text{pressure gradient} \div \text{resistance}$; with the gradient unchanged, halving resistance doubles flow.
- 3 The bed fills with blood arriving *directly from the arterial side*, which is fully oxygenated and therefore bright red.
WHY: Oxyhemoglobin is scarlet; deoxyhemoglobin is dark purple-blue. Colour follows the saturation of the haemoglobin actually present.
- 4 Because flow is fast, oxygen extraction per unit volume is *low* — blood leaves the bed still well oxygenated.
WHY: Transit time is short, so each red cell gives up less of its oxygen before it exits.
- 5 The large volume of warm core blood delivered per minute raises the surface temperature of the tissue.
WHY: Blood is the body's main convective heat carrier; more flow means more heat delivered to the periphery.

NET RESULT A tissue that is **bright red (erythema)**, **warm**, and **slightly swollen**, with **increased metabolism and function**. Two of the five cardinal signs of inflammation — *rubor* and *calor* — are simply active hyperemia.

Typical settings

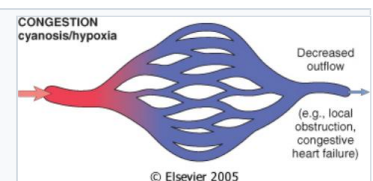
- **Exercising skeletal muscle** — local metabolites accumulate and dilate arterioles, matching supply to demand. Flow to a working muscle can rise more than twentyfold.
- **Blushing** — emotional stimuli withdraw sympathetic vasoconstrictor tone from facial skin.
- **Sites of acute inflammation** — histamine and other mediators dilate arterioles, delivering leukocytes, antibodies and complement to the injured tissue.
- **Post-ischemic reactive hyperemia** — after a temporary occlusion is released, accumulated metabolites cause a transient flow overshoot.

Hyperemia is beneficial. It is a physiological, adaptive response that raises delivery of oxygen, nutrients and immune effectors exactly where they are needed, and it resolves when the stimulus stops. This is a genuine conceptual difference from congestion, which is always pathological.

1.3 Congestion — the Passive, Venous Process

DEFINITION

Congestion (venous or passive hyperemia) is a **passive** increase in local blood volume caused by **impaired venous outflow** from a tissue.



Nothing is actively dilating here. Blood simply cannot leave, so it backs up. This is the key to every downstream consequence: the stagnant blood in the bed has already given up its oxygen to the tissue, so the bed is filled with **deoxygenated** haemoglobin.

Fig 1.3 Congestion. Outflow is obstructed, so the bed fills with deoxygenated blood held up on the venous side — hence cyanosis and local coolness.

Etiology — three headings

TABLE 1.1 CAUSES OF CONGESTION, FROM THE VESSEL OUTWARDS

Category	Mechanism	Representative examples
Compression on the vein	An external mass squeezes the thin, low-pressure venous wall shut. Veins collapse far more easily than arteries because their walls are thin and their intraluminal pressure is low.	Tumour, gravid uterus on the IVC, tight cast or bandage, constricting scar, volvulus and hernia strangulating mesenteric veins
Obstruction within the vein	The lumen itself is plugged, most often by thrombus.	Deep vein thrombosis, portal vein thrombosis, hepatic vein thrombosis (Budd–Chiari syndrome), IVC tumour thrombus from renal cell carcinoma

Category	Mechanism	Representative examples
Cardiac failure	The pump cannot clear the blood delivered to it, so pressure rises retrogradely through the entire venous bed behind the failing chamber. This is the only systemic (generalised) cause.	Left heart failure → pulmonary congestion; right heart failure → hepatic, splenic, renal and lower-limb congestion

○ MECHANISM — WHY CONGESTED TISSUE IS BLUE, COOL AND SWOLLEN

- 1 Venous outflow is obstructed (compression, thrombus, or a failing heart behind it).
WHY: The venous side is the only low-resistance escape route for capillary blood; block it and the blood has nowhere else to go.
- 2 Blood accumulates and pressure rises retrogradely into the capillaries and venules.
WHY: Pressure in a closed hydraulic system transmits backwards from the point of obstruction.
- 3 The bed distends with blood that has *already perfused* the tissue and is therefore desaturated.
WHY: Deoxyhaemoglobin is dark blue-purple; ≥ 5 g/dL of it in the capillary bed is visible through the skin as cyanosis.
- 4 Because blood is stagnant, transit time is long and the tissue extracts a *greater* fraction of the oxygen still present, deepening the desaturation further.
WHY: Slow flow gives each red cell more time at the tissue interface — extraction per unit volume rises even as total delivery falls.
- 5 Delivery of warm core blood per minute falls, so the tissue cools.
WHY: Convective heat delivery is proportional to flow, and flow is now reduced despite the increased standing volume.
- 6 The raised capillary hydrostatic pressure drives water and salts across the capillary wall into the interstitium.
WHY: This is the Starling equation acting directly — see §1.7. It is why congestion and edema almost always travel together.
- 7 Chronic stagnation causes hypoxia of the parenchymal cells, capillary rupture with small haemorrhages, and ultimately scarring.
WHY: Cells at the far (most distal) end of the perfusion path see the lowest oxygen tension and die first; their loss is repaired by fibrosis, not regeneration.

NET RESULT A tissue that is **blue-red (cyanotic)**, **cool**, **swollen and heavy**, with **reduced metabolism and function**, and — if the process persists — **haemorrhage, hemosiderin deposition, parenchymal atrophy and fibrosis**.

Gross morphology of any congested organ

Four features, and each follows directly from the mechanism above:

- **Increased in size and weight** — the organ is holding extra blood and extra interstitial fluid.
- **Firm (solidified) consistency** — distended vessels and edematous interstitium resist deformation.
- **Dark red / blue-purple colour** — the pooled blood is deoxygenated.
- **Tense, stretched capsule** — the swollen parenchyma pushes outward against an inelastic capsule; this is why an acutely congested liver is *painful* (capsular stretch).

On sectioning, a congested organ oozes dark blood freely from the cut surface.

CLINICAL CORRELATION — VARICOSE VEINS

Chronically raised intraluminal pressure in the superficial leg veins stretches them until their valves no longer meet. Once a valve is incompetent, standing blood is no longer broken into short columns by the valve cusps; the full hydrostatic column from the right atrium to the ankle now bears on the vein wall, which dilates further — a self-reinforcing loop. The result is chronic lower-limb congestion with **stasis dermatitis**, brown haemosiderin staining of the skin, and poorly-healing **varicose ulcers**. Note the exam point: embolism from *superficial* varicose veins is rare, whereas embolism from *deep* vein thrombosis is common and dangerous.

1.4 Hyperemia versus Congestion

TABLE 1.2 HYPEREMIA VS CONGESTION — THE COMPARISON YOU WILL BE EXAMINED ON

Feature	Hyperemia (active)	Congestion (passive)
Primary event	Arteriolar dilation → increased inflow	Venous obstruction → decreased out-flow
Nature	Active — an energy-requiring, signalled vasodilation	Passive — blood simply backs up
Blood in the bed	Oxygenated arterial blood	Deoxygenated venous blood
Colour	Bright red — erythema / rubor	Blue-purple — cyanosis
Temperature	Increased (more warm blood per minute)	Decreased (less flow per minute)
Distribution	Always local	Local (venous obstruction) or systemic (heart failure)
Effect on tissue	Beneficial — increased metabolism and function	Harmful — hypoxia, reduced function
If prolonged	Resolves completely when the stimulus stops	Haemorrhage → haemosiderin → parenchymal atrophy → fibrosis
Associated edema	Minimal (inflammatory hyperemia excepted)	Frequent and often marked
Classic examples	Exercising muscle; blushing; acute inflammation	Heart failure; DVT; tumour compressing a vein

MNEMONIC

A-R-T → **Active, Red, Toasty** = hyper**A**emia is arterial. Everything else (blue, cool, passive) is congestion.

EXAM TRAP

Both processes are sometimes called "hyperemia" in older texts (hence "*venous* hyperemia" for congestion). If a question offers both "hyperemia" and "congestion" as options for a cyanotic, cool, swollen leg with a DVT, the answer is **congestion**. Read the qualifier: *active* = arterial = hyperemia; *passive/venous* = congestion.

1.5 The Congested Lung

The pulmonary circulation drains into the **left atrium**. Anything that raises left atrial pressure — failing left ventricle, mitral stenosis, mitral regurgitation — is transmitted backwards into the pulmonary veins and capillaries. Pulmonary congestion is therefore the signature lesion of **left**-sided heart disease.

Acute pulmonary congestion



Fig 1.4 Acutely congested lungs: heavy, dark red-purple, dripping frothy blood-tinged fluid from the cut surface.

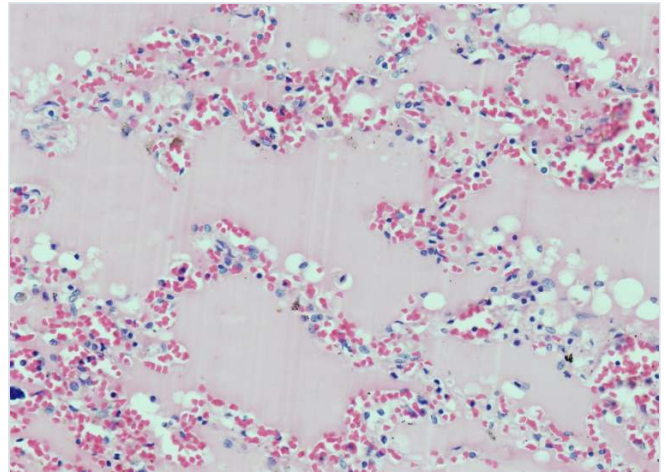


Fig 1.5 Histology: alveolar septal capillaries are dilated and engorged with red cells; pale protein-poor edema fluid fills some alveoli; scattered red cells have escaped into the airspaces.

Gross: heavy, wet, dark red-purple lungs. The cut surface oozes frothy, blood-tinged fluid — frothy because edema fluid is being whipped with air in the airways.

Microscopic — three findings: (i) alveolar septal capillaries are **dilated and packed with red cells**; (ii) **protein-poor edema fluid** (transudate) fills alveolar spaces, staining pale pink and homogeneous; (iii) **scattered red cells** lie free within alveoli where capillaries have leaked or ruptured.

Chronic pulmonary congestion and the heart failure cell

When left heart failure persists for months, the lung changes from a passively wet organ into a structurally remodelled one. The classical name is **brown induration of the lung**: *brown* from haemosiderin, *induration* (hardening) from fibrosis.

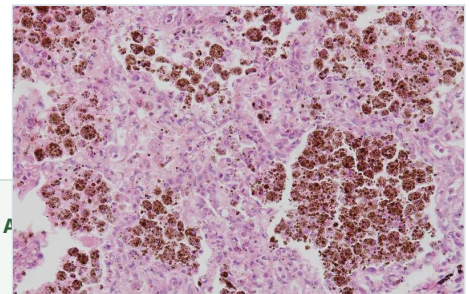


Fig 1.6 Chronic pulmonary congestion. Alveolar septa are **thickened and fibrotic**; alveolar spaces contain **haemosiderin-laden macrophages** — the heart failure cells — recognisable as large cells stuffed with coarse golden-brown granules.

○ MECHANISM — HOW A FAILING LEFT VENTRICLE PRODUCES HEART FAILURE

- 1 The left ventricle fails, so it cannot eject its full end-diastolic volume. Residual volume accumulates and left ventricular end-diastolic pressure rises.
WHY: A weakened pump ejects a smaller fraction of what it receives; what is not ejected stays behind and raises filling pressure.
- 2 That pressure is transmitted backwards: left ventricle → left atrium → pulmonary veins → pulmonary capillaries.
WHY: During diastole the mitral valve is open, so the ventricle, atrium and pulmonary veins form one continuous column of fluid; pressure equilibrates along it.
- 3 Pulmonary capillary hydrostatic pressure rises above the plasma oncotic pressure, and fluid is forced into the alveolar septa and then into the airspaces.
WHY: Net filtration = (hydrostatic gradient) – (oncotic gradient). Once hydrostatic pressure exceeds ~25 mmHg, the oncotic pull of albumin can no longer hold the fluid in.
- 4 The same pressure over-distends thin-walled septal capillaries until small numbers of them rupture, releasing red blood cells into the alveolar space.
WHY: An alveolar capillary wall is ~0.5 µm thick — built for gas exchange, not for pressure. Chronic over-distension causes microscopic breaks.
- 5 Alveolar macrophages phagocytose the extravasated red cells.
WHY: Free red cells in the airspace are debris; clearing them is the resident macrophage's job.
- 6 Inside the phagolysosome, haemoglobin is degraded. The globin is recycled and the porphyrin ring is opened, but the **iron cannot be excreted**, so it is stored as ferritin aggregates that condense into **haemosiderin**, a coarse, golden-brown, refractile pigment.
WHY: The body has no route for excreting iron in quantity; the only options are re-export or local storage. A macrophage sitting in an alveolus cannot export it efficiently, so it accumulates.
- 7 These pigment-stuffed macrophages are the **heart failure cells** (siderophages). They accumulate in alveoli and can be coughed up in sputum.
WHY: They are simply macrophages that have eaten too much blood; the name reflects the clinical context in which they are found, not a unique cell type.
- 8 In parallel, chronic hypoxia and the mechanical stress of persistent edema stimulate septal fibroblasts to lay down collagen, and the septa thicken.
WHY: Hypoxia and stretch are both profibrotic stimuli; the lung repairs recurrent low-grade injury with scar because alveolar architecture cannot be perfectly regenerated.

NET RESULT Brown induration of the lung — firm, brown, heavy lungs with rust-coloured spots on the cut surface; microscopically, thickened fibrotic alveolar septa plus haemosiderin-laden macrophages in the alveoli. Functionally, the thickened septum impairs gas diffusion, so the patient becomes dyspnoeic.



Figure 4.42. Chronic pulmonary congestion.

Fig 1.7 Chronic pulmonary congestion, gross. The pleural surface and cut surface show **rust-coloured spots**, each a focus of old alveolar haemorrhage now converted to haemosiderin.

HIGH-YIELD — THREE FEATURES OF CHRONIC PULMONARY CONGESTION

1. **Thickened, fibrotic alveolar septa** (the "induration").
2. **Haemosiderin-laden macrophages (heart failure cells)** in the alveolar spaces (the "brown").
3. **Dilated, congested septal capillaries** with variable interstitial edema.

Grossly: firm, heavy, brown lungs with **rust-coloured spots**. This is the lecture's named learning objective — expect it verbatim.

CLINICAL CORRELATION

A patient with chronic left heart failure or **mitral stenosis** presents with exertional dyspnoea, orthopnoea, paroxysmal nocturnal dyspnoea and a cough productive of **rust-coloured or blood-tinged sputum** containing heart failure cells. In long-standing mitral stenosis, the chronically raised pulmonary venous pressure is eventually transmitted to the pulmonary arteries, producing pulmonary hypertension, right ventricular hypertrophy and finally *right* heart failure — left-sided disease that has "crossed over" to become right-sided. Note the sequence: chronic **congestion** of the lung also predisposes to **red infarction** after a pulmonary embolus, because the compensatory bronchial supply is already inadequate (see §5.3).

1.6 The Congested Liver and Nutmeg Liver

The hepatic veins drain into the **inferior vena cava** and thence into the **right atrium**. Hepatic congestion is therefore the signature lesion of **right-sided** heart failure (and of any obstruction of hepatic venous outflow, such as Budd–Chiari syndrome).

The essential anatomy — why the centre dies first

Blood enters the liver lobule at its **periphery**, from the portal triads (portal venule + hepatic arteriole), flows inward along the sinusoids, and exits at the **centre** through the **central vein**. Oxygen is extracted all the way along that path. Therefore:

Zone 1 (periportal) — highest pO_2 — **Zone 2** — **Zone 3** (centrilobular) — lowest pO_2 , **dies first**

Two facts follow, and between them they explain the whole of nutmeg liver: (i) the **centrilobular** (zone 3) hepatocytes are the most hypoxia-vulnerable cells in the lobule because they see blood that is already partly deoxygenated; and (ii) the **central vein** is the first structure to feel back-pressure from the right heart, because it is the most downstream point in the lobule.



Fig 1.8 A bisected nutmeg (the spice) — dark radiating lines on a paler ground. The name of the lesion comes from this resemblance.

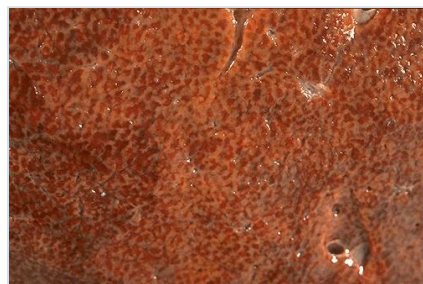


Fig 1.9 Nutmeg liver, cut surface: a fine mottled pattern of dark red foci (congested centrilobular zones) alternating with pale tan zones (viable, often fatty, periportal parenchyma).



Fig 1.10 Fixed specimen, chronic passive hepatic congestion. The mottling is present throughout the organ, which is enlarged with a tense capsule.

○ MECHANISM — HOW THE NUTMEG PATTERN IS BUILT, STEP BY STEP

- 1** The right ventricle fails (or the hepatic veins are obstructed), so systemic venous pressure rises.
WHY: *The right heart is the end of the systemic venous circuit; when it cannot empty, everything behind it becomes engorged.*
- 2** Pressure is transmitted backwards: right atrium → inferior vena cava → hepatic veins → **central veins** of the lobules.
WHY: *These form an uninterrupted low-pressure column; there is no valve between the right atrium and the hepatic sinusoids to interrupt transmission.*
- 3** The central veins and the sinusoids immediately around them dilate and fill with dark venous blood.
WHY: *They are the thinnest-walled, most distensible and most downstream vessels in the lobule, so they yield to the raised pressure first.*

- 4 Sinusoidal flow slows to a crawl, so oxygen delivery to zone 3 falls below the level the hepatocytes need.

WHY: Those cells were already at the lowest pO_2 in the lobule; any further reduction crosses their threshold for survival while zone 1 cells, bathed in fresh portal and arterial blood, are still comfortable.

- 5 Centrilobular hepatocytes undergo hypoxic injury: first fatty change and atrophy, then, if the hypoxia is severe or prolonged, **centrilobular haemorrhagic necrosis**.

WHY: Hypoxia halts oxidative phosphorylation; fatty acids can no longer be β -oxidised and accumulate as fat, and ATP depletion eventually causes irreversible membrane failure and cell death. The necrotic zone is haemorrhagic because it is simultaneously full of pooled blood.

- 6 Meanwhile the **periportal** (zone 1) hepatocytes survive, because they receive blood first. They commonly develop **fatty change** from milder hypoxia and from the chronic illness itself, which turns them yellow-tan.

WHY: Fatty change is the reversible, "lesser" hypoxic injury; it stops short of necrosis and gives the surviving tissue a pale yellow colour that contrasts sharply with the dark centre.

- 7 The cut surface therefore shows **dark red centrilobular foci** set in a **pale yellow-tan periportal background**, repeated in every lobule across the whole organ.

WHY: The lobule is a repeating unit, so a lesion that targets one zone of the lobule produces a regular, uniform mottling of the entire cut surface rather than a patchy lesion.

NET RESULT A cut surface that looks like a **bisected nutmeg**. The dark red = congested, necrotic centrilobular zones. The pale tan = surviving, often fatty, periportal parenchyma. If congestion persists for years, collagen is laid down around the central veins — **cardiac sclerosis**, sometimes progressing to true **cardiac cirrhosis**.

EXAM TRAP — WHICH PART OF THE NUTMEG IS DARK?

Students routinely invert this. The **dark red** areas are **centrilobular** (around the central vein) — congested and necrotic. The **pale** areas are **periportal** — surviving. The logic is unambiguous once you remember that blood *backs up from* the central vein, so congestion must start there. A second trap: nutmeg liver is a lesion of **right** heart failure; brown induration of the lung is a lesion of **left** heart failure. Do not mix the sides.

HIGH-YIELD — THREE FEATURES OF NUTMEG LIVER

1. Dilated, congested central veins and perivenular sinusoids.
2. Centrilobular hepatocyte atrophy, degeneration and haemorrhagic necrosis.
3. Periportal hepatocytes preserved, commonly with fatty change — producing the pale component of the mottling.

Clinically: tender hepatomegaly, raised jugular venous pressure, peripheral edema, ascites, and mildly raised transaminases. In acute severe right heart failure the transaminase rise can be dramatic ("ischemic hepatitis" / shock liver).

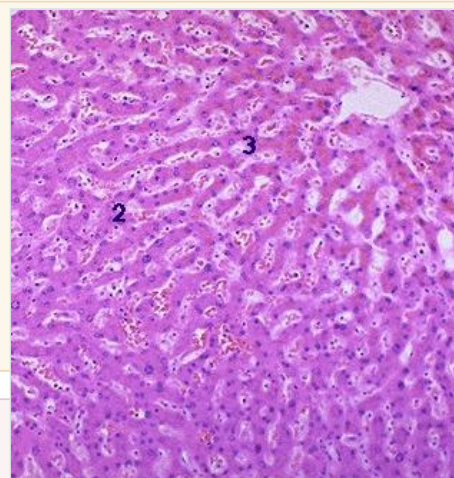


Fig 1.11 Chronic hepatic congestion, H&E. 1 Dilated central vein and congested perivenular sinusoids packed with red cells. 2 Atrophic, degenerating centrilobular hepatocyte cords compressed between the engorged sinusoids. 3 Intact periportal hepatocytes, often showing fatty vacuolation.

1.7 Edema — Starling Forces and the Physiology of Leakage

DEFINITION

Edema is an **increase in fluid within the interstitial tissue spaces**, or within body cavities. It is an *extravascular* accumulation — the water has left the vessel; the blood cells have not.

Roughly 60% of body weight is water, and about two-thirds of that is *intracellular*. Edema concerns the remaining extracellular compartment, and specifically the balance between its two sub-compartments — plasma (inside the vessel) and interstitium (outside it). The wall of the capillary is freely permeable to water and small solutes but relatively impermeable to plasma proteins. Movement of fluid across that wall is governed by four **Starling forces**:

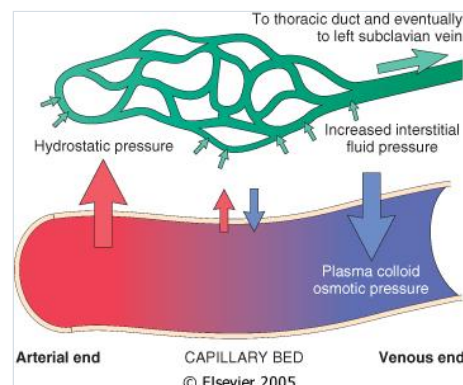


Fig 1.12 Starling forces across a capillary. Hydrostatic pressure (red arrow) pushes fluid out and falls from the arterial to the venous end; plasma colloid osmotic pressure (blue arrow) pulls fluid in and is nearly constant. Net filtration occurs at the arterial end, net reabsorption at the venous end, and the small unabsorbed surplus is removed by lymphatics.

TABLE 1.3 THE FOUR STARLING FORCES

Force	Direction	Comment
Capillary hydrostatic pressure (P_c)	OUT	The blood pressure inside the capillary. ~32 mmHg at the arterial end, ~12 mmHg at the venous end. The force most often deranged in disease.
Plasma colloid osmotic (oncotic) pressure (π_c)	IN	Generated almost entirely by albumin , which is too large to cross. ~25 mmHg, essentially constant along the capillary.
Interstitial hydrostatic pressure (P_i)	IN	Normally near zero or slightly negative; rises as the interstitium fills, which is a natural brake on further edema.
Interstitial oncotic pressure (π_i)	OUT	Normally low, because lymphatics continuously remove the few proteins that escape. Rises sharply in inflammation and in lymphatic obstruction.

$$\text{Net filtration} = K_f \times [(P_c - P_i) - \sigma(\pi_c - \pi_i)] \rightarrow \text{edema whenever this exceeds what the lymphatics can carry away}$$

In health there is a small **net filtration** of fluid into the interstitium — roughly 2–4 litres per day across the whole body — and the **lymphatic system** returns exactly that amount to the circulation via the thoracic duct. Edema therefore requires one of two things: an increase in filtration that exceeds lymphatic capacity, or a failure of lymphatic drainage. The lymphatics can up-regulate their flow perhaps tenfold, which is why modest increases in capillary pressure produce no visible edema at all.

HIGH-YIELD

Every cause of edema you will ever be asked about works by altering exactly one of four variables: $\uparrow P_c$, $\downarrow \pi_c$, $\uparrow$ lymphatic resistance, or $\uparrow$ capillary permeability (which raises π_i by letting protein out). Sodium retention is best understood as a mechanism that raises P_c by expanding plasma volume. If you can classify a scenario into one of these four, you have answered the question.

1.8 The Four Mechanisms of Non-inflammatory Edema

① Increased hydrostatic pressure

Anything that impedes venous return raises P_c upstream of the obstruction.

- **Local:** deep vein thrombosis → unilateral leg edema; constrictive pericarditis; a tumour compressing a vein; prolonged standing or immobility.
- **Generalised:** congestive heart failure — the single commonest cause of edema in clinical practice.

○ MECHANISM — THE VICIOUS CIRCLE OF EDEMA IN CONGESTIVE HEART FAILURE

- 1 Cardiac output falls because the ventricle cannot eject effectively.

WHY: This is the primary defect; everything downstream is compensation or consequence.

- 2 Blood dams up behind the ventricle, raising venous and hence capillary hydrostatic pressure. Fluid begins to filter into the interstitium.

WHY: The Starling balance is tipped outward by the raised P_c .

- 3 Renal perfusion falls, because cardiac output is low and because fluid has moved out of the vascular compartment.

WHY: The kidney senses arterial underfilling, not total body water; it cannot distinguish "not enough blood" from "blood in the wrong compartment".

- 4 The juxtaglomerular apparatus releases **renin** → angiotensin I → angiotensin II (via ACE) → **aldosterone**; ADH is also released.

WHY: This is the body's standard response to perceived hypovolaemia — retain salt and water to restore circulating volume.

- 5 The kidney retains sodium and water, expanding plasma volume.

WHY: Aldosterone drives Na^+ reabsorption in the distal nephron; water follows osmotically, assisted by ADH-driven aquaporin insertion.

- 6 But the heart still cannot pump the extra volume, so the added fluid simply raises venous pressure further.

WHY: Expanding the volume in front of a broken pump does not improve forward flow; it only increases the pressure behind it. **This is the vicious circle.**

- 7 Edema worsens, arterial underfilling persists, renin release continues.

WHY: The compensation is now the disease. This is exactly why diuretics and ACE inhibitors — which interrupt steps 4–5 — are cornerstone treatments in heart failure.

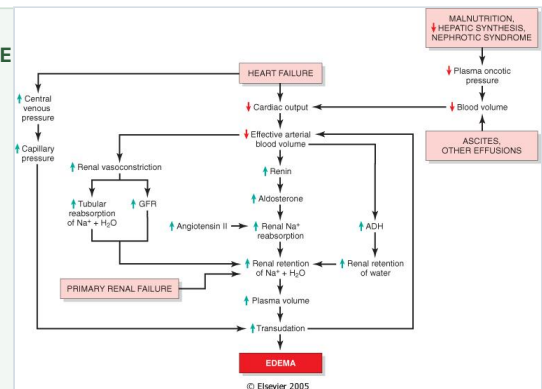


Fig 1.13 Why heart failure and renal failure both end in edema. Note the two arms in heart failure: the mechanical arm ($\uparrow$ hydrostatic pressure) and the neurohormonal arm (renin-angiotensin-aldosterone → Na^+ and water retention → $\uparrow$ plasma volume → further $\uparrow$ hydrostatic pressure). The second arm is the reason a failing heart produces far more edema than the pump defect alone would predict.

NET RESULT Progressive, dependent, pitting edema with raised jugular venous pressure, hepatomegaly and — eventually — ascites and anasarca.

② Reduced plasma oncotic pressure (hypoalbuminaemia)

Albumin is the molecule that holds water inside the vessel. When serum albumin falls below roughly **2.5 g/dL**, the inward oncotic pull is no longer sufficient to balance even a normal hydrostatic pressure, and fluid leaves the circulation everywhere at once. The result is **generalised** edema, most obvious where tissues are loose — periorbital, scrotal, and eventually anasarca.

- **Excessive loss — nephrotic syndrome:** damaged glomerular capillary walls leak albumin into the urine (>3.5 g/day). The classic presentation is periorbital edema worst on waking.
- **Reduced synthesis — cirrhosis** (the diseased liver cannot make albumin) and **protein malnutrition** (kwashiorkor: no substrate to make it from). The protuberant abdomen of a child with kwashiorkor is ascites on a background of muscle wasting.
- **Protein-losing enteropathy** — albumin is lost across a diseased gut mucosa.

The circle closes here too. Loss of albumin reduces intravascular volume → renal hypoperfusion → renin-angiotensin-aldosterone activation → salt and water retention. But the retained fluid cannot be held in the vessel either, because albumin is still low. So it too leaks out, worsening the edema without correcting the underlying deficit.

③ Lymphatic obstruction — lymphedema

Here filtration is normal but **drainage** has failed. Because the lymphatics are also the route by which escaped plasma protein is returned to the blood, obstruction causes protein to accumulate in the interstitium. That raises interstitial oncotic pressure (π_i), which pulls yet more water out — so lymphedema is **self-perpetuating and protein-rich**, which is why it becomes firm and non-pitting over time rather than staying soft.

- **Surgical** — axillary lymph node dissection in breast cancer surgery; classic cause of arm lymphedema.
- **Neoplastic** — tumour infiltrating lymphatic channels or nodes. In breast carcinoma, dermal lymphatic invasion tethers the skin and produces the dimpled **peau d'orange** ("orange peel") appearance — the skin is edematous but remains anchored at the hair follicles, which therefore appear as pits.
- **Post-radiation fibrosis; post-inflammatory scarring and thrombosis** of lymphatics.
- **Filariasis** — *Wuchereria bancrofti*. Adult nematodes live in and obstruct lymphatic channels, provoking fibrosing inflammation of inguinal nodes and lymphatics. Massive chronic lymphedema of the leg and scrotum with thickened, verrucous skin is **elephantiasis**.
- **Milroy disease** — primary congenital lymphedema from agenesis or hypoplasia of lymphatics.

*If obstructed lymphatics rupture, milky lymph accumulates in cavities: **chylous ascites, chylothorax, chylopericardium**.*

④ Sodium and water retention

Salt is osmotically obliged to carry water with it. Retaining sodium therefore expands the extracellular fluid volume, which raises capillary hydrostatic pressure and simultaneously *dilutes* plasma proteins, lowering oncotic pressure. Both Starling forces move in the direction of edema at once — which is why this mechanism is so potent.

- **Primary renal failure** — the damaged kidney simply cannot excrete the filtered sodium load (acute glomerulonephritis is the classic example; children present with periorbital edema, hypertension and cola-coloured urine).

- **Secondary** — the renin–angiotensin–aldosterone activation described above in heart failure, cirrhosis and nephrotic syndrome.
- **Excessive salt intake with renal insufficiency**; some drugs (NSAIDs, corticosteroids, some antihypertensives).

MNEMONIC — THE FOUR CAUSES

"Pressure Out, Protein Low, Lymph Blocked, Salt Held"

Pressure ($\uparrow$ hydrostatic) · Protein ($\downarrow$ oncotic) · Lymph (obstruction) · Salt (Na^+ retention). The fifth, inflammatory edema, belongs to a different category — see below.

1.9 Transudate versus Exudate

The four mechanisms above all produce a **transudate**: the capillary wall is intact, so only water and small solutes escape. **Inflammatory edema** is mechanistically different — chemical mediators (histamine, bradykinin, leukotrienes, C3a/C5a) make the endothelium of post-capillary venules contract and pull apart, opening interendothelial gaps. Now **protein and cells escape too**, and the fluid is an **exudate**.

TABLE 1.4 TRANSUDATE VS EXUDATE — AND WHAT EACH TELLS YOU ABOUT THE WALL

Feature	Transudate	Exudate
Underlying defect	Hydrostatic / oncotic imbalance. Vessel wall is intact .	Increased vascular permeability. Wall is leaky (inflammation).
Protein content	Low ($<2.5\text{--}3$ g/dL); fluid/serum protein ratio <0.5	High (>3 g/dL); fluid/serum protein ratio >0.5
Specific gravity	<1.012	>1.020
LDH	Low; fluid/serum LDH <0.6	High; fluid/serum LDH >0.6 (Light's criteria)
Cells	Few; mainly mesothelial cells	Many; neutrophils, lymphocytes, sometimes malignant cells
Fibrinogen / clotting	Absent — does not clot	Present — clots on standing
Appearance	Clear, straw-coloured, watery	Turbid, cloudy, may be purulent or haemorrhagic
Typical causes	Heart failure, cirrhosis, nephrotic syndrome, hypoalbuminaemia	Infection (pneumonia, TB), malignancy, pulmonary embolism, pancreatitis, autoimmune serositis

CLINICAL CORRELATION — WHY THE DISTINCTION MATTERS AT THE BEDSIDE

When you aspirate a pleural effusion, the transudate/exudate call redirects the entire work-up. A **transudate** says "the vessel is fine — go and look at the heart, liver or kidney." An **exudate** says "something is wrong with the pleura itself — think infection, malignancy, pulmonary embolism or connective tissue disease," and mandates cytology, culture and often further imaging. **Light's criteria** (any one of: fluid/serum protein >0.5, fluid/serum LDH >0.6, fluid LDH > $\frac{2}{3}$ the upper limit of normal serum LDH) define an exudate.

EXAM TRAP

Pulmonary edema of **cardiogenic** origin is a transudate (raised hydrostatic pressure, intact wall). Pulmonary edema of **ARDS** is an exudate (diffuse alveolar damage, leaky wall). Same organ, same clinical picture of hypoxaemia and bilateral infiltrates, opposite mechanism and opposite treatment. If the vignette mentions sepsis, aspiration, trauma or pancreatitis with a *normal* heart, think ARDS.

1.10 Morphology and Clinical Patterns of Edema

Named fluid collections

TABLE 1.5 TERMINOLOGY FOR FLUID IN CAVITIES

Term	Cavity	Note
Hydrothorax (pleural effusion)	Pleural cavity	Compresses lung → atelectasis, dyspnoea, dullness to percussion, absent breath sounds
Hydropericardium	Pericardial cavity	A large or rapidly accumulating effusion impairs diastolic filling → tamponade
Hydroperitoneum (ascites)	Peritoneal cavity	Commonest cause is cirrhosis with portal hypertension + hypoalbuminaemia
Anasarca	Whole body, subcutaneous	Severe generalised edema with profound subcutaneous swelling; implies advanced heart, liver or renal failure



Fig 1.14 Hydrothorax. Chest radiograph shows homogeneous opacification of the lower hemithorax with a meniscus at its upper border.



Fig 1.15 Ascites: tense, distended abdomen with everted umbilicus and shifting dullness.

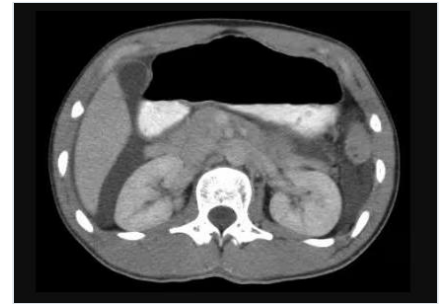


Fig 1.16 CT of the same process — free fluid of water density surrounding the abdominal viscera.

Subcutaneous edema and the pitting sign

Dependent edema is the distribution characteristic of raised hydrostatic pressure: gravity adds to capillary pressure in whichever part of the body is lowest. In an ambulant patient this is the **legs and ankles**; in a bedbound patient it is the **sacrum**. Always examine the sacrum in a patient who has been lying down — leg edema may be absent there.

Periorbital edema, by contrast, points to hypoalbuminaemia or acute glomerulonephritis: the periorbital tissue is so loose that it distends with even small amounts of fluid, and it is not gravity-dependent. This is why **nephrotic syndrome** classically presents with puffy eyelids on waking.

Pitting versus non-pitting. Low-protein transudate is mobile and pits readily. Protein-rich lymphedema is viscous, and chronic lymphedema deposits collagen in the interstitium, so it becomes firm, woody ("brawny induration") and **non-pitting**. Myxoedema in hypothyroidism is also non-pitting, but for a different reason: the interstitium is filled with hydrophilic glycosaminoglycans, not free water.



Fig 1.17 Pitting edema. Firm finger pressure displaces the low-viscosity, protein-poor interstitial fluid sideways; the depression persists for seconds to minutes because the fluid takes time to flow back.

Pulmonary edema

The lungs may weigh two to three times normal. Sectioning releases frothy, sometimes blood-tinged fluid — a mixture of air, edema fluid and extravasated red cells. Histologically, alveolar capillaries are engorged and the alveolar spaces contain a granular pink precipitate.

Causes: left ventricular failure (by far the commonest), mitral stenosis, renal failure with volume overload, ARDS, high-altitude exposure, and inflammatory or infectious lung disease. **Consequences:** fluid in the alveolus increases the diffusion distance for oxygen and dilutes surfactant, so the patient becomes hypoxaemic and the alveoli tend to collapse; it also creates an excellent culture medium, predisposing to secondary infection.

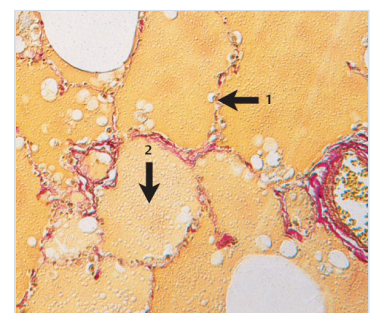


Fig 1.18 Pulmonary edema. **1** Dilated, congested septal capillaries. **2** Alveolar spaces filled with homogeneous, pale pink, protein-containing edema fluid.

Cerebral edema

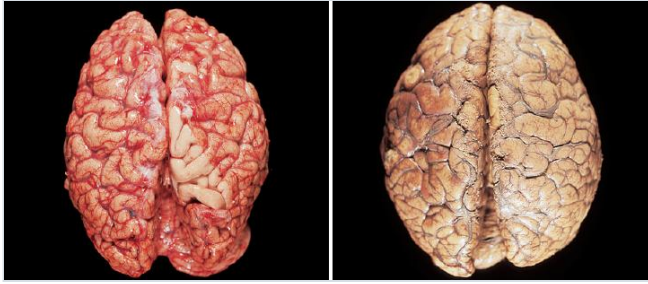


Fig 1.19 Normal brain (left) versus edematous brain (right). The gyri are **flattened and widened** and the sulci are **narrowed**, because the swollen brain is pressed against the unyielding inner table of the skull.

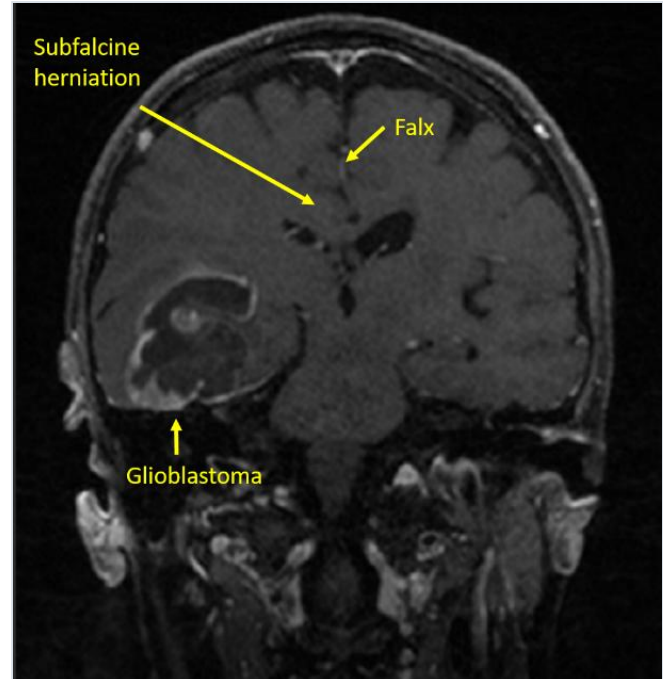


Fig 1.20 Subfalcine (cingulate) herniation. A space-occupying lesion with surrounding edema has pushed the cingulate gyrus **under the falx cerebri**, displacing the midline. Compression of the anterior cerebral artery against the falx can cause secondary infarction.

Cerebral edema is uniquely dangerous because the brain is enclosed in a rigid box. The **Monro-Kellie doctrine** states that the sum of brain, blood and cerebrospinal fluid volumes within the cranium is constant; if one increases, another must be displaced or intracranial pressure must rise. Initially CSF is displaced into the spinal subarachnoid space and venous blood is squeezed out — but these reserves are small, and once exhausted, pressure rises steeply.

Clinical features: headache (worse in the morning and on coughing or straining), nausea and vomiting, papilloedema, falling level of consciousness, and finally **herniation** — subfalcine, transtentorial (uncal) or tonsillar through the foramen magnum. Tonsillar herniation compresses the medulla and is rapidly fatal because it destroys the respiratory and cardiovascular centres.



Fig 1.21 Elephantiasis in filariasis. Chronic protein-rich lymphedema has led to massive, firm, non-pitting swelling with thickened, hyperkeratotic skin.

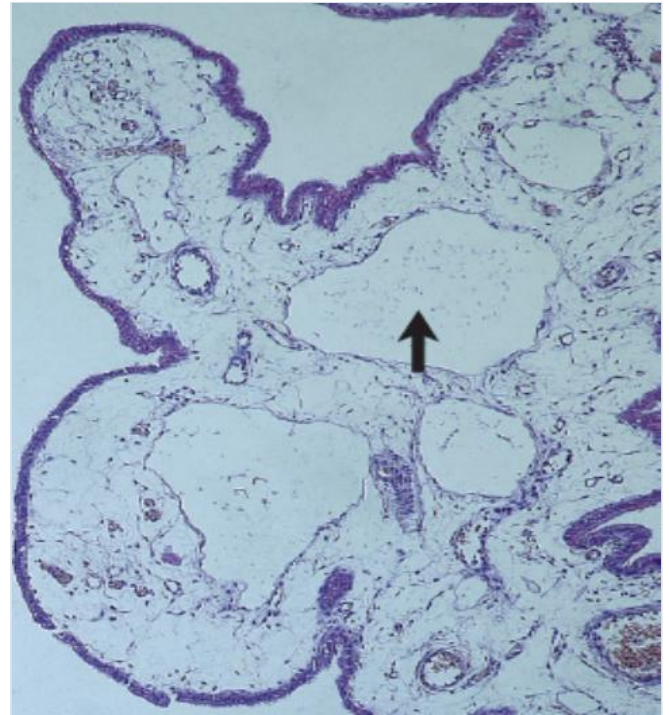


Fig 1.22 Dermal lymphatic channels distended by tumour (arrow) in inflammatory breast carcinoma — the histological basis of the peau d'orange sign.

HIGH-YIELD — READING AN EDEMA VIGNETTE IN THREE QUESTIONS

1. **Is it local or generalised?** Local → obstruction (venous or lymphatic) on that side. Generalised → heart, liver, kidney.
2. **Does it pit?** Pitting → low-protein transudate (hydrostatic or oncotic). Non-pitting → protein-rich lymphedema, or myxoedema.
3. **Where is it worst?** Legs/sacrum (dependent) → hydrostatic, i.e. heart failure. Periorbital → hypoalbuminaemia or acute glomerulonephritis. One arm after mastectomy → lymphatic.

PART TWO

02 Hemorrhage

Edema is water leaving the vessel through an intact wall. Hemorrhage is *whole blood* leaving it through a breach. The clinical question is never simply "is there bleeding?" but "how much, how fast, and where?" — because those three variables, not the diagnosis, determine whether the patient is inconvenienced or dead.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Define hemorrhage and distinguish rupture (*per rhexis*) from diapedesis.
- Apply the size-based terminology — petechia, purpura, ecchymosis — correctly and state the measurements.
- Explain the colour sequence of a resolving bruise from first principles of haem catabolism.
- Predict the consequence of a haemorrhage from its volume, rate and anatomical site.

2.1 Definition and the Two Mechanisms of Escape

DEFINITION

Hemorrhage is the **extravasation of blood** from the cardiovascular system into the extravascular space, a body cavity, or to the exterior.

Note what is and is not implied. Hemorrhage is not defined by volume — a petechia is a hemorrhage. It is not defined by visibility — a retroperitoneal bleed is invisible until the patient collapses. It is defined by *whole blood, including cells, having left the vascular compartment*. That is the difference from edema, where only the fluid phase escapes.

Two ways blood gets out

TABLE 2.1 RUPTURE VERSUS DIAPEDESIS

Mechanism	What happens to the wall	Typical result
Rupture (haemorrhage per rhexis)	A frank structural breach of the vessel wall — traumatic laceration, erosion by tumour or ulcer, or bursting of a wall weakened by aneurysm, atherosclerosis or hypertension.	Large, rapid, often life-threatening bleed. Aortic dissection, ruptured berry aneurysm, ruptured abdominal aortic aneurysm.
Diapedesis	No visible breach. Red cells squeeze one at a time between endothelial cells whose junctions have been loosened by severe congestion, hypoxia or inflammation.	Small, slow, multifocal bleeding. The scattered alveolar red cells of chronic pulmonary congestion; petechiae in severe hypoxia.

2.2 Causes of Hemorrhage

- **Trauma** — the commonest cause overall: laceration, crush, penetrating injury, fracture.
- **Vascular wall weakness** — **aneurysm** (berry aneurysm of the circle of Willis, abdominal aortic aneurysm, Charcot-Bouchard microaneurysms in chronic hypertension), **atherosclerosis**, **vasculitis**, and connective tissue disorders such as Marfan or Ehlers-Danlos syndrome.
- **Erosion by disease** — a peptic ulcer eroding the gastroduodenal artery; a bronchogenic carcinoma invading a pulmonary vessel (haemoptysis); tuberculous cavitation.
- **Bleeding disorders (haemorrhagic diatheses)** — three headings:
 - *Platelet* — thrombocytopenia (ITP, TTP, DIC, marrow failure, hypersplenism) or platelet dysfunction (uraemia, aspirin, Glanzmann thrombasthenia, Bernard-Soulier).
 - *Coagulation factor* — haemophilia A (factor VIII) and B (factor IX), von Willebrand disease, vitamin K deficiency, liver failure, warfarin.
 - *Vessel wall* — scurvy (defective collagen), Henoch-Schönlein purpura, amyloidosis, senile purpura, chronic corticosteroid use.
- **Severe congestion / hypoxia** — diapedetic haemorrhage as described above.

HIGH-YIELD — PLATELET PROBLEM OR FACTOR PROBLEM?

The **pattern** of bleeding tells you which arm of hemostasis has failed, before any laboratory test.

Primary hemostasis (platelet/vWF) defect → *mucocutaneous* bleeding: petechiae, purpura, epistaxis, gingival bleeding, menorrhagia, easy bruising. Bleeding starts **immediately** after injury because the platelet plug never forms.

Secondary hemostasis (coagulation factor) defect → *deep* bleeding: haemarthrosis, deep muscle haematomas, rebleeding **hours after** an initially controlled injury, because the weak platelet plug forms but is never reinforced with fibrin and then breaks down.

2.3 Classification by Size and Site

By size — the skin and mucosal series

TABLE 2.2 PETECHIA → PURPURA → ECCHYMOSIS. MEMORISE THE NUMBERS.

Term	Size	Typical setting	Note
Petechiae	1–2 mm	Thrombocytopenia, defective platelet function, vitamin C deficiency, raised intravascular pressure (e.g. straining, coughing, tourniquet)	Pinpoint haemorrhages into skin, mucous membrane or serosal surface. Do not blanch on pressure — this is how they are distinguished from an erythematous rash.
Purpura	3–5 mm (up to ~1 cm)	All causes of petechiae, plus trauma, vasculitis and increased vascular fragility (amyloidosis, steroids, scurvy)	Palpable purpura implies vessel wall inflammation (vasculitis, e.g. Henoch-Schönlein); flat purpura implies a platelet problem.
Ecchymosis	> 1–2 cm	Trauma (a "bruise"); also coagulopathy and the causes above	A large subcutaneous haematoma. Its evolving colour is diagnostic of its age — see §2.4.
Haematoma	Any size	Trauma, coagulopathy, vessel rupture	An accumulation of blood within a tissue , forming a discrete mass. May be trivial (a bruise) or fatal (a subdural or retroperitoneal haematoma).

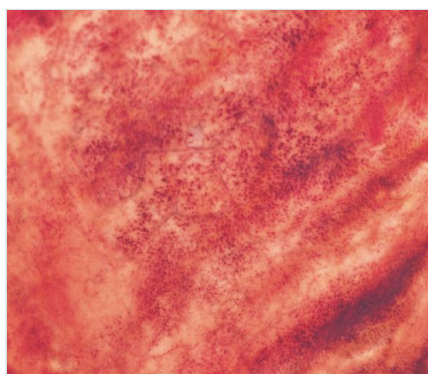


Fig 2.1 Petechiae of the gastric mucosa — pinpoint, 1–2 mm, non-blanching.



Fig 2.2 Purpuric haemorrhages of the testis, 3–5 mm — distinctly larger than petechiae and clearly visible against the pale tunica.



Fig 2.3 Ecchymoses, >1–2 cm — extensive subcutaneous haemorrhage after minor trauma in a patient with a coagulopathy.

By site — blood in a cavity

TABLE 2.3 NAMED CAVITY HAEMORRHAGES

Term	Cavity	Clinical significance
Haemothorax	Pleural cavity	The pleural space can hold litres. Rib fracture, penetrating trauma, ruptured aortic aneurysm. Causes both hypovolaemia and lung compression.
Haemopericardium	Pericardial cavity	The volume that kills is small. As little as 200–300 mL accumulating rapidly causes cardiac tamponade , because the parietal pericardium is fibrous and cannot stretch acutely. Causes: myocardial rupture after infarction, aortic dissection bleeding back into the sac, penetrating trauma.
Haemoperitoneum	Peritoneal cavity	Ruptured spleen or liver, ruptured ectopic pregnancy, ruptured abdominal aortic aneurysm. Large volumes can be lost before the abdomen looks distended.
Haemarthrosis	Joint cavity	Hallmark of haemophilia . Repeated bleeds into the knee, ankle or elbow cause synovial hypertrophy, cartilage destruction and crippling deformity.

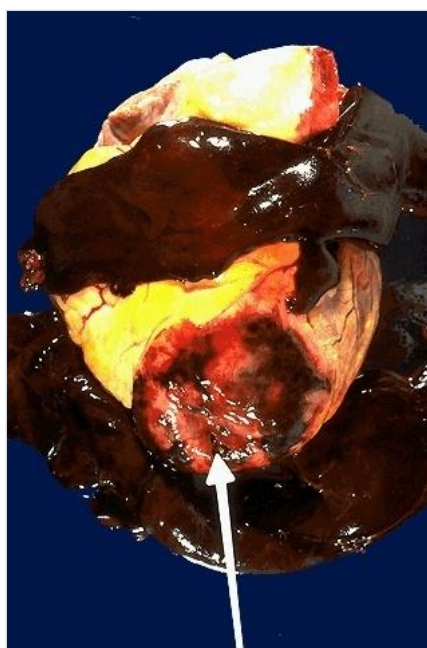


Fig 2.4 Haemopericardium. The opened pericardial sac is filled with clotted blood (arrow) following rupture of an infarcted left ventricular free wall — fatal tamponade.



Fig 2.5 Haemarthrosis of the right knee in haemophilia — a swollen, warm, exquisitely painful joint held in slight flexion.

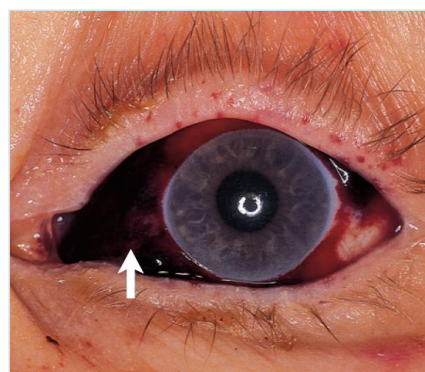


Fig 2.6 Subconjunctival haemorrhage (arrow) — alarming in appearance but usually benign, following a cough, sneeze or Valsalva.

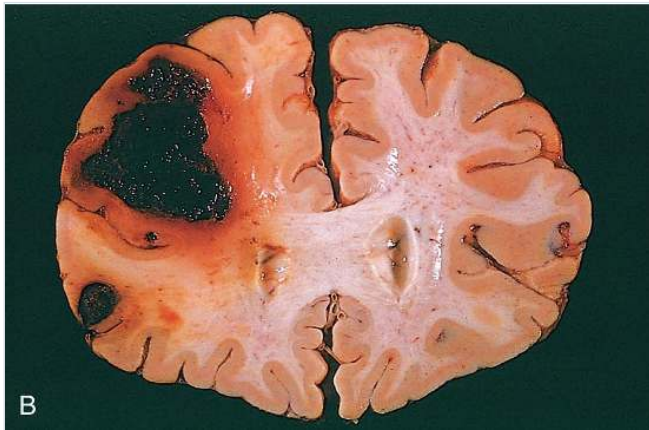


Fig 2.7 Intracerebral haemorrhage. A large haematoma has destroyed the basal ganglia and displaced the midline — the classic site of hypertensive haemorrhage, arising from ruptured Charcot–Bouchard microaneurysms of the lenticulostriate arteries.

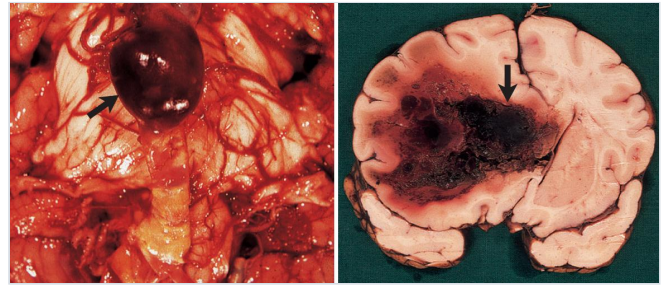


Fig 2.8 Fatal massive cerebral haemorrhage with rupture into the ventricular system. Here the lethal factor is not blood loss — it is the site.

2.4 Why a Bruise Changes Colour

The colour sequence of a resolving ecchymosis is not a curiosity — it is **haem catabolism happening in front of you**, and it lets you date an injury, which matters in forensic and paediatric practice.

MECHANISM — THE COLOUR CLOCK OF A BRUISE

- 1 **Day 0–1: red-purple.** Extravasated red cells are intact and their haemoglobin is still largely oxygenated, then desaturates.
WHY: You are simply looking at blood through skin. As the trapped blood loses its oxygen, the colour deepens from red to blue-purple.
- 2 **Day 1–5: blue-black.** Red cells lyse; free haemoglobin is phagocytosed by macrophages recruited to the site.
WHY: Extravasated red cells have no supporting environment and undergo haemolysis within a day or two; macrophages arrive as part of the normal clean-up response.
- 3 **Day 5–7: green.** Inside the macrophage, **haem oxygenase** opens the porphyrin ring of haem, releasing iron and carbon monoxide and producing **biliverdin**, which is green.
WHY: Haem oxygenase is the rate-limiting enzyme of haem degradation; biliverdin is its immediate, green-coloured product.
- 4 **Day 7–10: yellow-brown.** **Biliverdin reductase** converts biliverdin to **bilirubin**, which is yellow.
WHY: This is the same reaction that occurs in the spleen during normal red cell turnover — the bruise is simply doing it locally and visibly.
- 5 **Day 10–14+: golden-brown, fading.** The iron liberated in step 3 is stored as **haemosiderin**, a golden-brown pigment, while bilirubin diffuses away and is cleared.
WHY: Bilirubin is soluble and leaves; iron is not excretable and stays until macrophages slowly carry it off — which is why the brown stain is the last colour to disappear, and why chronic venous congestion of the leg leaves permanent brown skin staining.

NET RESULT Red-purple → blue-black → green → yellow → golden-brown. Haemoglobin → biliverdin → bilirubin → haemosiderin. The same pathway, in the same order, every time.

MNEMONIC

"Really Bad Grapes Yield Brown" — Red · Blue · Green · Yellow · Brown.

CLINICAL CORRELATION — MASSIVE HAEMORRHAGE AND JAUNDICE

A very large haematoma — a retroperitoneal bleed, or a huge subgaleal haemorrhage in a newborn — can produce clinically obvious **jaundice** a few days later. The reason follows directly from the mechanism above: breaking down a large mass of extravasated red cells generates a bilirubin load that outstrips the liver's conjugating capacity, giving an **unconjugated (indirect) hyperbilirubinaemia**. This is a favourite exam scenario in neonates with cephalohaematoma.

2.5 Clinical Consequences — Volume, Rate, Site

Three variables determine the outcome of any haemorrhage. Learn them as a triad, because a question will always be testing one of them.

① Volume

Total blood volume in an adult is about **5 L** (70 mL/kg).

TABLE 2.4 CLASSES OF ACUTE HAEMORRHAGE IN A 70 KG ADULT

Class	Loss	Physiology	Signs
I	<15% (<750 mL)	Fully compensated by venoconstriction and transcapillary refill	Essentially none; mild anxiety. <i>This is a blood donation.</i>
II	15–30% (750–1500 mL)	Sympathetic activation maintains blood pressure at the cost of vasoconstriction	Tachycardia, narrowed pulse pressure, cool clammy skin, anxiety. Systolic BP still normal — the classic trap.
III	30–40% (1500–2000 mL)	Compensation failing	Hypotension, marked tachycardia, tachypnoea, oliguria, confusion
IV	>40% (>2000 mL)	Decompensated haemorrhagic shock	Profound hypotension, anuria, obtundation; immediately life-threatening

② Rate

The *same* volume lost slowly and lost quickly produce entirely different pictures.

- **Rapid loss** → **hypovolaemic shock**. There is no time for compensation; the problem is loss of circulating volume and therefore of cardiac output. Note that haemoglobin concentration may initially be **normal**, because red cells and plasma are lost together — the haematocrit falls only after hours, as interstitial fluid shifts into the vascular space to refill it.
- **Slow, chronic loss** (occult colonic carcinoma, menorrhagia, peptic ulcer oozing) → **iron-deficiency anaemia**. Volume is replaced continuously, so the patient is never hypovolaemic; instead, iron stores are progressively depleted until erythropoiesis fails. The presentation is fatigue, pallor and a **microcytic, hypochromic** blood film — not shock. **In an adult male or postmenopausal female, iron-deficiency anaemia is gastrointestinal malignancy until proven otherwise.**

③ Site

This is the variable that overrides the other two. A volume of blood that would be trivial in the thigh is lethal in the wrong place, because the damage is **mechanical**, not volumetric.

- **Brainstem** — a few millilitres destroy the respiratory and cardiovascular centres. **Fatal.**
- **Pericardial sac** — 200–300 mL acutely causes tamponade and obstructive shock.
- **Intracranial (subdural, extradural, intracerebral)** — raised intracranial pressure and herniation.
- **Airway (retropharyngeal, laryngeal)** — asphyxia from a small volume.
- **Skin and subcutaneous tissue** — the same volume is merely a bruise.

EXAM TRAP

Do not reach for haemoglobin to assess an **acute** bleed. In the first hours it is falsely reassuring, because whole blood is lost — cells and plasma in the same proportion — so the *concentration* is unchanged. The early indicators are **heart rate** and **pulse pressure**, not haematocrit. Conversely, a low haemoglobin with a normal blood pressure and no tachycardia points to *chronic* loss.

PART THREE

03 Thrombosis

Hemostasis and thrombosis are the same process. The difference is entirely contextual: hemostasis is clot formation at a site of vascular injury, where it is needed; thrombosis is clot formation inside an intact vessel, where it is not. You cannot understand the pathology without first understanding the physiology, so this Part begins with normal hemostasis and only then asks what goes wrong.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Walk through the four stages of hemostasis and say what each contributes.
- Explain each arm of Virchow's triad with a clinical example, and say how each arm actually promotes clotting.
- Compare white, mixed, red and hyaline thrombi on composition, location and gross appearance.
- Recognise lines of Zahn and explain why they prove the thrombus formed in flowing blood before death.
- List the four fates of a thrombus and describe the mechanism of organization and recanalization.

3.1 Normal Hemostasis — the Four-Step Sequence

DEFINITION

Hemostasis is a precisely orchestrated process involving **platelets**, **clotting factors** and **endothelium** that occurs at a site of vascular injury and culminates in a blood clot that prevents or limits bleeding.

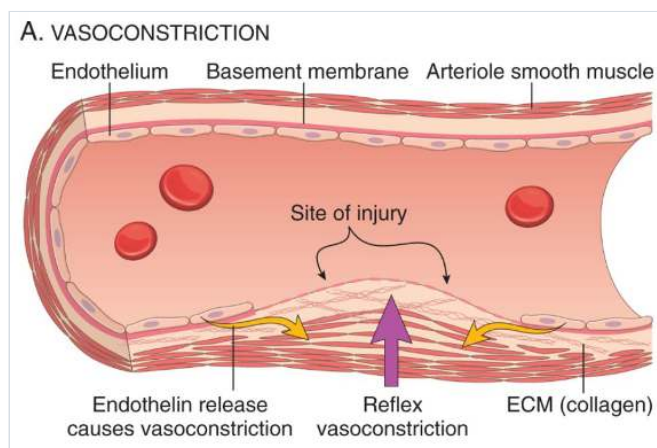


Fig 3.1 A Vasoconstriction. Reflex neurogenic vasoconstriction is augmented by **endothelin** released from injured endothelium. Flow through the damaged segment falls immediately.

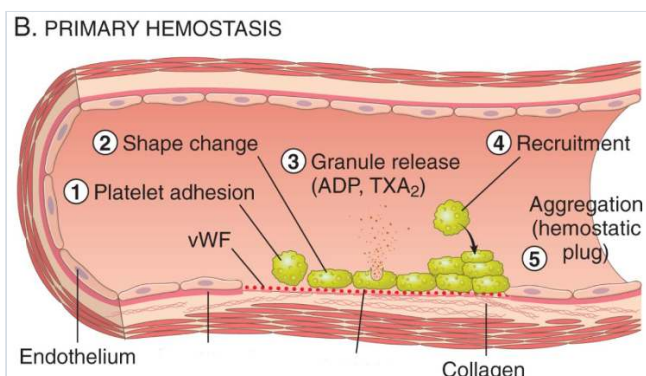


Fig 3.1 B Primary hemostasis. Platelets adhere to exposed subendothelial collagen via vWF, change shape, release their granules (ADP, TXA₂), recruit more platelets and aggregate into the **primary hemostatic plug**.

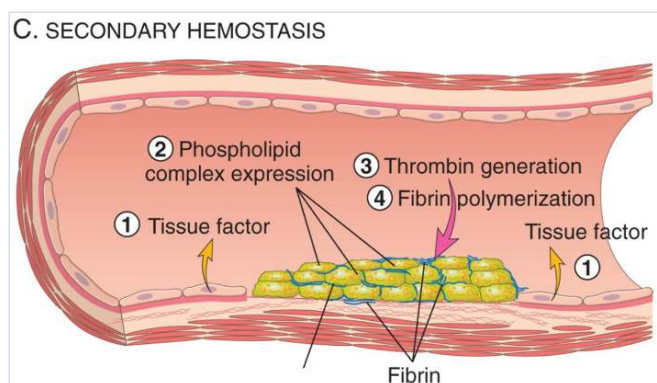


Fig 3.1 C Secondary hemostasis. Tissue factor exposed at the injury site, together with phospholipid on the activated platelet surface, drives thrombin generation; thrombin polymerises fibrin around and through the platelet plug.

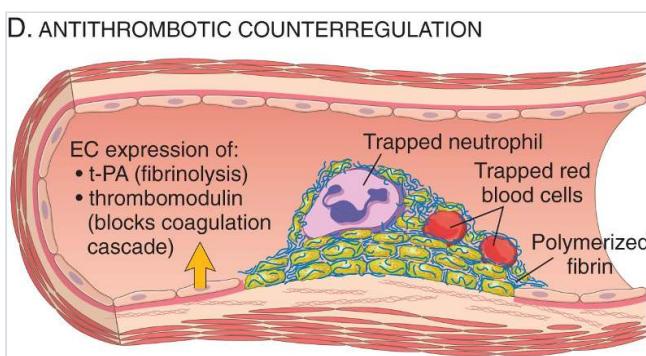


Fig 3.1 D Clot stabilisation and counter-regulation. Polymerised fibrin and platelet aggregates form a solid permanent plug; simultaneously t-PA and thrombomodulin released by surrounding intact endothelium confine the clot to the site of injury.

○ MECHANISM — THE FOUR STAGES, AND WHY EACH IS NECESSARY

- 1 **Arteriolar vasoconstriction.** Reflex neurogenic constriction is amplified by **endothelin** secreted by the injured endothelium.
WHY: Reducing flow buys time: it limits the rate of blood loss and, crucially, keeps activated platelets and clotting factors in the local area long enough to reach effective concentrations. Without this, everything downstream would be washed away. The effect is transient, so it alone is never sufficient.
- 2 **Primary hemostasis — the platelet plug.** Injury exposes subendothelial **collagen** and **von Willebrand factor**. Platelets adhere, change shape from smooth discs into flat plates with spiky protrusions, release their granules and aggregate.
WHY: The shape change is not cosmetic: it massively increases the surface area available both for platelet-to-platelet contact and for assembly of coagulation factor complexes. The plug forms within seconds — fast enough to stop bleeding from small vessels on its own.
- 3 **Secondary hemostasis — fibrin deposition.** **Tissue factor** exposed at the injury site activates the coagulation cascade, generating **thrombin**. Thrombin cleaves circulating fibrinogen into insoluble **fibrin**, creating a meshwork, and is itself a potent platelet activator.
WHY: The platelet plug alone is friable and would be swept away by arterial pressure. Fibrin is the reinforcing mesh — the steel in the concrete. Note thrombin's dual role: it makes fibrin and recruits more platelets, so the two arms of hemostasis amplify each other.
- 4 **Clot stabilisation and resorption.** Polymerised fibrin is covalently cross-linked by **factor XIIIa**; platelet contraction consolidates the mass. Counter-regulatory mechanisms (t-PA, thrombomodulin) then confine the clot to the site of injury and eventually dissolve it.
WHY: Cross-linking converts a reversible gel into a permanent solid plug. Counter-regulation is equally essential: without it, a clot triggered by a paper cut would propagate down the whole limb.

NET RESULT A **solid, permanent plug** that prevents further bleeding, localised precisely to the injured segment, and programmed for eventual removal once the wall has healed.

HIGH-YIELD

Thrombosis is hemostasis in the wrong place. The molecular machinery is identical. What differs is that in thrombosis, the trigger is an abnormality of the endothelium, the flow, or the blood itself — Virchow's triad — rather than a genuine wound.

3.2 Endothelium — the Switch

An intact endothelial monolayer is **actively** anti-thrombotic — not merely a passive barrier. This is a critical conceptual point: a resting endothelial cell is continuously working to keep blood fluid. When it is injured or activated, that work stops and the opposite programme begins. **The integrity and functional state of the endothelium determine whether clots form, propagate or dissolve.**

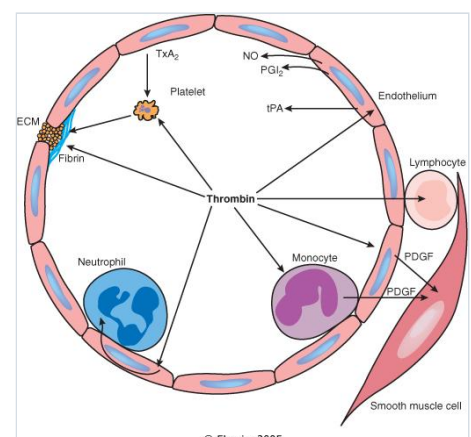


Fig 3.2 The endothelial cell sits at the centre of a balance, secreting both anti-thrombotic factors (NO, PGI₂, t-PA, thrombomodulin, heparin-like molecules) and, when activated, pro-thrombotic ones (tissue factor, vWF, PAI). Thrombosis is what happens when the balance tips.

TABLE 3.1 THE TWO FACES OF ENDOTHELIUM

Property	Anti-thrombotic (resting endothelium)	Pro-thrombotic (injured/activated)
Antiplatelet	An intact surface physically shields platelets from subendothelial collagen and vWF. Prostacyclin (PGI₂) , nitric oxide and adenosine diphosphatase (CD39) actively inhibit platelet adhesion and aggregation.	Loss of the barrier exposes collagen and vWF. Activated cells secrete vWF from Weibel–Palade bodies, promoting adhesion.
Anticoagulant	Heparin-like molecules accelerate antithrombin III; thrombomodulin binds thrombin and converts it into an activator of protein C; tissue factor pathway inhibitor (TFPI) blocks TF–VIIa.	Down-regulation of thrombomodulin and TFPI, plus expression of tissue factor in response to cytokines (TNF, IL-1), endotoxin and physical injury.
Fibrinolytic	Secretes tissue plasminogen activator (t-PA) , which converts plasminogen to plasmin and degrades fibrin.	Secretes plasminogen activator inhibitors (PAI) , which shut fibrinolysis down and thereby favour clot persistence.

HIGH-YIELD — ENDOTHELIAL INJURY DOES NOT REQUIRE DENUDATION

This is one of the most commonly missed points in the whole topic. A thrombus can form over an endothelium that is **morphologically intact but functionally activated** — no cells have been stripped away, yet the surface has switched from anti- to pro-thrombotic. Hypertension, turbulent flow, bacterial endotoxin, hypercholesterolaemia, cigarette smoke, homocysteine and inflammatory cytokines all do this. It explains why atherosclerotic plaques thrombose without visible ulceration, and why a "normal-looking" vessel on histology can still be the origin of a thrombus.

3.3 Primary Hemostasis — Adhesion, Activation, Aggregation

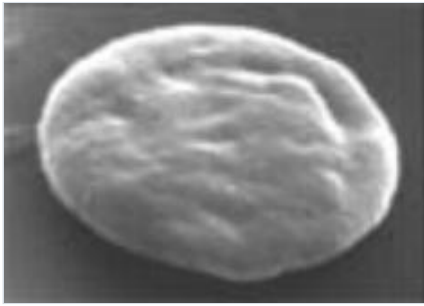


Fig 3.3 Resting platelet — a smooth, anucleate disc. Surface area is minimal.

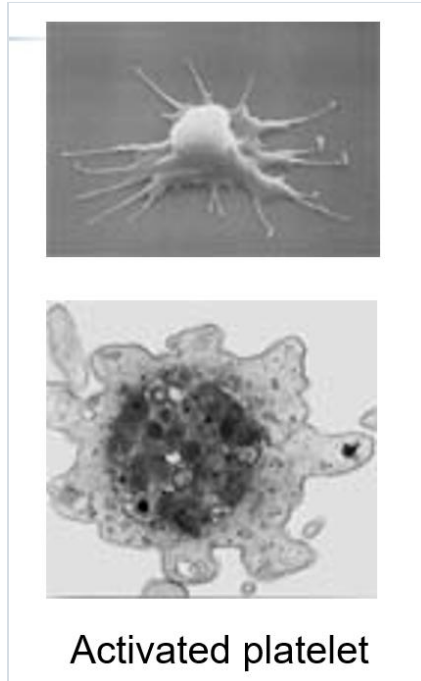


Fig 3.4 Activated platelet — a spiky "sea urchin" with long filopodia. The same cell, with a dramatically larger surface for contact and for assembling coagulation complexes.

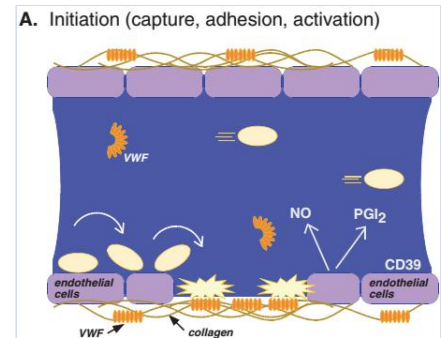


Fig 3.5 Initiation. vWF bridges exposed collagen to platelet GpIb. Resting endothelium nearby still releases NO, PGI₂ and CD39, confining the response.

MECHANISM — BUILDING THE PRIMARY HEMOSTATIC PLUG

- 1 Adhesion.** Subendothelial **collagen** is exposed. Plasma **von Willebrand factor** binds the collagen, changes conformation, and is then bound by platelet receptor **GpIb**.

WHY: vWF is a molecular bridge, and it is indispensable under high shear. Platelets cannot bind collagen directly at arterial flow rates — they would be torn off. vWF's elongation under shear creates a catch-bond that grips harder the faster the blood moves. This is why von Willebrand disease and Bernard-Soulier syndrome (absent GpIb) both cause mucocutaneous bleeding.
- 2 Activation and shape change.** The adherent platelet converts from a smooth disc into a flattened cell with spiky protrusions, and flips negatively charged **phosphatidylserine** to its outer membrane leaflet.

WHY: The spikes increase surface area for contact. The exposed phosphatidylserine is the platform on which the calcium-dependent coagulation complexes (tenase and prothrombinase) assemble — which is exactly how primary hemostasis hands over to secondary hemostasis.
- 3 Secretion (the release reaction).** α-granules release fibrinogen, factor V, vWF, fibronectin and PDGF; dense granules release **ADP**, calcium, ATP and serotonin (5-HT). The platelet also synthesises **thromboxane A₂ (TXA₂)** from membrane arachidonic acid via cyclo-oxygenase.

WHY: ADP and TXA₂ are the two key autocrine/paracrine amplifiers: they activate neighbouring platelets, turning a single adhesion event into a growing mass. This is also the pharmacological pressure point — **aspirin** irreversibly acetylates COX-1 and blocks TXA₂; **clopidogrel** blocks the P2Y₁₂ ADP receptor.
- 4 Aggregation.** ADP and TXA₂ induce a conformational change in platelet integrin **GpIIb/IIIa**, which now binds **fibrinogen**. Each fibrinogen molecule bridges two platelets.

WHY: Fibrinogen is bivalent, so it physically links platelets into a mass. Glanzmann thrombasthenia (absent GpIIb/IIIa) blocks this step; so do the drugs abciximab, eptifibatide and tirofiban.

- 5 **Consolidation.** The first wave of aggregation is **reversible**. Thrombin, generated concurrently by the coagulation cascade, then (a) activates more platelets, (b) drives irreversible platelet contraction via the cytoskeleton, and (c) converts fibrinogen to fibrin and activates factor XIIIa, which covalently cross-links the fibrin.

WHY: *Reversibility early on is a safety feature — it allows an inappropriate aggregate to disperse. Thrombin is the commitment signal: once it arrives, the plug becomes the **definitive secondary hemostatic plug** and cannot disassemble.*

NET RESULT A contracted, fibrin-cemented platelet mass firmly anchored to the vessel wall.

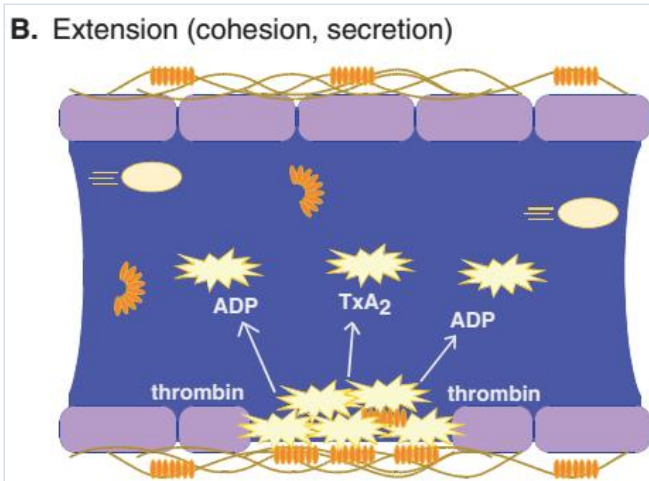


Fig 3.6 Extension (cohesion, secretion). Activated platelets release ADP and TxA_2 , recruiting and activating additional platelets; thrombin generation begins on the platelet surface.

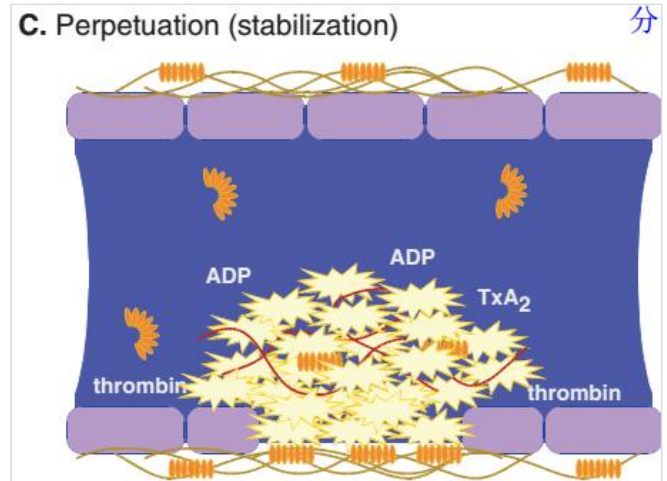


Fig 3.7 Perpetuation (stabilisation). Thrombin and ADP sustain the aggregate; fibrin cross-linking and platelet contraction make it permanent.

3.4 Secondary Hemostasis — the Coagulation Cascade

The cascade is a series of **amplifying proenzyme-to-enzyme conversions**. Each activated factor cleaves many molecules of the next, so a tiny initiating stimulus produces an explosive burst of thrombin. Each reaction requires a **phospholipid surface** (supplied by the activated platelet) and **calcium** — which is precisely how the cascade is confined to the site where platelets have already gathered.

EXTRINSIC: Tissue factor + VII → VIIa → **X**

INTRINSIC: XII → XIIa → XI → XIa → IX → IXa (+ VIIIa, PL, Ca²⁺) → **X**

COMMON: Xa (+ Va, PL, Ca²⁺) → Prothrombin (II) → **Thrombin (IIa)** → Fibrinogen (I) → **Fibrin** → (XIIIa) → Cross-linked fibrin

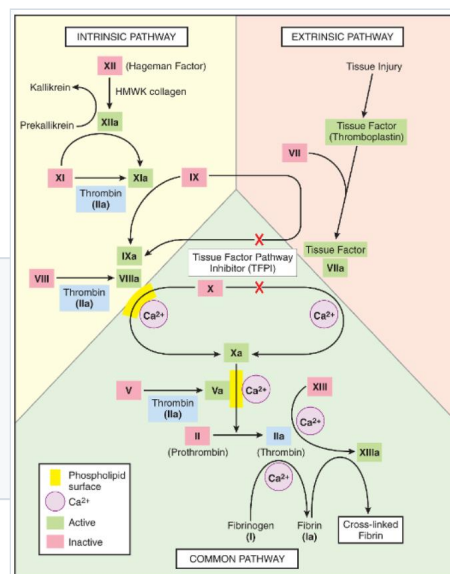


Fig 3.8 The coagulation cascade. The **extrinsic** pathway (right) begins with tissue factor and factor VII; the **intrinsic** pathway (left) with factor XII and a negatively charged surface. Both converge on factor **X**, forming the **common** pathway: Xa + Va + phospholipid + Ca²⁺ convert prothrombin to thrombin, and thrombin converts fibrinogen to fibrin.

Why two pathways, and what the laboratory tests actually measure

The division into "intrinsic" and "extrinsic" is an *in vitro* construct — it reflects how the two clotting-time assays are performed, not two independent biological routes. **In vivo, the tissue factor (extrinsic) pathway is the dominant initiator**; the intrinsic pathway's main physiological role is amplification, with thrombin itself feeding back to activate factors XI, VIII and V. Nevertheless the division is clinically indispensable, because it is what makes PT and PTT interpretable.

TABLE 3.2 PT VERSUS PTT

Test	How it is performed	Factors assessed	Clinical use
PT prothrombin time	Tissue factor , phospholipid and calcium are added to plasma; time to fibrin clot is recorded.	Extrinsic + common: VII, X, V, II (prothrombin), fibrinogen	Monitors warfarin (as INR). Prolonged in liver disease and vitamin K deficiency — factor VII has the shortest half-life , so PT rises first.
PTT partial thromboplastin time	Clotting is initiated by negatively charged particles (e.g. ground glass, kaolin) plus phospholipid and calcium.	Intrinsic + common: XII, XI, IX, VIII, X, V, II, fibrinogen	Monitors unfractionated heparin . Prolonged in haemophilia A (VIII), haemophilia B (IX), von Willebrand disease and antiphospholipid antibody.

MNEMONIC

PeT the ex-TRINSIC dog; PiTT the IN-TRINSIC cat.

PT → exTrinsic. **PTT** → inTrinsic (two T's, the longer pathway with more factors). Vitamin K-dependent factors: **II, VII, IX, X** plus proteins **C** and **S** — "**1972**" read backwards, or "**Two, Seven, Nine, Ten — and C and S again**".

Thrombin — the central enzyme

Of all the coagulation factors, **thrombin is the most important**. It is not merely the enzyme that makes fibrin; it is the hub through which coagulation, platelet activation, inflammation and even repair are coordinated. Six actions worth knowing:

1. Converts **fibrinogen** → **fibrin**, and activates **factor XIII** to cross-link it.
2. **Activates platelets** through protease-activated receptors (PAR-1) — the most potent physiological platelet agonist.
3. **Amplifies its own generation** by activating factors V, VIII and XI.
4. **Pro-inflammatory** — activates leukocytes and endothelium via PARs, linking thrombosis to inflammation.
5. **Anticoagulant, when bound to thrombomodulin** on intact endothelium: the complex activates **protein C**, which (with protein S) inactivates factors Va and VIIIa. The same molecule thus switches from clot-promoting to clot-limiting depending on where it is.
6. **Promotes repair** — stimulates fibroblast proliferation and smooth muscle migration, which is the first step of thrombus organization (§3.12).

3.5 Keeping the Clot Local — the Natural Anticoagulants

A cascade that amplifies this powerfully would be lethal without brakes. Four systems restrain it, and — importantly — an inherited deficiency of any of them produces a hypercoagulable state (§3.7).

TABLE 3.3 THE FOUR NATURAL ANTICOAGULANT SYSTEMS

System	How it works	Clinical relevance
Antithrombin III	A serine protease inhibitor that irreversibly binds and inactivates thrombin, IXa, Xa, XIa and XIIa . Its activity is accelerated 1000-fold by heparin-like molecules on endothelium.	This is how heparin works . Antithrombin deficiency causes recurrent venous thrombosis and apparent heparin resistance.
Protein C / Protein S	Thrombin bound to endothelial thrombomodulin activates protein C; activated protein C, with protein S as cofactor, proteolytically inactivates factors Va and VIIIa .	Factor V Leiden is resistant to this cleavage → the commonest inherited thrombophilia. Protein C/S deficiency explains warfarin-induced skin necrosis .
Tissue factor pathway inhibitor (TFPI)	Secreted by endothelium; directly inhibits the tissue factor-VIIa-Xa complex.	Shuts off the initiating step, forcing continued coagulation to depend on thrombin-driven amplification.
Fibrinolytic system	t-PA (released by endothelium, and most active when bound to fibrin) converts plasminogen to plasmin , which digests fibrin into fibrin degradation products , including D-dimer .	t-PA acting preferentially on fibrin-bound plasminogen is why fibrinolysis is largely confined to the clot. D-dimer is the clinical marker of active clot breakdown — elevated in DVT, PE and DIC.

3.6 Virchow's Triad — the Three Roads to Thrombosis

DEFINITION

Thrombosis is the **formation of an adherent clotted mass of blood within the cardiovascular system of a living body**. The mass itself is a **thrombus**.

Every word in that definition is load-bearing. Adherent — it is attached to the wall, which distinguishes it from a post-mortem clot. Within the cardiovascular system — it can arise in cardiac chambers, on valve cusps, in arteries, veins or capillaries. In a living body — a clot formed after death is not a thrombus.

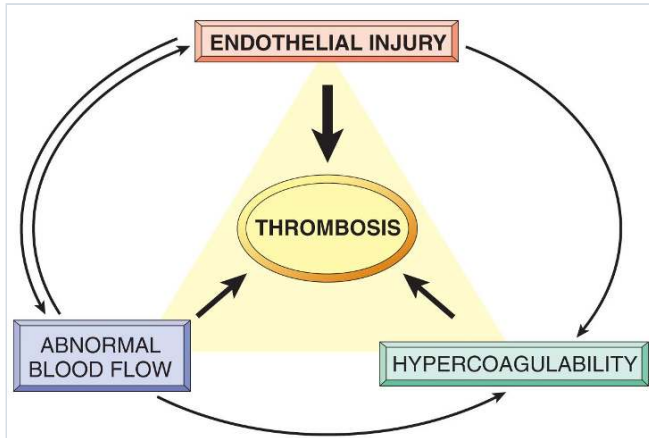


Fig 3.9 Virchow's triad. Endothelial injury, abnormal blood flow and hypercoagulability. The arrows are bidirectional for a reason: the three factors are not independent but reinforce one another — turbulence injures endothelium, injured endothelium disturbs flow, and both promote local hypercoagulability.

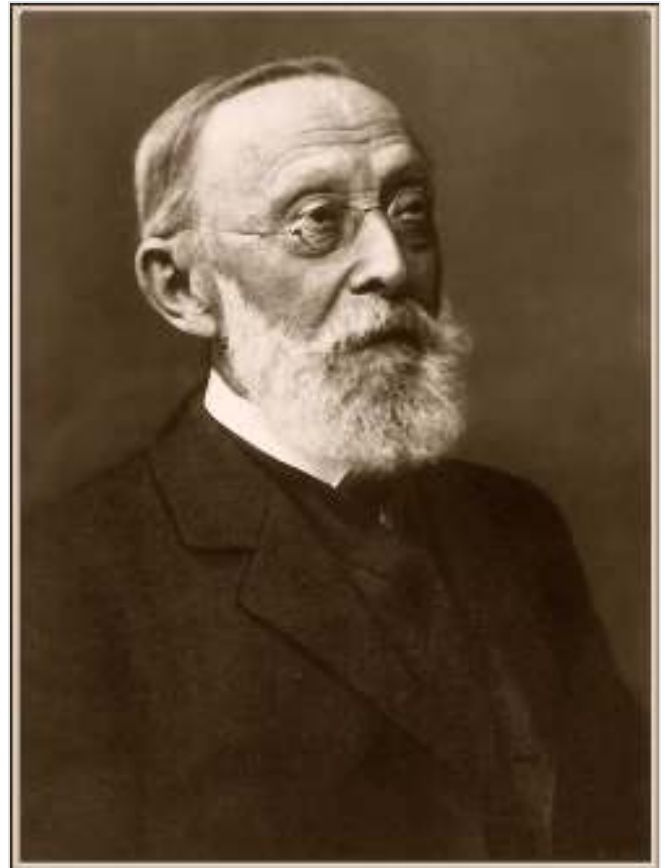


Fig 3.10 Rudolf Virchow (1821–1902), who formulated the triad and, more broadly, established that disease originates in cellular abnormality.

① Endothelial injury

The **dominant** influence, and the one that can cause thrombosis *by itself*, without help from the other two. It is particularly important in the **heart and arteries**, where rapid flow would normally prevent platelets adhering and would dilute activated clotting factors — so something must actively overcome that protection.

○ MECHANISM — HOW ENDOTHELIAL INJURY INITIATES A THROMBUS

- 1 The anti-thrombotic surface is lost or switched off (physical denudation, or functional activation by cytokines, endotoxin, hyperlipidaemia, smoking, hypertension, homocysteine or radiation).

WHY: Recall from §3.2 that the endothelium is actively anti-thrombotic. Losing that activity alone is enough to shift the balance.

- 2 Subendothelial **collagen** and **von Willebrand factor** become accessible.
WHY: These are the ligands for platelet GpIa/IIa and GpIb — the physical trigger for adhesion.
- 3 Platelets adhere, activate, release ADP, TXA₂ and serotonin, and aggregate.
WHY: This is the release reaction of §3.3, now occurring where there is no wound to seal.
- 4 **Tissue factor** is expressed by the injured endothelium and by any exposed subendothelial cells, triggering the extrinsic cascade; the activated platelet surface provides the phospholipid for the cascade complexes.
WHY: Tissue factor is the physiological initiator; the platelet supplies the assembly platform. Neither works efficiently without the other, which is why the two arms of hemostasis are obligatorily coupled.
- 5 Thrombin is generated; fibrinogen is converted to fibrin, which traps more platelets and red cells; the mass enlarges and becomes adherent.
WHY: Once fibrin cross-links, the thrombus is anchored to the wall and can no longer be swept away — it is now committed to grow or to be organized.

CLINICAL CAUSES OF ENDOTHELIAL INJURY

- **Ulcerated atherosclerotic plaque** — the single most important cause of arterial thrombosis; plaque rupture exposes the highly thrombogenic lipid core.
- **Myocardial infarction** — injured endocardium overlying the infarct is a nidus for mural thrombus.
- **Valvular injury** — rheumatic disease, infective endocarditis, prosthetic valves.
- **Vasculitis; trauma and surgery; indwelling catheters.**
- **Hypertension and turbulent flow** — chronic mechanical stress activates endothelium without stripping it.
- **Metabolic and toxic** — hypercholesterolaemia, homocystinuria, cigarette smoke, radiation, bacterial endotoxin.

② Abnormal blood flow — stasis and turbulence

Normal blood flow is **laminar**: it moves in concentric layers, fastest in the centre, and the cellular elements are confined to that central stream. A cell-free **plasmatic zone** separates them from the endothelium. Anything that destroys laminar flow removes that separation.

○ MECHANISM — SIX WAYS ABNORMAL FLOW PROMOTES THROMBOSIS

- 1 **It disrupts laminar flow and brings platelets into contact with the endothelium.**
WHY: The protective plasmatic zone is abolished, so platelets can now physically reach the wall.
- 2 **It prevents dilution of activated clotting factors.**
WHY: In normal flow, activated factors are swept away and diluted below the threshold needed to propagate the cascade. In stasis they accumulate locally until that threshold is crossed.
- 3 **It retards the inflow of clotting factor inhibitors.**
WHY: Antithrombin III, protein C and TFPI must be delivered by flowing blood. If fresh blood does not arrive, the brakes are never applied.
- 4 **It permits the build-up of platelet aggregates and nascent fibrin** in the sluggish stream or in pockets of stasis.
WHY: Small aggregates that would normally be dispersed by shear are allowed to consolidate.

- 5 It promotes endothelial hypoxia and injury**, causing platelet and fibrin deposition and reducing release of t-PA.
WHY: Endothelial cells depend on flowing blood for oxygen; hypoxic endothelium switches to the pro-thrombotic phenotype and stops secreting fibrinolytic t-PA.
- 6 Turbulence itself is a mechanism of endothelial injury.**
WHY: Disturbed shear stress alters endothelial gene expression, up-regulating adhesion molecules and tissue factor. This is why the triad's arms are interlinked rather than separate.

TABLE 3.4 TURBULENCE VERSUS STASIS

	Where	Clinical examples
Turbulence	Mainly arteries and cardiac chambers	Ulcerated atherosclerotic plaque; aneurysm sac; stenotic or prosthetic valve; arterial bifurcations
Stasis	Mainly veins and dilated chambers	Immobility (bed rest, long-haul flight, plaster cast, post-operative state); atrial fibrillation (stasis in the dilated left atrial appendage); dilated cardiomyopathy ; ventricular aneurysm after MI; hyperviscosity (polycythaemia vera, sickle cell disease, multiple myeloma)

③ Hypercoagulability

Any alteration of the blood — most often of the clotting mechanism — that predisposes to thrombosis. Of all the coagulation factors, **thrombin is the most important**. Hypercoagulability is the *least* common of the three arms as a sole cause, but it is the one you must actively look for in a young patient or an unusual site of thrombosis.

3.7 Hypercoagulable States

TABLE 3.5 PRIMARY (INHERITED) AND SECONDARY (ACQUIRED) THROMBOPHILIA

Condition	Mechanism	Note
PRIMARY — INHERITED		
Factor V Leiden	A point mutation replaces arginine with glutamine at position 506, the site where activated protein C cleaves factor Va. Factor Va therefore cannot be inactivated and continues to drive thrombin generation.	The commonest inherited thrombophilia — 2–15% of Caucasians; present in up to 60% of patients with recurrent DVT.
Prothrombin G20210A	A mutation in the 3' untranslated region increases prothrombin transcript stability → higher plasma prothrombin → more thrombin.	Second commonest; ~1–2% of the population.
Antithrombin III deficiency	Loss of the principal inhibitor of thrombin and factors IXa, Xa, XIa, XIIa.	Presents as apparent heparin resistance : PTT fails to rise appropriately with heparin.
Protein C or S deficiency	Factors Va and VIIIa are not inactivated.	Risk of warfarin-induced skin necrosis : protein C has a short half-life, so starting warfarin transiently produces a net <i>pro-coagulant</i> state.

Condition	Mechanism	Note
Hyperhomocysteinaemia	Homocysteine is directly toxic to endothelium and inhibits antithrombin III and thrombomodulin.	Causes both arterial and venous thrombosis. From MTHFR mutation, cystathionine β -synthase deficiency, or B ₆ /B ₁₂ /folate deficiency.
SECONDARY — ACQUIRED		
Prolonged immobilisation	Stasis — loss of the calf muscle pump.	Bed rest, long flights, plaster cast, post-operative period.
Surgery and trauma	Massive tissue factor release, endothelial injury, immobility and an acute-phase rise in fibrinogen — all three arms of the triad at once .	Orthopaedic surgery (hip, knee) carries the highest risk of all.
Malignancy	Tumours release tissue factor and other pro-coagulants.	Trousseau syndrome — migratory thrombophlebitis, classically with pancreatic, lung or gastric adenocarcinoma. May be the first sign of an occult cancer.
Antiphospholipid antibody syndrome	Autoantibodies against phospholipid-binding proteins activate platelets and endothelium.	The paradox: prolonged PTT in vitro but thrombosis in vivo. Recurrent arterial and venous thrombosis plus recurrent fetal loss; often with SLE.
Heparin-induced thrombocytopenia (HIT)	IgG against the heparin–platelet factor 4 complex binds platelets via Fc γ RIIA and activates them.	The second paradox: a low platelet count with <i>thrombosis</i> , not bleeding — platelets are consumed by being activated. Stop heparin; start a direct thrombin inhibitor.
Oral contraceptives, pregnancy, puerperium	Oestrogen increases hepatic synthesis of fibrinogen and factors II, VII, IX, X and reduces protein S; the gravid uterus compresses pelvic veins.	Risk multiplies dramatically if combined with smoking or factor V Leiden.
Nephrotic syndrome	Urinary loss of antithrombin III along with albumin.	Classically causes renal vein thrombosis .
Disseminated intravascular coagulation	See §3.11.	Simultaneous thrombosis and bleeding.
Others	Severe burns, advanced age, obesity, smoking, sickle cell disease, polycythaemia vera, prosthetic cardiac valves.	

HIGH-YIELD — WHEN TO SUSPECT INHERITED THROMBOPHILIA

Investigate for a primary hypercoagulable state when thrombosis occurs: **under 50 years of age**; **without provocation**; **recurrently**; at an **unusual site** (cerebral, hepatic, portal or mesenteric vein); with a **strong family history**; or alongside **recurrent fetal loss**.

3.8 Morphology — Lines of Zahn and the Four Thrombus Types

Lines of Zahn

Named after the Swiss–German pathologist Friedrich Wilhelm Zahn, these are **alternating pale and dark laminations** within a thrombus: pale layers of **platelets and fibrin** alternating with dark, red-cell-rich layers.

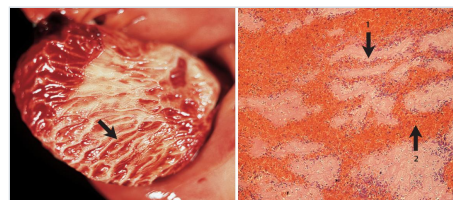


Fig 3.11 Lines of Zahn, gross. Alternating pale and dark laminations are visible on the cut surface of the thrombus.

MECHANISM — WHY THE LAMINATIONS FORM, AND WHY THEY MATTER

- 1 A thrombus begins at a point of endothelial injury and grows into the lumen against **flowing blood**.
WHY: Growth is not uniform — it happens in waves, driven by the pulsatile nature of flow.
- 2 Platelets, being small and light, are preferentially deposited on the leading surface during rapid flow, forming a **pale layer**.
WHY: At high shear, platelets marginate toward the vessel wall and the growing thrombus surface, while the bulkier red cells remain in the central stream.
- 3 Fibrin then forms across that platelet layer, trapping the red cells that happen to be passing when flow momentarily slows — producing a **dark layer**.
WHY: Red cells are trapped passively, in proportion to how slow the stream is at that instant.
- 4 The cycle repeats with each variation in flow, building up visible laminations.
WHY: The alternation is a direct recording of the pulse — which is exactly why it is diagnostic.

NET RESULT Lines of Zahn prove the thrombus formed in flowing blood in a living body — i.e. it is a **true ante-mortem thrombus, not a post-mortem clot**. They are most conspicuous in the heart and aorta, where flow is fastest; in slow venous flow they are present but poorly developed.



Fig 3.12 Lines of Zahn, histology. Pale eosinophilic bands of platelets and fibrin alternate with darker bands packed with red cells.

The four morphological types

A venous thrombus classically grows as a **propagating** structure with three parts: the **head** attached to the wall (white thrombus), the **body** (mixed thrombus) and the free-floating **tail** (red thrombus). The tail is the dangerous part — it is poorly anchored and most likely to break off as an embolus.

TABLE 3.6 WHITE, MIXED, RED AND HYALINE THROMBI

Type	Composition	Location	Gross appearance and notes
White thrombus (pale / conglutination)	Mainly platelets and fibrin ; few red cells	The head of a thrombus, firmly attached to the wall. Arteries, cardiac chambers, valve cusps — sites of rapid flow .	Dry, friable, grey-white mass. On transection, darker grey lines of Zahn are seen. Histologically shows platelet trabeculae — coral-like branching bands of platelets with neutrophils at their margins.

Type	Composition	Location	Gross appearance and notes
Mixed thrombus	Alternating platelet/fibrin layers and red-cell layers	The body of a propagating venous thrombus	White thrombus intermingled with red. This is where lines of Zahn are most clearly seen — it is literally the alternation made visible.
Red thrombus (stasis / coagulation thrombus)	Large numbers of red cells enmeshed in fibrin, with some platelets	The tail , and the whole of an occlusive venous thrombus. Venous system, at sites of stasis .	Dark red, moist, elastic when fresh; becomes dry, friable and grey-brown with age. Laminations poorly developed because flow is slow. Often forms a long cast of the vein lumen .
Hyaline thrombus (fibrin microthrombus)	Almost pure fibrin with entrapped platelets; no laminations	Microcirculation — capillaries, especially of kidney, lung, brain, liver and adrenal. Seen in DIC .	Invisible to the naked eye — microscopic only . Appears as homogeneous, glassy, pink ("hyaline") plugs filling capillary lumina.

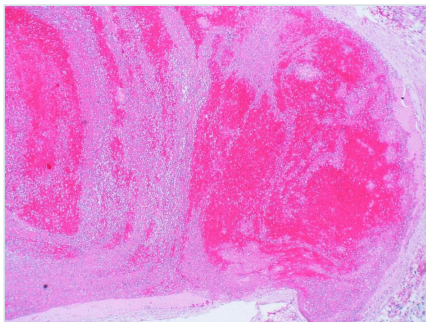


Fig 3.13 Mixed thrombus, histology — laminated bands of platelets/fibrin and red cells.

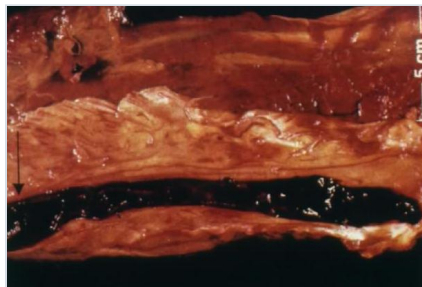


Fig 3.14 Red (stasis) thrombus — dark red, moist, forming a cast of the vein lumen.

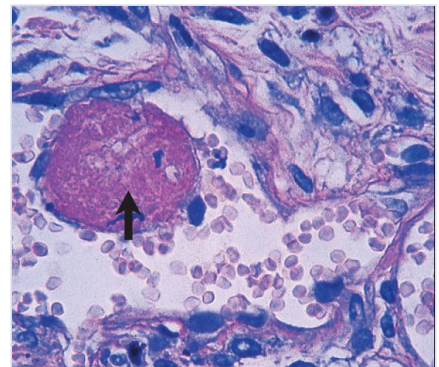


Fig 3.15 Hyaline (fibrin) microthrombus (arrow) occluding a glomerular capillary in DIC.

EXAM TRAP — "WHITE THROMBUS" VERSUS "WHITE INFARCT"

They share an adjective and nothing else. A **white thrombus** is pale because it is made of *platelets* rather than red cells. A **white infarct** is pale because red cells have been *lysed and cleared* from a zone of necrosis in a solid organ. One is a clot; the other is dead tissue.

3.9 Arterial versus Venous Thrombi

TABLE 3.7 ARTERIAL (MURAL / OCCLUSIVE) VS VENOUS (PHLEBOTHROMBOSIS)

Feature	Arterial / cardiac thrombus	Venous thrombus
Dominant triad arm	Endothelial injury and turbulence	Stasis and hypercoagulability
Composition	Platelet-rich (white / mixed)	Red-cell-rich (red / stasis)
Occlusion	Often mural (attached to one wall, lumen partly patent) in the heart and aorta; occlusive in smaller arteries	Almost invariably occlusive , forming a long cast of the lumen

Feature	Arterial / cardiac thrombus	Venous thrombus
Direction of growth	Retrograde — backwards, towards the heart, against flow	Antegrade — with the flow, towards the heart
Commonest sites	Coronary > cerebral > femoral arteries; left ventricle after MI; left atrium in atrial fibrillation; abdominal aortic aneurysm	Deep veins of the lower extremity — 90%. Less common: upper extremity, periprostatic plexus, ovarian and peri-uterine veins, dural sinuses, portal vein
Main consequence	Ischaemia and infarction of the tissue downstream	Congestion and edema upstream, plus pulmonary embolism
Treatment aimed at	Platelets — aspirin, clopidogrel, GpIIb/IIIa inhibitors	Coagulation cascade — heparin, warfarin, direct oral anticoagulants



Fig 3.16 Mural thrombus in the left ventricle overlying a healed myocardial infarct. It adheres to one wall and does not occlude the chamber — but it is a reservoir for systemic emboli.

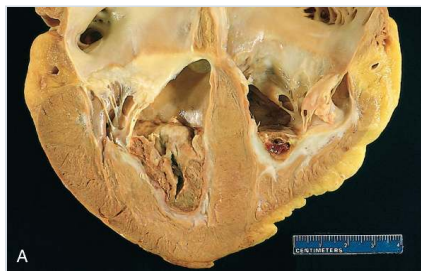


Fig 3.17 Another mural thrombus, filling the apex of a dilated left ventricle where stasis is greatest.

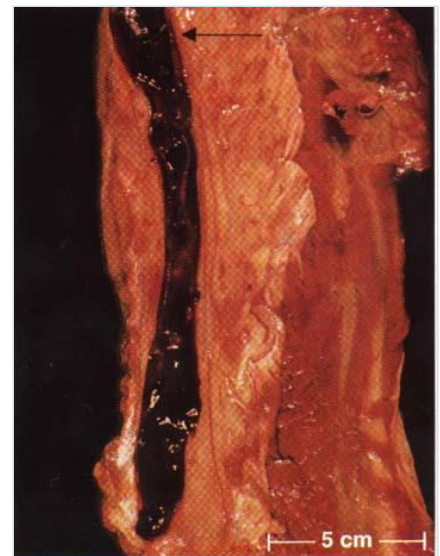


Figure 3.22. Occlusive (coagulation) thrombus in a femoral vein.

Fig 3.18 Occlusive (coagulation) thrombus filling a femoral vein — the classic source of a pulmonary embolus.

3.10 Vegetations — Thrombi on Heart Valves

Thrombotic masses on valve cusps are called **vegetations**. Three kinds matter:

TABLE 3.8 THREE KINDS OF VALVULAR VEGETATION

Type	Nature	Key features
Infective endocarditis	Thrombus containing organisms , formed on a damaged or prosthetic valve	Large, bulky, friable, destructive — erodes the cusp and can perforate it. Emboli are septic and produce abscesses and mycotic aneurysms.
Non-bacterial thrombotic (marantic) endocarditis	Sterile fibrin–platelet deposits on an essentially normal valve, in a hypercoagulable patient	Small, bland, non-destructive , along the line of closure. Associated with advanced malignancy (especially mucin-secreting adenocarcinoma) and DIC. Still embolises.
Libman–Sacks endocarditis	Sterile verrucous vegetations in systemic lupus erythematosus	Small, warty, and characteristically on both surfaces of the cusp. Driven by immune complex and complement deposition along the endocardium, with secondary platelet aggregation; strongly associated with antiphospholipid antibodies .

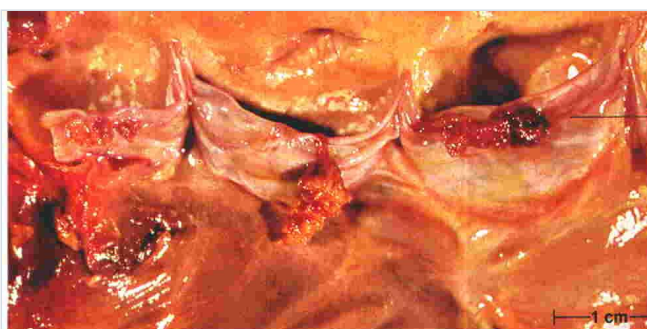
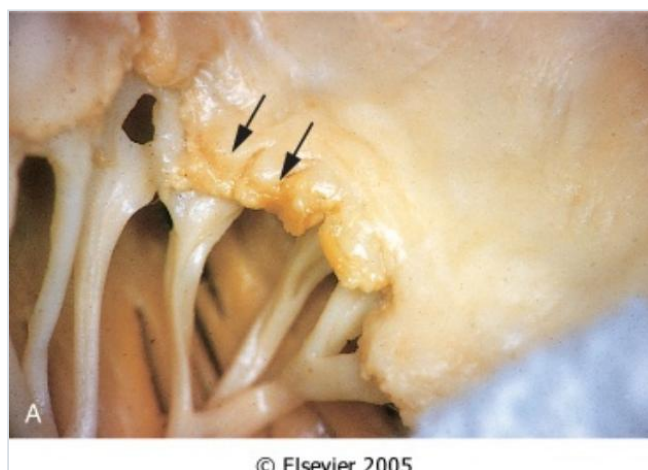


Figure 2.49. Verrucous endocarditis of the aortic valve.

Fig 3.19 Verrucous (Libman–Sacks) endocarditis of the aortic valve — small warty vegetations along the cusps.

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Fig 3.20 Valvular vegetations (arrows) on the atrial surface of a mitral valve leaflet.

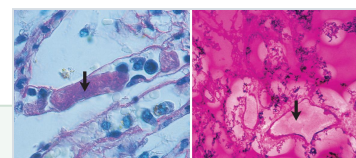
3.11 Disseminated Intravascular Coagulation

DIC is not a disease but a **complication** of another condition. It is the paradox of pathology: **widespread thrombosis that causes bleeding**.

MECHANISM — WHY DIC CAUSES THROMBOSIS AND BLEEDING

- 1 A trigger releases massive amounts of **tissue factor** or causes widespread endothelial injury.

WHY: Classic triggers: obstetric complications (abruptio placentae, amniotic fluid embolism, retained dead fetus), Gram-negative sepsis, massive trauma or burns, disseminated malignancy (especially acute promyelocytic leukaemia and mucinous adenocarcinoma), and severe transfusion reactions.

**Fig 3.21** DIC. Fibrin microthrombi (arrows) occlude capillaries in the lung. Thousands of such plugs form simultaneously throughout the microcirculation.

- 2 The coagulation cascade is activated **throughout the circulation at once**, not at a single site.
WHY: Normally the cascade is localised by the phospholipid surface requirement and by the natural anticoagulants. A systemic trigger overwhelms both.
- 3 **Fibrin microthrombi** (hyaline thrombi) form in capillaries and small vessels everywhere — especially kidney, lung, brain, liver, adrenal and skin.
WHY: Capillaries are where flow is slowest and surface-to-volume ratio highest, so they occlude first.
- 4 This causes **widespread microvascular ischaemia** and **microangiopathic haemolytic anaemia** — red cells are sheared into **schistocytes** as they are forced through fibrin strands.
WHY: A fibrin mesh across a capillary acts like a cheese wire on a passing red cell.
- 5 Meanwhile, **platelets and clotting factors are consumed** in making all that fibrin, and **secondary fibrinolysis** is activated, flooding the plasma with fibrin degradation products that further inhibit platelet function and fibrin polymerisation.
*WHY: The body's supply of coagulation substrate is finite. Once exhausted, the patient cannot clot at all — hence the alternative name, **consumption coagulopathy**.*
- 6 The patient now **bleeds** — from venepuncture sites, mucosae, surgical wounds and into the skin as petechiae and purpura — while simultaneously infarcting organs.
*WHY: Both phenomena have the same cause. This is why DIC is so difficult to treat: anticoagulating worsens the bleeding, transfusing clotting factors feeds the thrombosis. **Treatment must be directed at the underlying trigger.***

NET RESULT Laboratory picture: ↓platelets, ↑PT, ↑PTT, ↓fibrinogen, ↑↑D-dimer and fibrin degradation products, **schistocytes** on the blood film.

3.12 The Four Fates of a Thrombus

① Propagation

The thrombus enlarges by accreting more platelets and fibrin, often until it occludes a critical vessel. This is why anticoagulation is started immediately in DVT — not to dissolve the existing clot, but to **stop it growing** while the body's own fibrinolytic system works on it.

② Softening, liquefaction and resolution (lysis)

Plasmin, activated by t-PA, digests fibrin; leukocytic enzymes soften the mass; small thrombi may disappear completely. **Efficacy depends critically on age:** a fresh thrombus is rich in un-cross-linked fibrin and dissolves readily, whereas an old thrombus is extensively cross-linked by factor XIIIa and partly organized, and resists lysis. **This is the entire rationale for the "time is muscle / time is brain" principle** — thrombolytic therapy works only within a narrow window after clot formation.

③ Embolisation

The thrombus, or more usually its poorly-anchored tail, detaches and travels downstream. This is the bridge from Part 3 into Part 4, and the most feared outcome of a lower-limb DVT.

④ Organization and recanalization

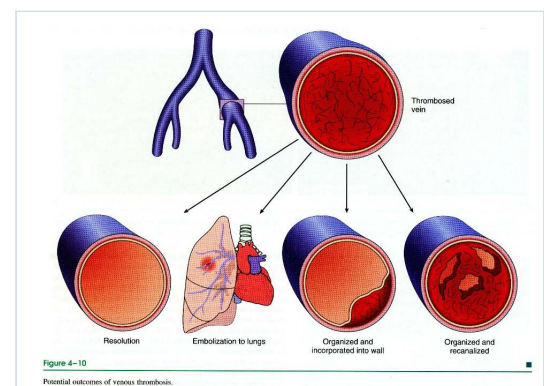


Fig 3.22 The potential outcomes of a venous thrombus: resolution (lysis), embolisation to the lung, and organization with recanalization.

MECHANISM — TURNING A CLOT BACK INTO A CHANNEL

- 1 Within 1–2 days, neutrophils and then macrophages infiltrate the thrombus and begin to digest it.
WHY: A thrombus adherent to the wall is treated by the body as any other injury requiring repair — the inflammatory phase comes first.
- 2 From day 2–3, **endothelial cells, smooth muscle cells and fibroblasts grow in from the adjacent vessel wall.**
WHY: Growth factors released by platelets in the thrombus itself — PDGF, TGF- β — and by thrombin recruit exactly these cell types. The thrombus provides its own scaffold and its own stimulus.
- 3 Over 1–2 weeks the ingrowing cells lay down collagen, converting the thrombus into a vascularised mass of **granulation tissue** — this is **organization**.
WHY: This is ordinary wound healing, simply occurring inside a vessel lumen instead of across a skin wound.
- 4 Capillary sprouts within the organizing mass elongate, anastomose and fuse into larger channels that eventually connect the two ends of the original lumen — **recanalization**.
WHY: Angiogenesis follows the pressure gradient across the occlusion; channels that connect a high-pressure to a low-pressure end carry flow and are retained, while blind sprouts regress.
- 5 The result is a fibrous plug crossed by several small channels — flow is partially restored but never normal.
WHY: The original single wide lumen has been replaced by several narrow ones; resistance is permanently higher. Clinically this is why **post-thrombotic syndrome** (chronic leg swelling, pain, stasis ulcers) follows a large DVT even after successful treatment.
- 6 Occasionally the organized thrombus undergoes **dystrophic calcification**, forming a hard nodule known as a **phlebolith**.
WHY: Necrotic and fibrotic tissue with a high local calcium–phosphate product mineralises even when serum calcium is normal. Phleboliths are commonly seen incidentally in the pelvis on X-ray.

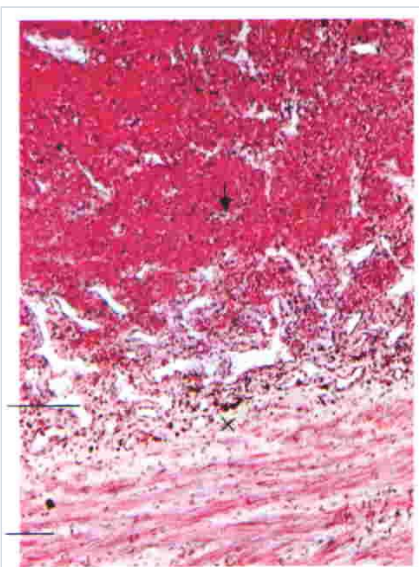


Figure 2.21 Organizing thrombus; HE stain.

Fig 3.23 Organizing thrombus, H&E. Fibroblasts and capillaries are growing into the fibrin mass from the vessel wall.

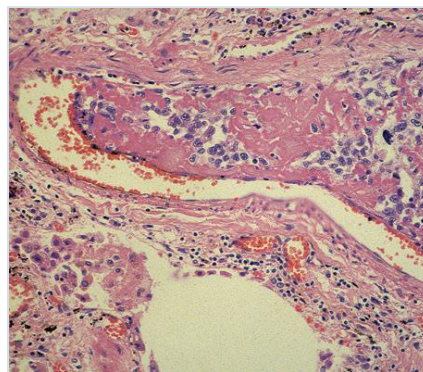


Fig 3.24 Recanalization — new endothelial-lined channels traverse the organized thrombus.



Fig 3.25 A fully recanalised vessel: the single original lumen has been replaced by several smaller channels within fibrous tissue.

3.13 Thrombus versus Post-mortem Clot

At autopsy this distinction decides whether you are looking at the cause of death or at an artefact of being dead. It is examined constantly.

TABLE 3.9 ANTE-MORTEM THROMBUS VS POST-MORTEM CLOT

Feature	Thrombus (ante-mortem)	Post-mortem clot
Attachment	Firmly adherent to the vessel wall — cannot be lifted out intact	Not adherent — lifts out easily as a cast
Lines of Zahn	Present (proving formation in flowing blood)	Absent
Surface and texture	Dry, dull, granular, friable	Wet, shiny, gelatinous, rubbery
Appearance on section	Uniform or laminated	Two layers by gravity sedimentation: a dark red dependent portion ("red currant jelly") and an overlying pale yellow portion ("chicken fat")
Shape	Conforms to the site of injury; may be mural	An exact cast of the vessel, including branch points

MNEMONIC

Chicken fat over currant jelly = dead man's clot. If you can lift it out whole and it shines, the patient died with it, not from it.

SUMMARY — THROMBOSIS IN THREE SENTENCES

1. Thrombus development depends on the relative contribution of the three components of **Virchow's triad**: endothelial injury, abnormal blood flow, and hypercoagulability.
2. Thrombi may **resolve, propagate, organize (with or without recanalization), calcify, or embolize**.
3. Thrombosis injures tissue in exactly two ways: by **local vascular occlusion**, or by **distal embolisation**.

PART FOUR

04 Embolism

A thrombus is dangerous where it forms. An embolus is dangerous somewhere else entirely — and that displacement is the whole point. Ninety-nine percent of emboli are dislodged thrombi, but the remaining one percent (fat, air, nitrogen, amniotic fluid, tumour, cholesterol, marrow) produce some of the most distinctive syndromes in pathology.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Trace an embolus from any starting point to its destination, including paradoxical and retrograde routes.
- Explain the three mechanisms by which a large pulmonary embolus kills.
- Compare thromboembolism, fat, gas and amniotic fluid embolism on setting, latent period, morphology and outcome.

4.1 Definition and the Four Routes of Travel

DEFINITION

Embolism is the **occlusion of part of the cardiovascular system by any detached intravascular mass carried there in the bloodstream from a distant site**. The transported mass is an **embolus**; it may be solid, liquid or gaseous.

99% of emboli are dislodged thrombi — hence the term thromboembolism. Unless a vignette gives you a specific reason to think otherwise (long bone fracture, labour, diving, central line), assume thromboembolism.

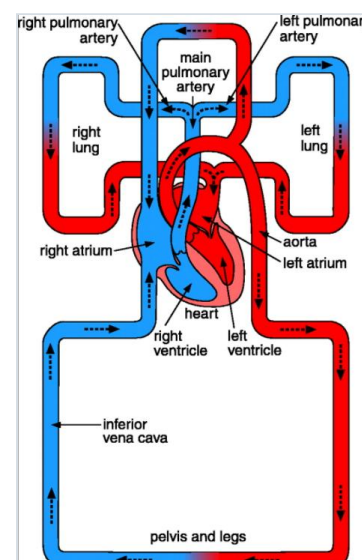


Fig 4.1 Routes of embolus transport. An embolus travels with the blood and lodges where the vessel becomes too narrow to let it pass — so its destination is determined entirely by where it started.

TABLE 4.1 FOUR ROUTES, AND HOW TO WORK OUT THE DESTINATION

Route	Origin	Destination and reasoning
Venous embolism	Systemic veins — above all the deep veins of the leg ; also pelvic and periprostatic plexuses	The lung. Venous blood returns to the right heart and is then pumped into the pulmonary arteries, whose branches are the first capillary-sized bed it meets.
Arterial (systemic) embolism	Left heart — mural thrombus of the left ventricle or atrium, valve vegetations ; also aortic aneurysm and ulcerated aortic plaque	Lower limbs (75%), brain (10%), then intestine, kidney and spleen. The embolus follows the systemic arterial tree and lodges wherever the calibre falls below its diameter.
Paradoxical embolism	A venous thrombus, in a patient with a right-to-left shunt (patent foramen ovale, atrial or ventricular septal defect)	The systemic circulation. The embolus crosses from the right to the left heart through the defect, bypassing the pulmonary filter. Suspect it when a <i>young</i> patient has a stroke and a DVT.
Retrograde embolism	A thrombus in a large vein, e.g. the IVC	Against the direction of flow, when a sudden rise in intrathoracic or intra-abdominal pressure (coughing, straining, sneezing) transiently reverses flow. Explains, for example, vertebral metastasis from pelvic tumours via Batson's plexus.

HIGH-YIELD — THE ONE RULE THAT ANSWERS MOST EMBOLISM QUESTIONS

Follow the blood. Ask where the embolus started, then trace the circulation forwards until the vessel becomes narrower than the embolus. Veins → right heart → **lung**. Left heart or aorta → **systemic organs**. If a venous-origin embolus reaches a systemic organ, there *must* be a shunt — that is a paradoxical embolus by definition.

4.2 Pulmonary Thromboembolism

Over 95% of pulmonary emboli originate in the deep veins of the lower limb, most often the popliteal, femoral and iliac veins. Note that calf vein thrombi are common but rarely embolise dangerously; it is **proximal propagation above the knee** that creates a clinically significant embolic risk. This is why ultrasound surveillance concentrates on the proximal veins.

Outcome depends on the size of the embolus

Large ("saddle") embolus — sudden death

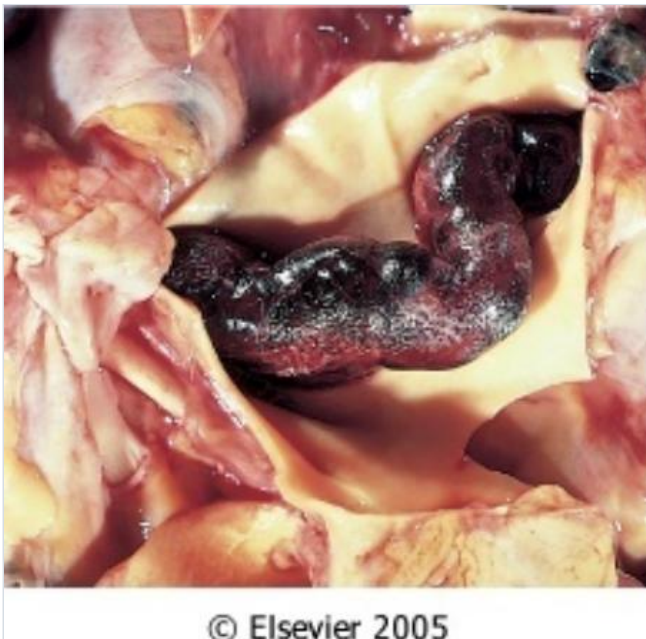
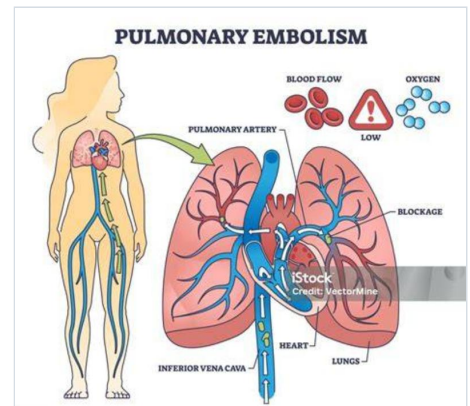


Fig 4.3 Large **saddle embolus** from a femoral vein lying astride the bifurcation of the main pulmonary artery into left and right branches. It has retained the shape of the vein in which it formed.

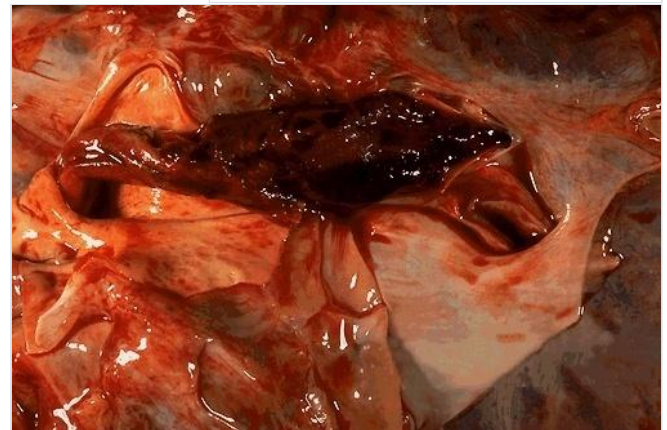


Fig 4.4 The pulmonary trunk and both main arteries opened at autopsy to reveal the saddle thromboembolus. An embolus of this size is almost always immediately fatal.

○ MECHANISM — THE THREE WAYS A MASSIVE PE KILLS

- 1 Mechanical obstruction of pulmonary outflow.** Occlusion of the main trunk or of >60% of the pulmonary vascular bed abruptly prevents blood reaching the left heart. Left ventricular preload collapses, cardiac output falls, and the patient becomes profoundly hypotensive.

WHY: The left ventricle can only pump what it receives. A completely obstructed pulmonary circuit means zero left-sided filling, which is electromechanical dissociation — the heart beats but ejects nothing.

- 2 Acute right heart strain — acute cor pulmonale.** Pulmonary vascular resistance rises abruptly, and the thin-walled right ventricle must suddenly generate far higher pressure.

WHY: *The right ventricle is built for a low-pressure circuit; it has no time to hypertrophy. It dilates acutely, its wall tension and oxygen demand rise while its coronary perfusion pressure falls, and it fails. The dilated RV also bows the interventricular septum leftwards, further impairing LV filling.*

- 3 Reflex vascular and bronchial spasm.** Serotonin, thromboxane and other mediators released from platelets within the embolus cause pulmonary arterial vasoconstriction and bronchoconstriction.

WHY: *This amplifies the mechanical obstruction chemically. It explains why an embolus that occludes only a fraction of the bed can still cause severe haemodynamic compromise, and why hypoxaemia is often worse than the degree of occlusion predicts.*

NET RESULT Sudden death, right heart failure, or cardiovascular collapse. Roughly two-thirds of fatal pulmonary emboli kill within the first hour.

Small emboli

Most pulmonary emboli are small and **clinically silent**: they occlude a peripheral branch, and the lung's **dual blood supply** — pulmonary artery plus bronchial artery — keeps the parenchyma alive while the embolus is lysed and organized. Such emboli are often found incidentally at autopsy.

But if the bronchial circulation is already inadequate — as in a patient with **chronic passive congestion of the lung** from left heart failure — the same small embolus produces **pulmonary haemorrhage or a red (haemorrhagic) infarct**. This is the direct link back to §1.5 and forward to §5.3, and it is a favourite examination connection.

Recurrent small emboli over months to years progressively obliterate the pulmonary vascular bed, producing **chronic thromboembolic pulmonary hypertension**, right ventricular hypertrophy and chronic cor pulmonale.

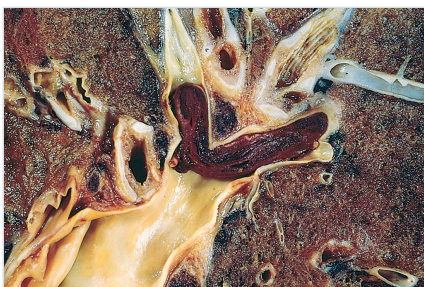


Fig 4.5 Thromboemboli within opened pulmonary arteries. Note the coiled, cast-like shape — they retain the form of the leg vein in which they formed.



Fig 4.6 Sectioned lung with thromboemboli filling branch pulmonary arteries.



Fig 4.7 Histology of an organizing thromboembolus within a pulmonary artery — fibrin and cellular debris being invaded by granulation tissue.

CLINICAL CORRELATION — RECOGNISING AND PREVENTING PE

Presentation: sudden dyspnoea, pleuritic chest pain, tachypnoea, tachycardia, and — if infarction has occurred — haemoptysis and a pleural rub. Hypotension and raised JVP indicate a massive embolus. The ECG classically (but uncommonly) shows S1Q3T3; sinus tachycardia is far more frequent. **D-dimer** is sensitive but not specific — useful to *exclude* PE in a low-probability patient. **CT pulmonary angiography** is the diagnostic standard.

Prevention is the real lesson. Because most PEs arise from leg DVT, and most DVTs arise from stasis during immobility, prophylaxis — early mobilisation, graduated compression stockings, intermittent pneumatic compression and low-molecular-weight heparin in at-risk inpatients — prevents far more deaths than treatment cures.

4.3 Systemic (Arterial) Thromboembolism

Origin: about 80% arise from **intracardiac mural thrombi** — two-thirds of these from left ventricular wall thrombus after myocardial infarction, and about a quarter from a dilated left atrium in **atrial fibrillation**. The remainder come from **valvular vegetations**, aortic aneurysms, ulcerated atherosclerotic plaques, and paradoxical emboli. In perhaps a third of cases no source is ever identified.

Destinations, in order of frequency: lower limbs (~75%), brain (~10%), then intestine, kidney and spleen.

Unlike pulmonary emboli, which lodge in a tissue with a dual blood supply, systemic emboli usually lodge in organs supplied by **end arteries**. The consequence is therefore **infarction** far more often than not — and this is the direct link into Part 5.

4.4 Fat and Marrow Embolism

Setting: fractures of the shafts of **long bones** (femur, tibia) — which have fatty marrow — and also soft tissue trauma, burns, orthopaedic procedures and liposuction. Fat and marrow enter the circulation through torn marrow sinusoids or ruptured adipose venules.

Crucially, fat embolism is common but fat embolism syndrome is rare. Fat globules can be found in the pulmonary circulation of up to 90% of patients with severe skeletal injury, yet fewer than 10% develop the clinical syndrome. The difference lies in the second, biochemical mechanism below.

MECHANISM — THE TWO-HIT PATHOGENESIS OF FAT EMBOLISM SYNDROME

- Hit 1 — mechanical.** Fat globules and marrow fragments released from the fracture enter torn venules and travel to the lung, physically obstructing pulmonary capillaries.

WHY: *Marrow sinusoids are thin-walled and held open by the surrounding bone, so they cannot collapse; a fracture opens them directly into the venous circulation, and the intramedullary pressure generated by the injury forces marrow contents in.*

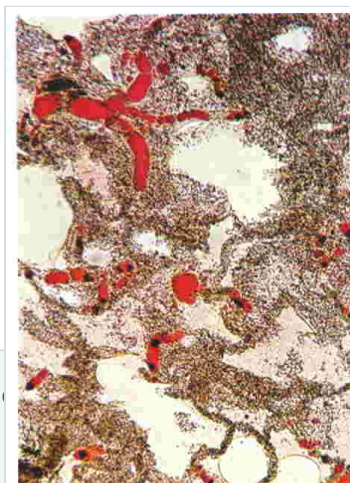


Figure 3.17 Fat embolism in the lung: Sudan stain.

Fig 4.8 Fat embolism in the lung, **Sudan stain** on a frozen section — fat globules stain orange-red within alveolar capillaries. Routine processing dissolves fat in xylene, so it is **invisible on a standard H&E paraffin section** and appears only as empty, clear vacuoles.

- 2 Hit 2 – biochemical.** Over 24–72 hours, lipases hydrolyse the neutral fat into **free fatty acids**, which are directly toxic to endothelium and pneumocytes, and platelets aggregate on the globule surfaces.
WHY: *This is why there is a **latent period**. The mechanical obstruction alone is usually tolerable; the delayed chemical injury is what produces the syndrome. Free fatty acids damage the alveolar–capillary membrane in exactly the way that causes ARDS.*
- 3** Endothelial injury and platelet consumption produce diffuse alveolar damage, pulmonary edema, haemorrhage, atelectasis and **thrombocytopenia**.
WHY: *Platelets adhere to fat globules and are consumed — hence the petechial rash and the falling platelet count.*
- 4** Small globules deform and pass through the pulmonary capillary bed into the systemic circulation, reaching **brain, kidney and skin**.
WHY: *Unlike a rigid thrombus, a fat droplet is deformable and can squeeze through a capillary. This is why fat embolism causes cerebral signs without a shunt — a key distinction from thromboembolism.*

NET RESULT The classic triad, 24–72 h after a long-bone fracture: ① **respiratory distress** (dyspnoea, tachypnoea, hypoxaemia), ② **neurological abnormality** (irritability, confusion, progressing to delirium or coma), ③ a **petechial rash** — characteristically over the upper anterior chest, axillae, neck and conjunctivae, and a valuable clue because it appears in no other embolic syndrome. Plus thrombocytopenia and anaemia. Mortality is around 10%. The older lecture terminology for the disseminated form is "**lung–brain–kidney–skin embolism syndrome**".

Quantitatively, the lecture notes that **9–20 g of fat** entering the pulmonary circulation may occlude up to **75% of the pulmonary arteries**, producing pulmonary edema, haemorrhage, atelectasis, asphyxia and right heart failure.

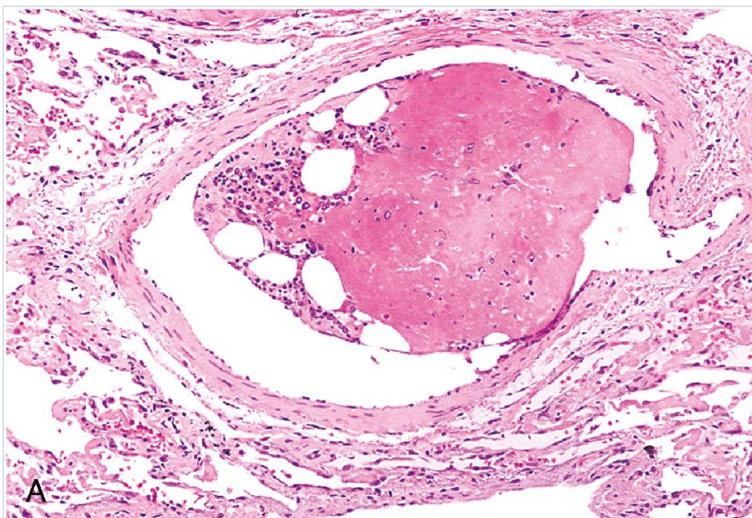


Fig 4.9 Fat emboli within a pulmonary arteriole appearing as clear, rounded, empty vacuoles on routine H&E — the fat itself has been dissolved out during processing.

4.5 Gas Embolism and Decompression Sickness

Air embolism

How air gets in: chest trauma; cardiothoracic, neurosurgical (especially sitting-position operations) and obstetric procedures; insertion, use or removal of a **central venous catheter**; haemodialysis; laparoscopic insufflation; and rupture of uterine or vaginal veins during delivery or criminal abortion.

MECHANISM — WHY GAS IS MORE DANGEROUS THAN AN EQUIVALENT VOLUME OF SOLID

- 1 Air enters a vein where local venous pressure is **below atmospheric** — most importantly in neck and thoracic veins during inspiration.
WHY: Inspiration generates negative intrathoracic pressure, which sucks air in through any open venous channel. This is why a patient having a central line removed is asked to hold their breath or lie head-down.
- 2 Bubbles are carried to the right ventricle, where the beating heart churns air with blood into a **compressible, frothy foam**.
*WHY: This is the critical step. The right ventricle is a pump designed for incompressible fluid. Foam compresses instead of being ejected — the ventricle contracts but nothing leaves. This is called an **air lock**.*
- 3 Outflow from the right heart ceases; cardiac output collapses.
WHY: Same end-point as a massive saddle embolus, reached by a different route.
- 4 Bubbles also mechanically injure endothelium and activate platelets and complement at the blood–gas interface.
WHY: A gas–liquid interface denatures plasma proteins, which is why gas emboli cause more inflammatory injury than their volume alone suggests.

NET RESULT Volume thresholds: small volumes are absorbed harmlessly; **>100 mL** causes acute distress; **>300 mL** entering rapidly is generally **fatal**. Management: place the patient in the **left lateral decubitus, head-down (Durant) position** so that air is trapped at the apex of the right ventricle away from the out-flow tract, give 100% oxygen, and aspirate air from a central line if one is in place.

Decompression sickness (caisson disease)

Who: deep-sea divers and **underwater construction workers** ascending too rapidly from high ambient pressure to normal; and **aviators** or high-altitude pilots ascending from normal to low pressure.

MECHANISM — WHY NITROGEN, AND WHY ONLY ON ASCENT

- 1 At depth, ambient pressure is high, so by **Henry's law** far more gas dissolves in blood and tissue than at the surface.
WHY: The solubility of a gas in a liquid is directly proportional to its partial pressure. At 30 m the ambient pressure is four atmospheres, so four times as much gas dissolves.
- 2 Oxygen and CO₂ are metabolised or buffered, but **nitrogen is inert and highly fat-soluble**, so it simply accumulates — preferentially in adipose tissue, bone marrow, nerve myelin and joint tissues.
WHY: Nitrogen cannot be consumed by any metabolic pathway, so its only exit is back through the lung. It concentrates in fat because it is about five times more soluble in lipid than in water.
- 3 On **rapid** ascent, ambient pressure falls faster than nitrogen can diffuse back into the lungs, so the dissolved gas comes out of solution as **bubbles** within blood, tissue fluid and fat.
WHY: Exactly like opening a carbonated drink. Slow ascent — or a decompression stop — allows nitrogen to be exhaled gradually and no bubbles form. This is why the rate of ascent, not the depth alone, determines the disease.
- 4 Bubbles obstruct capillaries, distend tissues and activate platelets and complement at the gas–blood interface.
WHY: The same interfacial injury as in air embolism, but generated throughout the body simultaneously rather than at one entry point.

NET RESULT Acute — "the bends": severe pain in joints and muscles from bubbles in periarticular tissue, forcing the sufferer to bend over. **"The chokes":** respiratory distress from pulmonary vascular bubbles causing edema, haemorrhage and atelectasis. **"The staggers":** vertigo and neurological deficits from

bubbles in the CNS. **Chronic (caisson disease proper):** repeated or persistent bubbles cause ischaemic necrosis of bone — **avascular necrosis of the femoral head, humeral head and tibia** — because these sites have fatty marrow and an end-arterial supply. Treatment is immediate **recompression in a hyperbaric chamber** followed by slow, controlled decompression, plus 100% oxygen.

4.6 Amniotic Fluid Embolism

Rare (about 1 in 40,000 deliveries) but catastrophic, with a mortality approaching 80% and accounting for roughly 10% of maternal deaths in developed countries. Survivors frequently have permanent neurological deficit.

○ MECHANISM

- 1 After rupture of the membranes, **vigorous uterine contractions** force amniotic fluid between the membranes and the uterine wall, up to the placental margin, and into the torn **uterine venous sinusoids**.

WHY: Two conditions must coexist: a breach in the fetal membranes (providing access) and a pressure gradient driving fluid into an open maternal vein (provided by contraction against a torn placental bed).

- 2 Amniotic fluid contents — **fetal squames, lanugo hairs, vernix caseosa fat, mucin and meconium** — travel to the pulmonary circulation.

WHY: The uterine veins drain into the systemic venous system, so the lung is the first capillary bed reached.

- 3 These particulates cause mechanical obstruction, but far more importantly they trigger an **anaphylactoid reaction** to fetal antigens, with massive release of vasoactive and procoagulant mediators.

WHY: The amount of particulate matter is too small to account for the clinical severity. The dominant injury is immunological — which is why the condition is now often called *anaphylactoid syndrome of pregnancy*.

- 4 Result: intense pulmonary vasoconstriction, right then left heart failure, non-cardiogenic pulmonary edema (diffuse alveolar damage), and — because amniotic fluid is rich in **tissue factor** — fulminant **DIC**.

WHY: The tissue factor content of amniotic fluid is what converts a cardiorespiratory catastrophe into a haemorrhagic one; the patient collapses and then bleeds uncontrollably from the uterus and every puncture site.

NET RESULT Sudden, during or immediately after labour: severe dyspnoea, cyanosis, hypotensive shock, seizures, coma, then profuse haemorrhage from DIC. **Autopsy finding (classic and examinable):** fetal **squamous cells, lanugo hairs, vernix fat and mucin** within pulmonary arteries and capillaries.

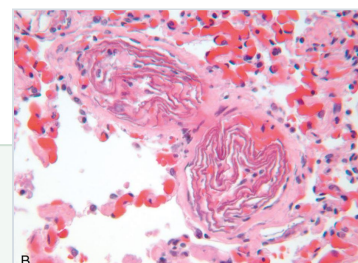


Fig 4.10 Amniotic fluid embolism. A pulmonary arteriole contains a mass of **fetal squamous cells** (squames) — the diagnostic finding. Lanugo hairs, vernix fat and mucin may also be present.

4.7 Other Emboli

- **Tumour embolism** — clusters of malignant cells entering the circulation. This is the mechanism of haematogenous **metastasis**. Renal cell carcinoma and hepatocellular carcinoma characteristically grow as solid tumour thrombus along the renal vein and IVC.
- **Septic embolism** — an infected thrombus fragment, typically from infective endocarditis. Produces **abscesses** at the site of lodgement rather than a bland infarct, and may weaken the arterial wall to form a **mycotic aneurysm**. See §5.5.

- **Atherosclerotic (cholesterol) embolism** — fragments of a ruptured plaque, recognisable microscopically by **needle-shaped cholesterol clefts** within the occluded vessel. Often follows angiography or vascular surgery; produces **blue toe syndrome**, livedo reticularis and renal impairment.
- **Bone marrow embolism** — intact fragments of haematopoietic marrow with fat, found in pulmonary vessels after fracture or vigorous cardiopulmonary resuscitation. Usually an incidental autopsy finding.
- **Foreign body embolism** — catheter fragments, intravenous drug-user talc or cotton fibres, bullet or shrapnel fragments.

4.8 All Embolism Types Compared

TABLE 4.2 MASTER COMPARISON OF EMBOLISM TYPES

Type	Typical setting	Timing	Key clinical features	Diagnostic morphology
Thromboembolism (pulmonary)	DVT of leg; immobility, surgery, malignancy, OCP, pregnancy	Sudden	Dyspnoea, pleuritic pain, tachycardia; massive PE → sudden death	Coiled saddle embolus ; lines of Zahn in the embolus
Thromboembolism (systemic)	LV mural thrombus post-MI; atrial fibrillation; vegetations	Sudden	Acute limb ischaemia (75%), stroke (10%), bowel/renal/splenic infarct	White (pale) infarct in a solid organ
Fat embolism	Long bone fracture , soft tissue trauma, burns	24–72 h latent	Triad: dyspnoea + confusion + petechial rash ; thrombocytopenia	Fat globules — Sudan-positive on frozen section , empty vacuoles on H&E
Air embolism	Central line, chest trauma, surgery, delivery	Immediate	>100 mL distress; >300 mL fatal; "mill-wheel" murmur	Frothy blood in the right ventricle (air lock)
Decompression sickness	Rapid ascent — divers, caisson workers, aviators	Minutes–hours; bone lesions years later	Bends (joint pain), chokes (respiratory), staggers (neurological)	Nitrogen bubbles; chronic → avascular necrosis of femoral/humeral head
Amniotic fluid embolism	Labour or immediate post-partum	Sudden	Dyspnoea, cyanosis, shock, seizures, then DIC ; ~80% mortality	Fetal squames, lanugo, vernix fat, mucin in pulmonary vessels
Septic embolism	Infective endocarditis	Variable	Fever, new murmur, Janeway lesions, Osler nodes, Roth spots	Abscess at the lodging site; mycotic aneurysm
Cholesterol embolism	After angiography or vascular surgery; severe atherosclerosis	Days after procedure	Blue toe syndrome , livedo reticularis, renal failure, eosinophilia	Needle-shaped cholesterol clefts in the vessel

SUMMARY — EMBOLISM IN THREE SENTENCES

1. An embolus is any detached solid, liquid or gaseous mass carried by the blood to a site distant from its origin.
2. Pulmonary emboli derive primarily from lower-extremity deep vein thrombosis, and their effect depends on the **size** of the embolus.
3. Systemic emboli derive primarily from cardiac mural or valvular thrombi, aortic aneurysms or atherosclerotic plaque; whether one causes infarction depends on the **site** of embolisation and the adequacy of the **collateral circulation**.

PART FIVE

05 Infarction

This is where the story ends: tissue that has been cut off from its blood supply and died. Nearly every clinically important consequence of the previous four Parts — myocardial infarction, stroke, bowel infarction, gangrene — arrives here. The two questions worth asking of any infarct are: *why did it happen at all* (many occlusions cause none), and *why is it that colour*.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- List the four factors that determine whether a vascular occlusion produces an infarct.
- Explain mechanistically why some infarcts are white and others red.
- Describe the gross and microscopic evolution of an infarct over hours, days and weeks.
- Differentiate dry, wet and gas gangrene.
- Trace a complete thrombosis → embolism → infarction sequence in a single patient.

5.1 Definition and Causes

DEFINITION

An **infarct** is an area of **ischaemic necrosis** of a tissue or organ caused by occlusion of either its **arterial supply** or its **venous drainage**. The process of its formation is **infarction**.

Note the "or". Most infarcts are arterial, but venous occlusion can also cause one — see §5.3, where it is the key to understanding the red infarct.

Causes, in order of frequency

- **Thrombotic arterial occlusion** — usually on a ruptured atherosclerotic plaque. The commonest cause overall; this is myocardial infarction and most strokes.
- **Arterial embolism** — from cardiac mural thrombus, valvular vegetation, aortic plaque or a paradoxical venous embolus.
- **Vasospasm** — cocaine-induced coronary spasm, Prinzmetal angina.
- **Extrinsic compression** of a vessel — tumour, **volvulus**, **intussusception**, incarcerated **hernia**, or a testicular/ovarian **torsion**. Note that twisting compresses the thin-walled *veins* first, which is why these lesions produce **haemorrhagic** infarcts.
- **Vessel wall expansion or dissection** — atheromatous plaque haemorrhage, aortic dissection occluding a branch ostium.
- **Venous thrombosis** — where there is no alternative outflow, e.g. ovarian, testicular or mesenteric vein thrombosis, or superior sagittal sinus thrombosis.
- **Vasculitis**, sickle cell crisis, and external pressure (a tight cast, prolonged pressure causing a pressure sore).

5.2 The Four Determinants of Whether an Infarct Develops

Occlusion does not automatically mean infarction. A surgeon can tie off the splenic artery and the spleen may survive on short gastric collaterals; a pulmonary artery branch can be blocked and the lung remains viable. Four variables decide the outcome.

TABLE 5.1 WILL AN OCCLUSION CAUSE AN INFARCT?

Factor	How it works	Illustration
① Nature of the vascular supply	The single most important factor. An alternative route means the tissue survives. Two protective arrangements exist: effective anastomoses , and a genuine dual blood supply .	Protected: lung (pulmonary + bronchial arteries), liver (portal vein + hepatic artery), forearm and hand (radial + ulnar), brain (circle of Willis). Vulnerable — end-arterial: kidney, spleen, heart, retina, testis.
② Rate of development of the occlusion	Slowly developing occlusions permit collateral vessels to enlarge and new ones to grow . A sudden occlusion gives no time for this adaptation.	A coronary artery narrowed over years by atherosclerosis may be totally occluded without infarction, because collaterals have developed. The same artery occluded suddenly by plaque rupture causes a full-thickness infarct.
③ Vulnerability of the tissue to hypoxia	Cells differ enormously in how long they survive without oxygen, largely according to their dependence on oxidative metabolism and their capacity for anaerobic glycolysis.	Neurons: 3–4 minutes. Cardiac myocytes: 20–30 minutes. Renal tubular epithelium: ~30 minutes. Skeletal muscle: several hours. Fibroblasts: many hours. Hence the order of death within any ischaemic field.

Factor	How it works	Illustration
④ Oxygen-carrying capacity of the blood	Even a partially obstructed vessel may deliver enough oxygen if the blood is fully saturated — but not if it is anaemic or de-saturated.	An anaemic or cyanotic patient tolerates arterial insufficiency far less well than a normal person. The same degree of coronary stenosis that causes only stable angina in a healthy adult can cause infarction in a patient with severe anaemia, carbon monoxide poisoning or congestive heart failure.

MNEMONIC — ORGANS WITH A DUAL BLOOD SUPPLY

"LLH" — Lung, Liver, Hand (and forearm). Lung = pulmonary + bronchial arteries. Liver = hepatic artery + portal vein. Hand/forearm = radial + ulnar arteries via the palmar arches. These are the tissues that resist infarction and, when they do infarct, do so **red**.

5.3 White versus Red Infarct

White (pale, anaemic) infarct

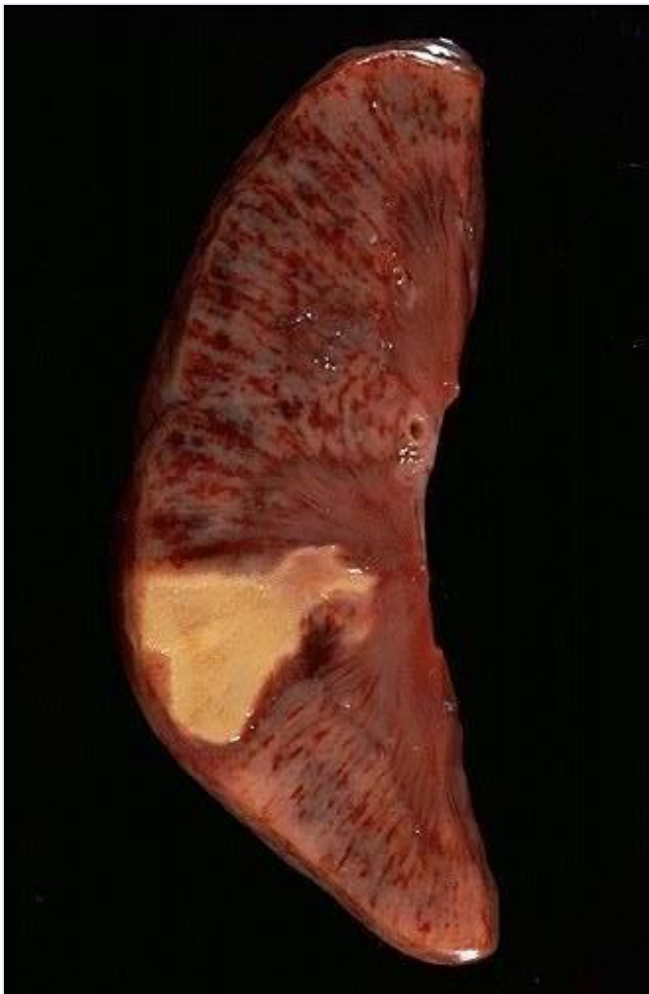


Fig 5.1 White infarct of the spleen. A sharply demarcated, **wedge-shaped**, pale yellow-white area with its base on the capsule and its apex pointing towards the occluded arterial branch — the geometry of an end-arterial territory.



Fig 5.2 A second splenic infarct. The pale wedge contrasts sharply with the dark red viable parenchyma around it — the red cells within the necrotic zone have been lysed and cleared.

○ MECHANISM — WHY SOME INFARCTS TURN PALE

- 1 An **end artery** supplying a **solid, compact** organ is occluded. The organ has little or no effective collateral circulation.
WHY: Kidney, spleen, heart and retina are supplied by end arteries with minimal anastomoses, so occlusion removes the only supply to that territory.
- 2 Inflow stops completely. Only the small amount of blood already inside the territory remains.
WHY: There is no second route to push more blood in — this is the decisive difference from a dual-supply organ.
- 3 The tissue undergoes **coagulative necrosis**: cells die but their structural outlines persist for days.
WHY: Ischaemia denatures both the structural proteins and the lysosomal hydrolases that would otherwise digest the cell, so the architecture is preserved as a "ghost" outline. The brain is the sole exception — see the trap box below.

- 4 The red cells trapped in the territory are **lysed**, and their haemoglobin is degraded and removed by macrophages.
WHY: Dead red cells have no way to maintain their membranes; haemolysis follows, and phagocytes clear the pigment. Nothing new arrives to replace it, because inflow is blocked.
- 5 The necrotic zone therefore progressively **decolorises**, becoming pale yellow-white and sharply demarcated from the surrounding congested viable tissue.
WHY: A tissue is red because of haemoglobin. Remove the haemoglobin and prevent replacement, and what is left is the pale colour of the protein matrix itself.

NET RESULT A **pale, firm, wedge-shaped** infarct with the **apex at the occluded vessel** and the **base at the organ surface**, often with a thin **hyperaemic rim** of inflamed viable tissue at its edge. Typical organs: **kidney, spleen, heart**.

Red (haemorrhagic) infarct

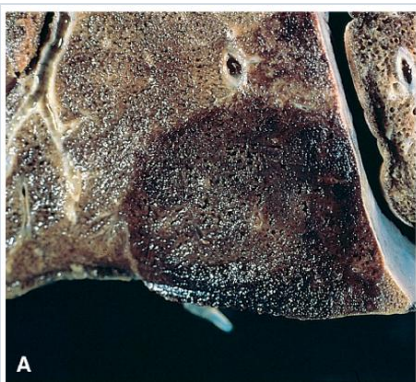


Fig 5.3 Red infarct of the lung. A firm, dark red-brown area with its base on the pleural surface. The infarct is haemorrhagic throughout because bronchial arterial flow continued to fill a territory the pulmonary artery could no longer oxygenate.



Fig 5.4 An older haemorrhagic infarct in a fixed lung specimen. As the extravasated red cells are broken down, the lesion turns from red to **brown** — the same haemoglobin → haemosiderin conversion described in §2.4.



Fig 5.5 Haemorrhagic infarction of small bowel following volvulus. The affected loop is dark purple-black and grossly distinct from the viable pink bowel beside it.

MECHANISM — THE THREE ROUTES TO A RED INFARCT

- 1 **Route A — venous occlusion.** Outflow is blocked while **arterial inflow continues**. Blood is pumped in but cannot leave, so the tissue engorges. Rising intravascular pressure eventually stops arterial inflow too, and the tissue dies — but it dies **full of blood**.
WHY: This is the central insight. A venous infarct is haemorrhagic by definition, because the blood was driven in under arterial pressure before the tissue died, and had no way out. Classic examples: testicular and ovarian torsion, mesenteric vein thrombosis, volvulus, intussusception, strangulated hernia.
- 2 **Route B — a loose, spongy tissue.** Lung and bowel have abundant interstitial space into which blood can pool.
WHY: A compact organ like the kidney has no room for blood to accumulate; a spongy organ like the lung does. The same volume of haemorrhage disappears in one and is conspicuous in the other.

- 3 Route C — dual circulation or collateral reperfusion.** After the main artery is occluded, blood still enters the dead zone from the second supply or from an adjacent bed — but not enough to keep it alive.

WHY: *In the lung, the bronchial arteries continue to perfuse a territory whose pulmonary artery is blocked. That flow is sufficient to fill the necrotic zone with blood but not sufficient to oxygenate it, because bronchial arterial flow is only ~1% of cardiac output. Haemorrhage is compounded by ischaemic damage to the capillary walls, which then leak. The same happens after **reperfusion** — for example thrombolysis or angioplasty restoring flow to an already-necrotic myocardium, which converts a white infarct into a haemorrhagic one.*

NET RESULT A **dark red to purple-black, firm, wedge-shaped** infarct, often with an overlying fibrinous exudate on the serosal surface. Typical organs: **lung, small intestine, testis, ovary** — and any organ after reperfusion. As the extravasated red cells are broken down over days, the lesion becomes brown from **haemosiderin**.

TABLE 5.2 WHITE VERSUS RED INFARCT

Feature	White (pale / anaemic) infarct	Red (haemorrhagic) infarct
Vascular event	Arterial occlusion	Venous occlusion, or arterial occlusion in a dual-supply or reperfused organ
Tissue architecture	Solid, compact — no room for blood to pool	Loose, spongy — abundant interstitial space
Collateral supply	End-arterial , minimal collaterals	Dual supply or good collaterals
Typical organs	Kidney, spleen, heart , retina	Lung, intestine, testis, ovary , brain (venous)
Colour	Pale yellow-white, becoming paler with time	Dark red to purple-black, becoming brown (haemosiderin)
Demarcation	Sharply demarcated	Less sharply demarcated
Shape	Both are wedge-shaped , with the apex at the occluded vessel and the base at the organ surface — the geometry of any arterial territory.	
Necrosis type	Coagulative in all organs except the brain , where it is liquefactive .	

EXAM TRAP — THE BRAIN IS DIFFERENT, TWICE OVER

First, cerebral infarcts undergo **liquefactive**, not coagulative, necrosis. The reason is mechanistic: the brain has very little supporting collagenous stroma and a very high lipid content, and microglia release abundant hydrolytic enzymes, so the dead tissue digests itself into a fluid-filled cyst rather than retaining its outline. Second, the brain is on the *vulnerable* list for hypoxia (neurons die in 3–4 minutes) despite having the circle of Willis — because that anastomosis protects only against proximal occlusion, not against occlusion of the small penetrating end arteries beyond it.

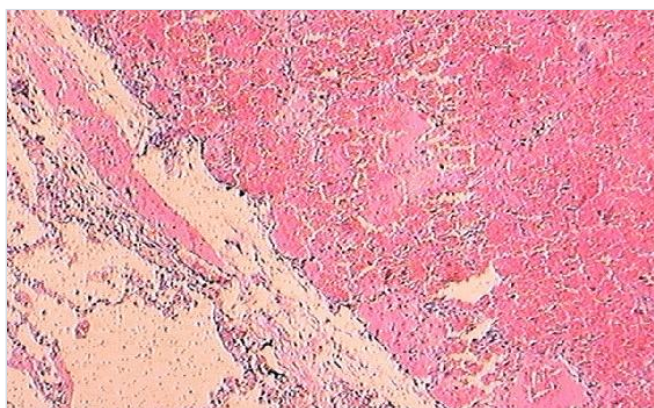


Fig 5.6 Red infarct of lung, histology. The alveolar spaces on the right are filled with red cells; the necrotic alveolar walls have lost their nuclei but retain their outlines — **coagulative necrosis**.

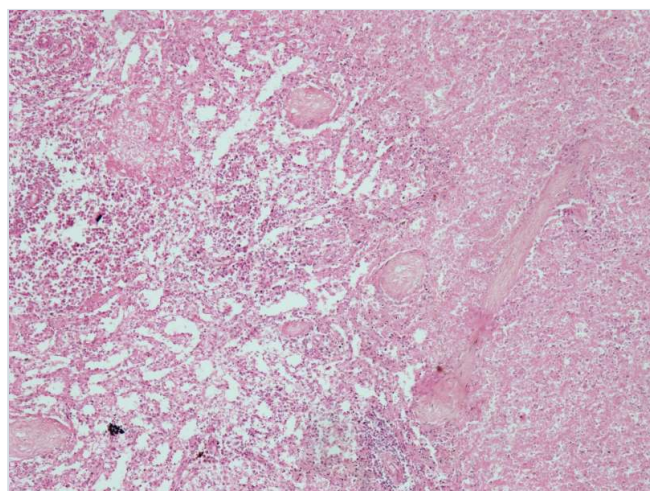


Fig 5.7 Coagulative necrosis at higher power: cells are anucleate "ghosts" with preserved outlines, and a band of acute inflammatory cells marks the viable margin.

5.4 Morphology and Time Course

The evolution below uses myocardial infarction as the model, because its timings are the most examined, but the sequence is the same in any organ.

TABLE 5.3 EVOLUTION OF AN INFARCT

Time	Gross	Microscopic	Clinical relevance
0–4 h	No change visible	None by light microscopy; possible wavy fibres at the border	Diagnosis is clinical + ECG + troponin , not morphological

Time	Gross	Microscopic	Clinical relevance
4–24 h	Dark mottling	Coagulative necrosis begins: hyperosinophilic cytoplasm, pyknosis and karyolysis; early neutrophil margination	Highest risk of arrhythmia and sudden death
1–3 days	Pale yellow-tan centre with a hyperaemic border	Dense neutrophil infiltrate ; complete loss of nuclei	Fibrinous pericarditis may appear over a transmural infarct
3–7 days	Yellow centre, softening ; red-brown margin	Neutrophils die; macrophages arrive and begin phagocytosing debris	Maximum risk of rupture — the wall is at its weakest. Free wall rupture → tamponade; septal rupture → VSD; papillary muscle rupture → acute mitral regurgitation
1–2 weeks	Depressed red-grey border	Granulation tissue : capillaries, fibroblasts, early collagen	Risk of rupture falls; ventricular aneurysm may begin to form
2–8 weeks	Grey-white, firm, contracted scar	Dense collagenous scar ; no regeneration of myocytes	Cardiac muscle is permanent tissue — it cannot regenerate, so function is permanently lost. Scar predisposes to arrhythmia and aneurysm

General principles that apply to all infarcts: they are **wedge-shaped** with the apex at the occluded vessel; the margins become better defined over time; the dominant histological change is **coagulative necrosis** (except brain); an acute inflammatory response begins within hours and is followed by macrophages and then by **organization and fibrous scar**; and if the organ's cells are permanent (cardiac myocytes, neurons), the lost function is never recovered.

5.5 Septic Infarct

When an infarct is caused by an **infected (septic) embolus** — classically from infective endocarditis — or when bacteria seed a bland infarct, the lesion does not heal by scarring. Instead it is converted into an **abscess**.

Why: necrotic tissue is an ideal culture medium — it is protein-rich, has no blood supply to deliver leukocytes, antibodies or antibiotics, and no mechanism to clear organisms. Bacteria therefore multiply unchecked, drawing in a massive neutrophil response that liquefies the tissue and walls it off with a pyogenic membrane. The inflammatory response is far more intense than in a bland infarct, and healing requires drainage rather than simple organization.

Additional consequence: organisms in the embolus may destroy the arterial wall at the lodging site, producing a **mycotic aneurysm** that can rupture catastrophically — a classic cause of fatal intracerebral haemorrhage in endocarditis.

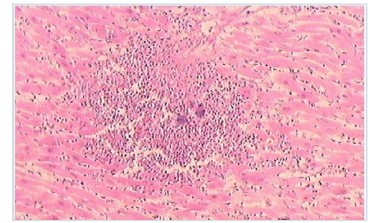


Fig 5.8 Septic infarct. Necrotic tissue is densely infiltrated by neutrophils and colonised by bacteria — the infarct has become an **abscess**.

5.6 Gangrene

Gangrene is not a distinct type of necrosis but a **clinical term** for necrosis of a substantial mass of tissue, usually a limb or a segment of bowel, with superimposed changes.

TABLE 5.4 DRY, WET AND GAS GANGRENE

Type	Mechanism	Appearance	Typical setting	Outcome / management
Dry gangrene	Arterial occlusion with intact venous drainage . Coagulative necrosis; the tissue desiccates because fluid drains away and none arrives.	Dry, shrunken, black, mummified , with a sharp line of demarcation from viable tissue	Toes and feet in diabetes mellitus and atherosclerotic peripheral vascular disease ; Buerger disease	Slow progression; infection is limited because the tissue is dry. May auto-amputate . Elective amputation at the demarcation line.
Wet gangrene	Venous (or combined) obstruction with superimposed bacterial infection . Liquefactive change superimposed on coagulative necrosis.	Swollen, soft, wet, black-green, foul-smelling ; no clear demarcation	Bowel (volvulus, strangulated hernia, intussusception); diabetic foot with infection	Rapid spread, systemic toxæmia and septic shock. Surgical emergency — resection and antibiotics.

Type	Mechanism	Appearance	Typical setting	Outcome / management
Gas gangrene	Infection of necrotic muscle by <i>Clostridium perfringens</i> and related anaerobes. Bacterial α-toxin (a lecithinase) lyses membranes; fermentation of tissue carbohydrate produces gas .	Swollen, discoloured muscle with crepitus (palpable gas) and gas on radiograph; thin, foul, brownish discharge	Deep contaminated wounds — war injuries, crush injuries, soil-contaminated trauma, septic abortion	Life-threatening and rapidly progressive. Urgent radical debridement, penicillin plus clindamycin, hyperbaric oxygen.

MNEMONIC

Dry = Drains (venous outflow open). Wet = Waterlogged (venous outflow blocked) + Wicked bugs. The presence or absence of venous drainage is the whole distinction.

5.7 Fate of an Infarct, and the Complete Chain

Fate

- **Small infarct** — softens, is digested by macrophages and completely **absorbed**; may leave no trace.
- **Large infarct** — undergoes **organization**: granulation tissue replaces necrotic tissue and matures into a contracted **fibrous scar**. The organ surface becomes depressed over the scar.
- **Septic infarct** — becomes an **abscess**, requiring drainage.
- **Infarction of brain, myocardium or lung** — may cause **sudden death**, from arrhythmia, herniation or acute right heart failure respectively.
- **Brain infarct specifically** — liquefies and leaves a fluid-filled **cystic cavity** bounded by reactive gliosis, not a collagenous scar.

The complete chain — one patient, five sections

WORKED EXAMPLE

A 68-year-old woman undergoes total hip replacement. She is immobile for four days post-operatively.

1. **Thrombosis (Part 3).** Immobility → **stasis** in the calf veins; surgery → massive **tissue factor** release and an acute-phase rise in fibrinogen → **hypercoagulability**; the operation itself injures pelvic vein endothelium. All three arms of **Virchow's triad** are satisfied simultaneously, and a red (stasis) thrombus forms in the femoral vein, growing antegradely with a poorly anchored tail.
2. **Congestion and edema (Part 1).** The occluded femoral vein raises capillary hydrostatic pressure upstream. Her leg becomes **swollen, cyanotic, cool and painful** — passive congestion plus pitting edema.
3. **Embolism (Part 4).** On day 6 she sits up. The tail detaches, travels via the iliac vein, IVC and right heart, and lodges in a segmental branch of the right pulmonary artery — a **venous thromboembolism** to the lung.
4. **Infarction (Part 5).** She has long-standing left heart failure, so her **bronchial circulation is already compromised by chronic passive pulmonary congestion** (Part 1 again). The dual supply is therefore inadequate, and a **red (haemorrhagic) wedge infarct** forms — spongy tissue, dual supply, incomplete reperfusion.
5. **Hemorrhage (Part 2).** The infarcted lung bleeds into the alveoli, and she coughs up blood — **haemoptysis**. Fibrinous exudate over the infarcted segment irritates the pleura, producing **pleuritic chest pain** and an audible rub.

SUMMARY — INFARCTION IN THREE SENTENCES

1. Infarcts are areas of ischaemic, usually **coagulative**, necrosis caused by occlusion of arterial supply or, less commonly, venous drainage.
2. They are most commonly caused by occlusive **arterial thrombi** or by **embolisation** of arterial or venous thrombi.
3. Infarcts caused by **venous** occlusion, or occurring in **loose tissues with a dual blood supply**, are typically **haemorrhagic (red)**; those caused by **arterial** occlusion in **compact end-arterial tissues** are **pale (white)**.

06

Review Questions, Master Tables and Glossary

Twenty-eight clinical vignettes covering all five sections, each with a fully explained answer that also says *why the distractors are wrong* — because most marks are lost by eliminating the right answer, not by failing to recognise it. Then the compressed master tables and a glossary for the night before.

HOW TO USE THIS SECTION

- Attempt all 28 questions **closed-book** and write your answers down before checking.
- Read **every** explanation, including those for questions you answered correctly — the distractor analysis teaches as much as the answer.

6.1 Twenty-Eight USMLE-Style Vignettes

PART 1 — HYPEREMIA, CONGESTION AND EDEMA (Q1-7)

Q1 CONGESTION

A 67-year-old man with a 10-year history of hypertension presents with progressive exertional dyspnoea, orthopnoea and a cough productive of rust-coloured sputum. Cytological examination of the sputum shows large mononuclear cells containing coarse golden-brown granules that stain blue with Prussian blue. Which of the following best explains the presence of these cells?

- A. Circulating monocytes have taken up inhaled carbon particles
- B. Alveolar epithelium has undergone squamous metaplasia with keratin accumulation
- C. Alveolar macrophages have phagocytosed red cells that escaped from over-distended septal capillaries
- D. Type II pneumocytes have accumulated surfactant in response to chronic hypoxia
- E. Neutrophils have released myeloperoxidase into the alveolar space

Q2 CONGESTION

A 58-year-old woman with long-standing tricuspid regurgitation is found at autopsy to have an enlarged, tense liver. The cut surface shows a fine, regular mottling of dark red foci on a pale yellow-tan background, resembling a bisected nutmeg. The dark red areas correspond to which anatomical zone, and why?

- A. Randomly distributed, because hepatic congestion is not zonal
- B. Periportal (zone 1), because portal blood arrives there first
- C. Centrilobular (zone 3), because back-pressure is transmitted first to the central vein and zone 3 has the lowest oxygen tension
- D. Midzonal (zone 2), because it is furthest from both blood supplies
- E. The portal tracts, because of fibrous expansion

Q3 EDEMA

A 6-year-old boy is brought in with puffy eyelids that are worst on waking. Urinalysis shows 4+ proteinuria with fatty casts; serum albumin is 1.8 g/dL and there is no haematuria. Which change in Starling forces is the primary driver of his edema?

- A. Increased capillary permeability to protein
- B. Increased interstitial hydrostatic pressure
- C. Lymphatic obstruction
- D. Increased capillary hydrostatic pressure
- E. Decreased plasma colloid osmotic pressure

Q4 EDEMA

A 54-year-old woman who underwent right mastectomy with axillary node clearance three years ago has persistent swelling of the right arm. The skin is thickened and firm, and does not pit on pressure. Which feature of the interstitial fluid best explains why this edema is non-pitting?

- A. It is protein-rich and has provoked interstitial fibrosis
- B. It has a low specific gravity
- C. It contains predominantly neutrophils
- D. It is confined to the subcutaneous fat
- E. It has a fluid-to-serum LDH ratio below 0.6

Q5 TRANSUDATE/EXUDATE

Thoracentesis in a 71-year-old man with bilateral pleural effusions yields clear straw-coloured fluid. Protein is 1.6 g/dL (serum 6.4), LDH 80 U/L (serum 250), specific gravity 1.008, and there are few cells. The fluid does not clot on standing. Which diagnosis is most consistent?

- A. Bacterial pneumonia with parapneumonic effusion
- B. Metastatic adenocarcinoma of the pleura
- C. Pulmonary embolism with infarction
- D. Tuberculous pleuritis
- E. Congestive heart failure

Q6 HYPEREMIA

During a 400-metre sprint, blood flow to the quadriceps rises approximately twentyfold. The muscle is warm and erythematous. Which statement most accurately describes this process?

- A. Reactive hyperemia from prior arterial occlusion
- B. Active hyperemia from arteriolar dilation, delivering oxygenated blood
- C. Passive congestion from impaired venous return during muscle contraction
- D. Exudative edema from increased vascular permeability
- E. Hyperviscosity from haemoconcentration

Q7 CEREBRAL EDEMA

A 45-year-old man with a glioblastoma develops worsening morning headache, vomiting and papilloedema. MRI shows midline shift with the cingulate gyrus displaced beneath the falx cerebri. Compression of which vessel by this herniation would most likely cause secondary infarction?

- A. Anterior cerebral artery
- B. Posterior cerebral artery
- C. Basilar artery
- D. Middle meningeal artery
- E. Superior sagittal sinus

PART 2 — HEMORRHAGE (Q8–11)

Q8 BLEEDING PATTERN

A 7-year-old boy has had three episodes of a swollen, painful knee after minor falls, and bled heavily 12 hours after a dental extraction despite initial haemostasis being achieved. Platelet count and bleeding time are normal; PTT is prolonged and PT is normal. Which defect is most likely?

- A. Deficiency of factor VII
- B. Absence of platelet GpIIb/IIIa
- C. Vitamin C deficiency
- D. Deficiency of factor VIII
- E. Deficiency of von Willebrand factor

Q9 HEMORRHAGE — SITE

A 62-year-old man is brought to the emergency department 40 minutes after collapsing. He had a myocardial infarction 5 days earlier. He is hypotensive with distended neck veins and muffled heart sounds. Pericardiocentesis yields 250 mL of blood. Which factor best explains why this relatively small volume produced cardiovascular collapse?

- A. Pericardial blood directly compresses the coronary ostia
- B. Loss of 250 mL triggers a vasovagal reflex
- C. The fibrous parietal pericardium cannot stretch acutely, so pressure rises steeply and impairs diastolic filling
- D. 250 mL represents more than 20% of total blood volume
- E. The blood in the pericardium causes rapid haemolysis and hyperkalaemia

Q10 HEMORRHAGE — CLASSES

A 24-year-old man involved in a motorcycle crash has a pulse of 122/min, blood pressure 118/96 mmHg, cool clammy skin and mild anxiety. Haemoglobin is 13.8 g/dL. Which statement is correct?

- A. He has lost less than 15% of blood volume, as shown by the normal systolic pressure
- B. The normal haemoglobin excludes acute blood loss
- C. He is in class IV haemorrhagic shock
- D. The narrowed pulse pressure indicates sepsis rather than haemorrhage
- E. He has lost 15–30% of blood volume; the normal haemoglobin does not exclude significant haemorrhage

Q11 HEME CATABOLISM

A large ecchymosis on the thigh is green-yellow on day 6 after injury. The pigment responsible for the green colour is produced by which enzymatic step?

- A. Haem oxygenase opening the porphyrin ring to form biliverdin
- B. Biliverdin reductase converting biliverdin to bilirubin
- C. Ferritin polymerisation to haemosiderin
- D. UDP-glucuronyl transferase conjugating bilirubin
- E. Myeloperoxidase generating hypochlorous acid

PART 3 — THROMBOSIS (Q12–19)**Q12**

VIRCHOW'S TRIAD

A 70-year-old woman is immobile in bed for 6 days after hip replacement. On day 7 her left calf is swollen, warm and tender. Which combination of factors best accounts for the thrombus in her femoral vein?

- A. Deficiency of von Willebrand factor
- B. Stasis from immobility, hypercoagulability from surgical tissue factor release, and endothelial injury from the operation
- C. Turbulence alone
- D. Reduced plasma oncotic pressure
- E. Lymphatic obstruction

Q13

ZAHN LINES

At autopsy, a mass is found within the left ventricle. On sectioning it is dry, grey-white, friable and shows alternating pale and dark laminations. Which conclusion is best supported?

- A. It represents a myxoma
- B. It is a fibrin microthrombus of DIC
- C. It is a post-mortem clot, because it is friable
- D. It formed in flowing blood before death
- E. It formed after death by gravity sedimentation

Q14

THROMBOPHILIA

A 26-year-old woman on a combined oral contraceptive has her second unprovoked DVT. Her mother had a pulmonary embolus at 40. Laboratory testing shows failure of activated protein C to prolong her clotting time. Which molecular abnormality is most likely?

- A. Autoantibodies against heparin–platelet factor 4
- B. Deficiency of protein S
- C. A point mutation in factor V that prevents its cleavage by activated protein C
- D. Deficiency of antithrombin III
- E. A mutation increasing prothrombin transcript stability

Q15

HIT

A 68-year-old man receiving unfractionated heparin after vascular surgery has a platelet count fall from 240 to $68 \times 10^9/L$ on day 6. He then develops an ischaemic right foot. What is the mechanism?

- A. Heparin directly suppresses megakaryocytes
- B. Heparin causes disseminated intravascular coagulation by releasing tissue factor
- C. Antithrombin III is consumed, causing a rebound hypercoagulable state
- D. The platelet fall is dilutional following intra-operative fluid resuscitation
- E. IgG against heparin–platelet factor 4 complexes binds and activates platelets, consuming them while promoting thrombosis

Q16

DIC

A 29-year-old woman develops profuse bleeding from her episiotomy and every venepuncture site two hours after a placental abruption. Platelets $32 \times 10^9/L$, PT and PTT both prolonged, fibrinogen 0.7 g/L, D-dimer markedly raised, schistocytes on the film. Which best explains the simultaneous bleeding and organ ischaemia?

- A. Widespread microvascular fibrin thrombi have consumed platelets and clotting factors
- B. An isolated deficiency of factor VIII
- C. Autoimmune destruction of platelets in the spleen
- D. Vitamin K deficiency from antibiotic therapy
- E. Hyperfibrinogenaemia of pregnancy

Q17

THROMBUS FATE

Eighteen months after an untreated iliofemoral DVT, a venogram shows the vein replaced by fibrous tissue traversed by several small channels. The patient has chronic leg swelling and a medial malleolar ulcer. Which process produced this appearance?

- A. Dystrophic calcification alone
- B. Organization with recanalization
- C. Complete fibrinolytic resolution
- D. Propagation
- E. Embolisation

Q18

ENDOTHELIUM

An intact endothelial monolayer prevents thrombosis by several mechanisms. Which pairing of molecule and action is correct?

- A. Plasminogen activator inhibitor — degrades fibrin
- B. Tissue factor — accelerates antithrombin III
- C. Thromboxane A_2 — inhibits platelet aggregation
- D. Thrombomodulin — binds thrombin and converts it into an activator of protein C
- E. von Willebrand factor — inhibits platelet adhesion to collagen

Q19

VEGETATIONS

A 41-year-old woman with SLE has small, warty, sterile vegetations on *both* surfaces of the mitral valve leaflets at autopsy. There is no valve destruction. What is the diagnosis?

- A. Non-bacterial thrombotic (marantic) endocarditis
- B. Carcinoid heart disease
- C. Libman-Sacks endocarditis
- D. Acute infective endocarditis
- E. Rheumatic valvulitis

PART 4 — EMBOLISM (Q20–24)

Q20

FAT EMBOLISM

A 22-year-old man sustains a mid-shaft femoral fracture. He is stable for 36 hours, then becomes acutely dyspnoeic and confused, with a petechial rash over the upper chest and axillae. Platelets $74 \times 10^9/L$. Which additional finding would best confirm the diagnosis?

- A. Lines of Zahn within a pulmonary arterial mass
- B. Squamous cells and lanugo hairs in pulmonary capillaries
- C. Frothy blood in the right ventricle
- D. Needle-shaped cholesterol clefts within arterioles
- E. Globules staining orange-red with Sudan on a frozen section of lung

Q21

PARADOXICAL EMBOLISM

A 31-year-old woman with a known patent foramen ovale develops sudden right-sided hemiparesis three days after a long-haul flight. Doppler shows a left popliteal DVT. Which route did the embolus most likely take?

- A. Leg vein → IVC → right atrium → across the septal defect → left atrium → systemic arteries → brain
- B. Leg vein → IVC → right ventricle → pulmonary artery → brain
- C. Carotid plaque → middle cerebral artery
- D. Retrograde flow up the vertebral venous plexus
- E. Left ventricular mural thrombus → aorta → brain

Q22

AMNIOTIC FLUID EMBOLISM

A 33-year-old woman collapses during the second stage of labour with sudden dyspnoea, cyanosis and seizures, followed by torrential uterine bleeding. She dies within an hour. Which autopsy finding is diagnostic?

- A. Bone marrow fragments in the right ventricle
- B. Fetal squamous cells, lanugo hairs and mucin within pulmonary arteries
- C. A saddle thromboembolus at the pulmonary bifurcation
- D. Fat globules in alveolar capillaries
- E. Nitrogen bubbles in the cerebral vessels

Q23

AIR EMBOLISM

During removal of a central venous catheter, a patient suddenly becomes hypotensive and cyanotic, and a "mill-wheel" murmur is heard. Which is the immediate mechanism of haemodynamic collapse?

- A. Free fatty acids damage the alveolar-capillary membrane
- B. Tissue factor from the catheter triggers DIC
- C. Vagal stimulation causes asystole
- D. Air churned into compressible foam in the right ventricle prevents ejection (air lock)
- E. Nitrogen comes out of solution in tissue fat

Q24 DECOMPRESSION

A 38-year-old commercial diver ascends rapidly from 40 metres. Within 30 minutes he has severe joint pain and dyspnoea. Five years later, radiographs show avascular necrosis of both femoral heads. Which gas is responsible, and why?

- A. Helium, because it has the lowest molecular weight
- B. Carbon monoxide, because it binds haemoglobin with high affinity
- C. Nitrogen, because it is metabolically inert and highly fat-soluble, so it accumulates and cannot be consumed
- D. Oxygen, because it is the most abundant gas in inspired air
- E. Carbon dioxide, because it is produced by tissue metabolism

PART 5 — INFARCTION (Q25–28)**Q25** WHITE VS RED

A patient with atrial fibrillation embolises to the left kidney. At autopsy the lesion is a sharply demarcated, pale yellow-white wedge with its base on the capsule. In the same patient, a pulmonary embolus produced a dark red wedge in the lower lobe. Which factor best explains the difference in colour?

- A. The kidney undergoes liquefactive necrosis and the lung coagulative necrosis
- B. Renal emboli are always septic and pulmonary emboli are always bland
- C. The lung infarct is older, so it has accumulated haemosiderin
- D. Renal tissue is more resistant to hypoxia than lung tissue
- E. The kidney is a compact end-arterial organ, whereas the lung is spongy with a dual blood supply that continues to fill the necrotic zone

Q26 BOWEL INFARCT

A 3-year-old boy undergoes laparotomy for intussusception. A 20 cm segment of ileum is dark purple-black, swollen, soft and foul-smelling, with no clear demarcation from viable bowel. Which single feature of the vascular compromise best explains both the colour and the wetness?

- A. Venous outflow was obstructed first while arterial inflow continued, so blood was driven into tissue that could not drain
- B. Arterial inflow was obstructed first, so the segment was drained of blood
- C. The bowel underwent caseous necrosis
- D. Lymphatic obstruction caused chylous accumulation
- E. Clostridial gas production distended the wall

Q27 MI TIMING

A 59-year-old man dies suddenly 5 days after an anterior myocardial infarction. Autopsy shows haemopericardium from rupture of the left ventricular free wall. Which histological finding at the rupture site best explains why the wall was weakest at this time point?

- A. Complete regeneration of cardiac myocytes
- B. Macrophage-rich tissue digesting necrotic myocytes, with collagen deposition not yet begun
- C. A dense collagenous scar
- D. Contraction band necrosis only
- E. Granulomatous inflammation with giant cells

Q28**INTEGRATION**

A 74-year-old woman with chronic left heart failure and known pulmonary congestion has a small pulmonary embolus lodge in a segmental artery. She develops pleuritic pain and haemoptysis, and a haemorrhagic infarct is found. In a patient with normal lungs, the same embolus would most likely have caused no infarct. Why?

- A. Emboli in normal lungs are always smaller
- B. Congested lungs have a higher oxygen tension
- C. Pulmonary alveoli regenerate faster in healthy lungs
- D. In a normal lung the bronchial arterial supply is adequate to keep the territory alive; chronic passive congestion compromises it
- E. Normal lungs contain more antithrombin III

6.2 Answer Key with Full Explanations

Q1 — C Alveolar macrophages that have phagocytosed extravasated red cells. Chronic left heart failure raises left atrial and hence pulmonary capillary hydrostatic pressure. The over-distended, 0.5 μm -thick septal capillaries rupture in small numbers, spilling red cells into the alveoli. Alveolar macrophages phagocytose them; haemoglobin is degraded, but the liberated iron cannot be excreted and is stored as haemosiderin — the coarse golden-brown, Prussian-blue-positive granules. These are **heart failure cells** (siderophages), and they are why the sputum is rust-coloured.

D Surfactant is not pigmented and does not stain with Prussian blue. **E** Myeloperoxidase produces no visible pigment and the cells described are mononuclear, not neutrophils. **A** Anthracotic carbon pigment is black and Prussian-blue *negative*, because it contains no iron — this is the key discriminator. **B** Keratin is not granular, not brown, and not iron-containing. See §1.5.

Q2 — C Centrilobular (zone 3). Two independent facts converge. First, blood *leaves* the lobule through the central vein, so back-pressure from a failing right heart is transmitted to the central vein and perivenular sinusoids before anywhere else. Second, blood *enters* at the portal tracts and gives up oxygen as it travels inward, so zone 3 sees the lowest pO_2 and is the first to die when flow slows. Congestion and hypoxic necrosis therefore coincide in zone 3, producing the dark red foci; the surviving, often fatty periportal hepatocytes give the pale tan background.

B Zone 1 receives oxygenated blood first and is the *protected* zone. **D** Zone 2 is intermediate, not extreme. **E** Portal tract fibrosis is the pattern of biliary disease, not cardiac congestion. **A** The lesion is strictly zonal — that is precisely what creates the regular nutmeg mottling. See §1.6.

Q3 — E Decreased plasma colloid osmotic pressure. Massive proteinuria with fatty casts and severe hypoalbuminaemia, without haematuria, defines nephrotic syndrome. Albumin generates almost all of the ~25 mmHg oncotic pressure that holds water inside the capillary; at 1.8 g/dL that inward pull can no longer balance even a normal hydrostatic pressure, so fluid leaves the circulation everywhere. The periorbital distribution and the worsening on waking reflect the extreme looseness of periorbital tissue and the absence of gravity dependence.

D Would give *dependent* edema (legs, sacrum) and is the mechanism of heart failure. **A** Would produce a protein-rich *exudate*; here the vessel wall is intact and the fluid is a transudate. **B** Rising interstitial pressure *opposes* edema — it is the natural brake. **C** Would cause localised, non-pitting, protein-rich swelling, not generalised periorbital edema. See §1.7–1.8.

Q4 — A Protein-rich fluid that has provoked interstitial fibrosis. Axillary node clearance severs the lymphatic channels that normally return escaped plasma protein to the blood. Protein accumulates in the interstitium, raising interstitial oncotic pressure and drawing out still more water — a self-perpetuating loop. Protein-rich fluid is viscous and displaces poorly under a finger; over years it also stimulates collagen deposition, producing brawny, woody induration. Both effects make the swelling non-pitting.

B Low specific gravity describes a transudate, which pits readily. **C** A neutrophil-rich fluid would be an inflammatory exudate — there is no infection here. **D** Anatomical location does not determine pitting. **E** Light's criteria classify effusions; they say nothing about pitting and would in any case describe a transudate.

See §1.8.

Q5 — E Congestive heart failure. Every number points to a transudate: fluid/serum protein 0.25 (<0.5), fluid/serum LDH 0.32 (<0.6), specific gravity 1.008 (<1.012), few cells, and no clotting because fibrinogen has not escaped. A transudate means the vessel wall is **intact** and the problem is a Starling-force imbalance — so look outside the pleura, at the heart, liver or kidney. Bilateral effusions strongly favour a systemic cause.

B, C, D, E all produce **exudates** (increased permeability from inflammation or tumour), with protein >3 g/dL, high LDH, high cellularity, turbid fluid and clotting on standing. A pulmonary embolus with infarction also typically gives a *unilateral*, often blood-stained exudate.

See §1.9.

Q6 — B Active hyperemia. Accumulating metabolites (adenosine, CO₂, H⁺, K⁺) relax arteriolar smooth muscle, dropping resistance and increasing inflow. The bed fills with blood arriving directly from the arterial side, which is fully oxygenated — hence erythema — and the high volume of warm core blood delivered per minute raises surface temperature.

C Congestion is blue and *cool* because the blood is deoxygenated and flow is reduced; that is the opposite of what is described. **D** There is no inflammation and no protein-rich exudate. **E** Haemoconcentration does not cause erythema and warmth. **A** Reactive hyperemia specifically *follows release of an occlusion*; here the stimulus is ongoing metabolic demand, not prior ischaemia.

See §1.2.

Q7 — A Anterior cerebral artery. In subfalcine (cingulate) herniation the cingulate gyrus is forced beneath the free edge of the falx cerebri. The anterior cerebral artery runs in exactly that location, along the medial surface of the hemisphere beside the falx, so it is compressed between the herniating gyrus and the rigid dural edge — producing infarction of the medial frontal and parietal cortex (contralateral leg weakness).

B The posterior cerebral artery is compressed in *uncal (transtentorial)* herniation, against the tentorial edge. **C** The basilar artery is involved in central/tonsillar herniation with Duret haemorrhages. **D** The middle meningeal artery is torn in *extradural* haematoma after temporal bone fracture — a cause, not a consequence, of raised pressure. **E** Sagittal sinus thrombosis causes venous infarction, but it is not compressed by subfalcine herniation.

See §1.10.

Q8 — D Factor VIII deficiency (haemophilia A). The pattern is textbook for a **secondary hemostasis** defect: deep bleeding into a joint (haemarthrosis), and *delayed* rebleeding hours after initial haemostasis was achieved. The explanation is mechanistic — the platelet plug forms normally (hence normal platelet count and bleeding time, and initial haemostasis) but is never reinforced by fibrin, so it disintegrates hours later. Factor VIII is in the intrinsic pathway, so PTT is prolonged and PT normal.

E von Willebrand disease impairs platelet adhesion, so it causes *mucocutaneous* bleeding with a *prolonged bleeding time* — both absent here (though PTT can be prolonged because vWF carries factor VIII). **A** Factor VII is extrinsic, so it would prolong **PT**, not PTT. **B** Glanzmann thrombasthenia is a platelet aggregation defect — mucocutaneous bleeding, prolonged bleeding time, normal PTT. **C** Scurvy causes perifollicular haemorrhage and bleeding gums with normal coagulation studies.

See §2.2.

Q9 — C The fibrous parietal pericardium cannot stretch acutely. This is the purest illustration of **site** overriding **volume**. 250 mL is only about 5% of blood volume — trivial as blood loss. But it is confined within a non-compliant fibrous sac, so intrapericardial pressure rises steeply, exceeds right atrial and ventricular diastolic pressure, and prevents the heart filling. The result is **obstructive shock** with Beck's triad: hypotension, distended neck veins and muffled heart sounds. Free wall rupture is a classic day 3–7 complication of transmural infarction, when macrophage digestion has weakened the wall most.

D 250 mL is ~5%, not >20%. **E** Haemolysis of that volume would not cause acute collapse. **A** Coronary ostia arise inside the aortic root, not in the pericardial space. **B** A vasovagal reflex causes bradycardia and does not produce raised JVP. See §2.3, §2.5.

Q10 — E Class II haemorrhage (15–30%), and the normal haemoglobin is uninformative. Tachycardia with a narrowed pulse pressure (118/96 = 22 mmHg) is the signature of class II: sympathetic vasoconstriction raises diastolic pressure and maintains systolic pressure, so blood pressure looks deceptively normal. In *acute* haemorrhage, whole blood is lost — red cells and plasma in the same proportion — so the **concentration** of haemoglobin is initially unchanged; it falls only over hours as interstitial fluid shifts in to refill the vascular space.

A Class I loss produces essentially no signs; he has marked tachycardia and peripheral shutdown. **B** This is the classic trap and is factually wrong, for the reason just given. **C** Class IV would show profound hypotension and obtundation. **D** A narrowed pulse pressure indicates vasoconstriction; *septic* shock classically gives a **widened** pulse pressure with warm peripheries. See §2.5.

Q11 — A Haem oxygenase producing biliverdin. Inside the macrophage phagolysosome, haem oxygenase opens the porphyrin ring, releasing iron and carbon monoxide and yielding **biliverdin**, which is green. This is the rate-limiting step of haem degradation and the source of the green phase of a bruise.

B Biliverdin reductase produces bilirubin, which is **yellow** — that is the *next* phase (day 7–10). **C** Haemosiderin is golden-brown and is the last phase. **D** Conjugation occurs in the liver, not in the bruise, and does not change the colour seen in skin. **E** Myeloperoxidase gives pus its green tint, but that is a neutrophil product in infection, not a feature of resolving haemorrhage. See §2.4.

Q12 — B All three arms of Virchow's triad act simultaneously. Immobility abolishes the calf muscle pump, producing venous **stasis**; major surgery releases large amounts of **tissue factor** and provokes an acute-phase rise in fibrinogen and factor VIII, producing **hypercoagulability**; and pelvic and femoral vein manipulation during hip replacement causes direct **endothelial injury**. This convergence is why orthopaedic surgery carries the highest DVT risk of any procedure and why prophylactic anticoagulation is routine.

C Turbulence is an arterial phenomenon; veins have slow, low-pressure flow where *stasis*, not turbulence, dominates. **D** and **E** cause edema, not thrombosis. **A** vWF deficiency causes *bleeding*, the opposite problem. See §3.6.

Q13 — D It formed in flowing blood before death. The laminations are **lines of Zahn** — alternating pale layers of platelets and fibrin with dark red-cell-rich layers. They can only form where blood is *moving*, because the alternation is produced by variation in flow during the cardiac cycle: platelets deposit preferentially during rapid flow, red cells are trapped when flow momentarily slows. Their presence therefore proves an ante-mortem thrombus. The dry, dull, friable texture confirms it.

E A post-mortem clot forms by gravity sedimentation and shows two layers ("chicken fat" over "red currant jelly"), not fine alternating laminations; it is also wet, shiny, gelatinous and non-adherent. **A** A myxoma is a gelatinous tumour mass with a stalk, not a laminated clot. **B** Fibrin microthrombi of DIC are microscopic only and occlude capillaries. **C** Friability is a feature of thrombus, not post-mortem clot — this option inverts the discriminator. See §3.8, §3.13.

Q14 — C Factor V Leiden. A point mutation substitutes glutamine for arginine at position 506 of factor V — the exact residue at which activated protein C cleaves and inactivates factor Va. The mutant factor Va is therefore resistant to cleavage and continues to support thrombin generation. The laboratory phenotype described, "failure of activated protein C to prolong the clotting time", is literally the APC-resistance assay. It is the commonest inherited thrombophilia, and the combined oral contraceptive multiplies the risk many-fold.

D Antithrombin III deficiency presents with apparent heparin resistance, not APC resistance. **E** The prothrombin G20210A mutation raises prothrombin levels; APC function is normal. **A** HIT is an acquired, heparin-dependent antibody disorder with thrombocytopenia. **B** Protein S deficiency reduces APC's *cofactor*, so adding exogenous APC would still work — the assay result described points specifically to a defect in the substrate, factor V.

See §3.7.

Q15 — E IgG against heparin–platelet factor 4 complexes. Heparin binds platelet factor 4, creating a neoantigen. IgG against this complex binds the platelet surface and cross-links FcγRIIA receptors, which **activates** the platelet. Activated platelets are cleared from the circulation (thrombocytopenia) and simultaneously drive thrombin generation (thrombosis). This is the paradox of HIT: a falling platelet count with **clotting, not bleeding**. Timing (day 5–10 of heparin) and a >50% fall are the clinical clues. Management: stop all heparin and start a direct thrombin inhibitor — warfarin alone is contraindicated acutely because it can precipitate venous limb gangrene.

A Marrow suppression would not cause thrombosis. **B** Heparin does not release tissue factor; and DIC would prolong PT and PTT and consume fibrinogen. **C** Antithrombin is not consumed in this way. **D** Dilution would occur on day 0–1, not day 6, and would not cause limb ischaemia.

See §3.7.

Q16 — A Widespread microvascular fibrin thrombi consuming platelets and clotting factors — DIC. Placental abruption releases massive amounts of tissue factor, activating coagulation throughout the circulation at once. Fibrin microthrombi form in capillaries everywhere, causing organ ischaemia and shearing red cells into **schistocytes** as they pass through fibrin strands. Manufacturing that fibrin exhausts platelets and clotting factors, and secondary fibrinolysis floods the plasma with fibrin degradation products that further inhibit haemostasis — hence bleeding. The laboratory panel (low platelets, prolonged PT and PTT, low fibrinogen, very high D-dimer, schistocytes) is diagnostic.

B Isolated factor VIII deficiency prolongs only PTT and does not lower platelets or fibrinogen. **C** ITP gives isolated thrombocytopenia with normal PT, PTT and fibrinogen. **D** Vitamin K deficiency prolongs PT then PTT but leaves platelets, fibrinogen and D-dimer normal. **E** Pregnancy raises fibrinogen; here it is consumed and low, which is the point.

See §3.11.

Q17 — B Organization with recanalization. Over days, leukocytes infiltrate the thrombus and endothelial cells, smooth muscle cells and fibroblasts grow in from the adjacent wall — recruited by PDGF and TGF-β released from platelets within the thrombus itself. Collagen is deposited and the thrombus becomes granulation tissue and then scar (**organization**). Capillary sprouts within it elongate and anastomose into channels that eventually reconnect the two ends of the lumen (**recanalization**). One wide lumen is thereby replaced by several narrow ones, so resistance is permanently higher — the anatomical basis of **post-thrombotic syndrome**: chronic swelling, pain and venous stasis ulceration.

C Complete lysis would leave a normal patent vein. **D** Propagation is enlargement, not fibrous replacement. **E** Embolisation removes the thrombus to a distant site. **A** Calcification alone produces a phlebolith, not multiple perfused channels.

See §3.12.

Q18 — D Thrombomodulin binds thrombin and converts it into an activator of protein C. This is one of the most elegant switches in physiology: thrombin, the most powerful procoagulant enzyme in the body, becomes an *anticoagulant* the moment it binds thrombomodulin on intact endothelium. The complex activates protein C, which with protein S inactivates factors Va and VIIIa. This is how a clot is confined to the injured segment — the surrounding healthy endothelium actively switches off any thrombin that drifts onto it.

E vWF *promotes* platelet adhesion by bridging collagen to GpIb. **A** PAI *inhibits* fibrinolysis; t-PA degrades fibrin. **B** Tissue factor *initiates* the extrinsic cascade; heparin-like molecules accelerate antithrombin III. **C** TXA₂ *promotes* aggregation; prostacyclin (PGI₂) *inhibits* it.

See §3.2, §3.5.

Q19 — C Libman–Sacks endocarditis. Three features together are specific: the patient has SLE, the vegetations are **sterile** and **non-destructive**, and they sit on **both** surfaces of the leaflet. The pathogenesis is deposition of immune complexes and complement along the endocardium with secondary platelet aggregation, and it is strongly associated with antiphospholipid antibodies.

D Infective endocarditis produces large, bulky, *destructive* vegetations containing organisms, and can perforate a cusp. **E** Rheumatic valvulitis gives small vegetations along the *line of closure only*, with Aschoff bodies, and heals with commissural fusion and mitral stenosis. **A** Marantic endocarditis is also sterile and bland, but occurs on the line of closure in patients with advanced malignancy or DIC, not SLE, and is not verrucous on both surfaces. **B** Carcinoid disease causes fibrous plaque-like thickening of *right-sided* valves.

See §3.10.

Q20 — E Sudan-positive fat globules on a frozen section. The vignette gives the complete fat embolism syndrome: long-bone fracture, a **24–72 hour latent period**, respiratory distress, neurological change, thrombocytopenia, and the highly specific **petechial rash** over the upper chest and axillae. The latency is the diagnostic clue and is mechanistic — the mechanical obstruction alone is tolerable, but over 1–3 days lipases hydrolyse neutral fat into free fatty acids that are directly toxic to the alveolar–capillary membrane. Fat must be demonstrated on a **frozen** section with a lipid stain, because routine paraffin processing dissolves it in xylene, leaving only empty vacuoles.

A Lines of Zahn indicate a thromboembolus; that would present suddenly, with no latency and no petechial rash. **B** Squames and lanugo are pathognomonic of amniotic fluid embolism. **C** Frothy right-ventricular blood is the finding in air embolism. **D** Cholesterol clefts indicate atheroembolism, which follows vascular instrumentation.

See §4.4.

Q21 — A Across the septal defect — a paradoxical embolus. Follow the blood. A thrombus in a leg vein must travel to the right heart. Normally it would then be pumped into the pulmonary arteries and be filtered there. In the presence of a patent foramen ovale, a transient rise in right atrial pressure (straining, coughing, or the pressure rise caused by the embolus itself) opens the flap and allows the embolus to cross into the left atrium, bypassing the pulmonary filter and entering the systemic circulation. **Whenever a venous-origin embolus reaches a systemic organ, a shunt must exist.**

B A pulmonary artery embolus cannot reach the brain; the pulmonary capillary bed is the end of that route. **C** and **E** are plausible stroke mechanisms in general but are excluded here by her age, the absence of cardiac or carotid disease, and the documented DVT and PFO. **D** Batson's plexus explains retrograde *metastasis* to vertebrae, not a hemispheric stroke.

See §4.1.

Q22 — B Fetal squamous cells, lanugo hairs and mucin in pulmonary vessels. These are the pathognomonic autopsy findings of amniotic fluid embolism, along with vernix caseosa fat. The clinical sequence is characteristic and follows the mechanism: particulate matter and — far more importantly — fetal antigens trigger an anaphylactoid reaction causing pulmonary vasoconstriction, cardiorespiratory collapse and seizures; then the high **tissue factor** content of amniotic fluid precipitates fulminant DIC, which is why catastrophic haemorrhage follows the collapse.

C A saddle thromboembolus would cause sudden death but not seizures followed by DIC-driven bleeding. **D** Fat embolism requires a fracture and has a 24–72 h latency. **E** Nitrogen bubbles require decompression. **A** Marrow emboli follow fracture or CPR and are usually incidental.

See §4.6.

Q23 — D Air churned into compressible foam — an air lock. During inspiration, intrathoracic venous pressure falls below atmospheric, so removing a central line can suck air into the vein. The bubbles reach the right ventricle, where the beating heart whips air and blood into foam. A ventricle is a pump designed for an *incompressible* fluid: foam simply compresses under systole instead of being ejected, so forward output ceases and the patient collapses. The churning also generates the classic "mill-wheel" murmur. Management: left lateral decubitus, head-down (Duran position), 100% oxygen, and aspiration through the line.

E That is decompression sickness, which requires a change in ambient pressure. **A** That is the second hit of fat embolism syndrome. **B** Catheters do not release tissue factor in amounts causing instantaneous DIC. **C** A vagal response causes bradycardia and would not produce a mill-wheel murmur.

See §4.5.

Q24 — C Nitrogen. At depth, high ambient pressure forces far more gas into solution (Henry's law). Oxygen is metabolised and CO₂ is buffered and excreted, but nitrogen is **metabolically inert** — no pathway consumes it — and it is about five times more soluble in lipid than in water, so it accumulates preferentially in fat, marrow, myelin and joint tissues. On rapid ascent, ambient pressure falls faster than nitrogen can diffuse back to the lungs, and it comes out of solution as bubbles: joint pain (*bends*), respiratory distress (*chokes*) and neurological signs (*staggers*). Chronically, bubbles in the fatty marrow of end-arterial bone cause **avascular necrosis of the femoral and humeral heads** — exactly the late finding described.

D and **E** are metabolically handled and do not accumulate. **A** Helium is used deliberately in deep mixes precisely *because* it is less soluble in lipid and less narcotic. **B** Carbon monoxide causes hypoxic injury, not bubbles.

See §4.5.

Q25 — E The kidney is a compact end-arterial organ; the lung is spongy with a dual supply. In the kidney, an embolus removes the *only* supply to that territory. No more blood can enter, the trapped red cells are lysed and their pigment cleared by macrophages, and — because nothing replaces them — the necrotic zone progressively decolorises to pale yellow-white. In the lung, the bronchial arteries continue to perfuse the territory whose pulmonary artery is blocked. That flow is enough to fill the dead zone with blood but not enough to oxygenate it (bronchial flow is ~1% of cardiac output), and ischaemically damaged capillaries leak — so the infarct is haemorrhagic. The spongy interstitium gives the blood somewhere to pool.

A Both undergo coagulative necrosis; only the brain liquefies. **B** Nothing in the vignette suggests infection, and septic infarcts become abscesses regardless of organ. **C** Age changes red to brown, not white to red, and the vignette gives no timeline difference. **D** Renal tubular epithelium is in fact highly susceptible to hypoxia.

See §5.3.

Q26 — A Venous outflow was obstructed first while arterial inflow continued. Intussusception (like volvulus, strangulated hernia and testicular torsion) compresses the mesenteric vessels. The thin-walled, low-pressure **veins collapse first**; the thicker, higher-pressure arteries keep pumping blood in. Blood is therefore driven into a segment that cannot drain, so the bowel engorges and becomes dark purple-black. Only when intravascular pressure finally equals arterial pressure does inflow stop and the tissue die — *full of blood*. That is a haemorrhagic (red) infarct. The wetness and foul smell indicate superimposed bacterial overgrowth from the gut lumen, making this **wet gangrene**, which is why there is no sharp demarcation and why it is a surgical emergency.

B Pure arterial occlusion in a drained tissue produces *dry*, pale, sharply demarcated gangrene. **C** Caseous necrosis is a feature of tuberculosis. **D** Chylous accumulation is milky lymph, not black blood. **E** Clostridial gas gangrene affects deep contaminated muscle wounds and produces crepitus.

See §5.3, §5.6.

Q27 — B Macrophage-rich tissue digesting necrotic myocytes, with collagen not yet deposited. Days 3–7 after infarction is the moment of **maximum mechanical weakness**, and the reason is a timing gap: neutrophils and macrophages have removed the dead myocytes and their supporting matrix, but fibroblasts have not yet laid down enough collagen to replace them. The wall is therefore at its thinnest and softest just as normal systolic pressure continues to load it. Hence the classic day 3–7 complications: free wall rupture (tamponade), septal rupture (VSD) and papillary muscle rupture (acute mitral regurgitation).

C A dense collagenous scar takes 2–8 weeks and is *strong*; rupture at that stage is rare. **D** Contraction band necrosis is a reperfusion finding at the infarct margin and does not explain rupture. **E** Granulomatous inflammation is not part of infarct healing. **A** Cardiac myocytes are permanent cells and do not regenerate.

See §5.4.

Q28 — D In a normal lung the bronchial supply is adequate; chronic congestion compromises it. This question integrates the whole session. The lung normally resists infarction because it has a **dual blood supply**: when a pulmonary arterial branch is blocked, the bronchial arteries continue to perfuse and oxygenate that segment, so the tissue survives while the embolus is lysed. In chronic left heart failure, however, sustained pulmonary venous hypertension raises the downstream pressure that bronchial arterial flow must overcome, and the septa are thickened and fibrotic, so bronchial perfusion becomes inadequate. The same small embolus now kills the tissue — and because blood continues to seep in from the bronchial circulation into damaged, leaky capillaries, the infarct is **red (haemorrhagic)**, producing haemoptysis and, through the overlying fibrinous pleural exudate, pleuritic pain.

E Antithrombin levels do not vary by pulmonary status. **A** Embolus size is unrelated to the health of the lung. **B** Congested lungs have *impaired* gas exchange and lower oxygen tension. **C** Alveolar regeneration has no role over this timescale. See §1.5, §4.2, §5.3.

ANSWER GRID

1 C · 2 C · 3 E · 4 A · 5 E · 6 B · 7 A · 8 D · 9 C · 10 E · 11 A · 12 B · 13 D · 14 C
15 E · 16 A · 17 B · 18 D · 19 C · 20 E · 21 A · 22 B · 23 D · 24 C · 25 E · 26 A · 27 B · 28 D

Cover this grid while you work. Each explanation also analyses the distractors — read those even for questions you answered correctly, because most marks are lost on plausible wrong options rather than on unfamiliar right ones.

6.3 Rapid-Revision Master Tables

These four tables compress the whole session. They are meant for the final pass, once the mechanisms are already understood — a table cannot teach you a mechanism, but it will let you retrieve one in seconds.

MASTER TABLE 1 THE FIVE DISORDERS SIDE BY SIDE

Disorder	What is where it should not be	Core mechanism	Signature lesion
Hyperemia	Too much oxygenated blood in a tissue	Active arteriolar dilation → ↑inflow	Red, warm, swollen tissue; <i>rubor</i> and <i>calor</i> of inflammation
Congestion	Too much deoxygenated blood in a tissue	Passive obstruction of venous outflow	Nutmeg liver (right HF); brown induration of lung with heart failure cells (left HF)
Edema	Water outside the vessel, cells inside	↑hydrostatic, ↓oncotic, lymphatic block, Na ⁺ retention, or ↑permeability	Pitting dependent edema; ascites; pulmonary edema; anasarca
Hemorrhage	Whole blood outside the vessel	Rupture (<i>per rhexis</i>) or diapedesis	Petechia 1–2 mm / purpura 3–5 mm / ecchymosis >1–2 cm; haematoma; cavity bleeds
Thrombosis	Solid blood inside a living vessel	Virchow's triad: endothelial injury, abnormal flow, hypercoagulability	Lines of Zahn ; white/mixed/red/hyaline thrombi; mural thrombus; vegetations
Embolism	A mass downstream of where it formed	Detachment and transport by the bloodstream	Saddle embolus ; fat globules; air lock; fetal squames

Disorder	What is where it should not be	Core mechanism	Signature lesion
Infarction	No blood at all where it is needed	Ischaemic necrosis after arterial or venous occlusion	Wedge-shaped white (kidney, spleen, heart) or red (lung, bowel, testis) infarct

MASTER TABLE 2 LEFT VERSUS RIGHT HEART FAILURE — THE TWO CONGESTION SYNDROMES

	Left heart failure	Right heart failure
Blood backs up into	Left atrium → pulmonary veins → lungs	Right atrium → vena cavae → systemic veins
Congested organ	Lung	Liver , spleen, kidneys, legs
Signature morphology	Brown induration : thickened fibrotic septa + heart failure cells	Nutmeg liver : congested necrotic centrilobular zones + fatty periportal zones
Clinical features	Dyspnoea, orthopnoea, PND, rust-coloured sputum, crackles	Raised JVP, tender hepatomegaly, ascites, peripheral pitting edema
Commonest cause	Ischaemic heart disease, hypertension, mitral/aortic valve disease	Left heart failure (commonest); then cor pulmonale from chronic lung disease
Downstream risk	Predisposes to red pulmonary infarct after even a small embolus	Cardiac sclerosis → cardiac cirrhosis if prolonged

MASTER TABLE 3 THROMBUS MORPHOLOGY AT A GLANCE

	White	Mixed	Red	Hyaline
Made of	Platelets + fibrin	Alternating layers	RBC + fibrin	Fibrin only
Position	Head (attached)	Body	Tail (free)	Capillaries
Where	Arteries, heart, valves	Veins	Veins, sites of stasis	Microcirculation in DIC
Gross	Dry, grey-white, friable	Laminated	Dark red, moist, elastic	Not visible
Lines of Zahn	Present	Best seen	Poorly developed	Absent

MASTER TABLE 4 THE HIGH-YIELD NUMBERS

Number	What it refers to
1–2 mm / 3–5 mm / >1–2 cm	Petechia / purpura / ecchymosis
~32 / ~12 / ~25 mmHg	Capillary hydrostatic pressure at arterial end / venous end; plasma oncotic pressure
<2.5 g/dL	Serum albumin below which generalised edema appears

Number	What it refers to
>3 g/dL protein; SG >1.020; ratios >0.5 / >0.6	Exudate criteria (protein, specific gravity, fluid/serum protein, fluid/serum LDH)
≥5 g/dL	Deoxyhaemoglobin needed for visible cyanosis
<15 / 15–30 / 30–40 / >40%	Haemorrhage classes I–IV
200–300 mL	Pericardial blood causing acute tamponade
>100 mL / >300 mL	Air causing acute distress / fatal air embolism
9–20 g fat → 75%	Fat load that may occlude three-quarters of the pulmonary arteries
24–72 hours	Latent period of fat embolism syndrome
>90–95%	Proportion of pulmonary emboli arising from lower-limb DVT
~75% / ~10%	Destinations of systemic emboli: lower limbs / brain
3–4 min / 20–30 min	Hypoxic tolerance of neurons / cardiac myocytes
Days 3–7	Period of maximum myocardial rupture risk after infarction
II, VII, IX, X + C, S	Vitamin K-dependent factors and proteins

6.4 Glossary of Key Terms

Arranged alphabetically for fast look-up. Each entry is written to stand alone, so it can be read the night before without re-reading the chapter.

Anasarca

Severe, generalised edema with profound subcutaneous swelling; implies advanced cardiac, hepatic or renal failure.

Anaemic infarct

Older term for a *white* or *pale* infarct — arterial occlusion in a compact, end-arterial organ.

Antithrombin III

Plasma protease inhibitor that inactivates thrombin and factors IXa, Xa, XIa and XIIa; accelerated about 1000-fold by heparin. Its deficiency causes apparent heparin resistance.

Ascites

Pathological accumulation of fluid in the peritoneal cavity (hydroperitoneum). Commonest cause is cirrhosis with portal hypertension and hypoalbuminaemia.

Bends, the

Acute decompression sickness: severe joint and muscle pain caused by nitrogen bubbles in periarticular tissue.

Brown induration of the lung

Chronic pulmonary congestion: *brown* from haemosiderin in heart failure cells, *indurated* from septal fibrosis. Lesion of chronic **left** heart failure.

Caisson disease

Chronic decompression sickness with avascular (ischaemic) necrosis of bone, typically the femoral and humeral heads.

Cardiac cirrhosis

True fibrous scarring of the liver resulting from very long-standing passive congestion; the end-stage of repeated centrilobular necrosis.

Coagulative necrosis

Necrosis in which cellular outlines are preserved as anucleate "ghosts", because ischaemia denatures both structural proteins and lysosomal enzymes. The pattern of infarcts in every organ *except the brain*.

Congestion

Passive increase in local blood volume from impaired venous outflow. Blue, cool, swollen, harmful; may be local or systemic.

Cor pulmonale

Right ventricular hypertrophy and eventual failure caused by pulmonary hypertension. *Acute* cor pulmonale is the abrupt right heart strain of a massive pulmonary embolus.

D-dimer

A fibrin degradation product released when cross-linked fibrin is digested by plasmin; a sensitive but non-specific marker of active clot breakdown.

Diapedesis

Escape of red cells one at a time between endothelial cells whose junctions have been loosened by severe congestion, hypoxia or inflammation — haemorrhage without a visible breach.

DIC

Disseminated intravascular coagulation: widespread microvascular fibrin thrombosis that consumes platelets and clotting factors, producing simultaneous organ ischaemia *and* bleeding. Also called consumption coagulopathy.

Ecchymosis

Subcutaneous haemorrhage larger than 1–2 cm; a bruise. Its colour sequence dates the injury.

Edema

Increased fluid in the interstitial tissue spaces or body cavities — an extravascular accumulation of the fluid phase of blood.

Elephantiasis

Massive chronic lymphedema with thickened, verrucous skin, classically from lymphatic obstruction by *Wuchereria bancrofti* (filariasis).

Embolus

Any detached intravascular solid, liquid or gaseous mass carried by the blood to a site distant from its origin. Over 99% are dislodged thrombi.

Exudate

Protein-rich, cellular, high-specific-gravity fluid produced by **increased vascular permeability** in inflammation; clots on standing.

Factor V Leiden

Point mutation rendering factor Va resistant to cleavage by activated protein C; the commonest inherited thrombophilia.

Fat embolism syndrome

Respiratory distress, neurological change and a petechial rash 24–72 h after long-bone fracture; caused by mechanical obstruction plus free fatty acid toxicity.

Gangrene

Clinical term for necrosis of a large mass of tissue. *Dry* = arterial occlusion with intact venous drainage; *wet* = venous obstruction plus bacterial infection; *gas* = clostridial infection of necrotic muscle.

Haemarthrosis

Bleeding into a joint cavity; hallmark of haemophilia and of secondary hemostasis failure generally.

Haematoma

Accumulation of blood within a tissue, forming a discrete mass. Its significance depends entirely on site.

Haemorrhage

Extravasation of blood from the cardiovascular system. Outcome depends on **volume, rate and site**.

Haemosiderin

Coarse, golden-brown, iron-containing pigment formed when ferritin aggregates after haemoglobin breakdown; Prussian-blue positive. Iron cannot be excreted, so haemosiderin is the last pigment to disappear.

Heart failure cells

Haemosiderin-laden alveolar macrophages (siderophages) found in the alveoli in chronic pulmonary congestion; the cause of rust-coloured sputum.

Hemostasis

The physiological sequence — vasoconstriction, platelet plug, fibrin deposition, clot stabilisation — that stops bleeding at a site of vascular injury.

Heparin-induced thrombocytopenia

IgG against heparin–PF4 complexes activates platelets, causing a falling platelet count together with **thrombosis**, not bleeding.

Hyperemia

Active increase in local blood volume from arteriolar dilation. Red, warm, beneficial, always local.

Hydropericardium / hydrothorax

Non-inflammatory fluid in the pericardial / pleural cavity.

Infarct

Area of ischaemic necrosis caused by occlusion of arterial supply or venous drainage; characteristically wedge-shaped with the apex at the occluded vessel.

Lines of Zahn

Alternating pale (platelet/fibrin) and dark (red cell) laminations within a thrombus; they prove it formed in **flowing blood before death**.

Liquefactive necrosis

Necrosis in which the tissue is digested to a fluid-filled cavity. Characteristic of the **brain** and of abscesses.

Lymphedema

Protein-rich edema from failed lymphatic drainage; becomes firm and **non-pitting** because the retained protein raises interstitial oncotic pressure and stimulates fibrosis.

Marantic endocarditis

Sterile, non-destructive fibrin–platelet vegetations on the line of valve closure, associated with advanced malignancy and DIC.

Mural thrombus

Thrombus adherent to one wall of a cardiac chamber or the aorta, leaving the lumen partly patent; a common source of systemic emboli.

Mycotic aneurysm

Localised arterial dilation caused by infective destruction of the wall, typically by a septic embolus from endocarditis; prone to fatal rupture.

Nutmeg liver

Chronic passive hepatic congestion: dark red congested and necrotic centrilobular zones alternating with pale, often fatty, periportal parenchyma. Lesion of **right** heart failure.

Paradoxical embolism

A venous embolus that crosses a right-to-left cardiac shunt and enters the systemic circulation, bypassing the pulmonary filter.

Peau d'orange

Dimpled, orange-peel appearance of skin over dermal lymphatic obstruction — the skin is edematous but tethered at hair follicles. Classic in inflammatory breast carcinoma.

Petechia

Pinpoint 1–2 mm haemorrhage into skin, mucosa or serosa; does not blanch on pressure.

Phlebolith

A calcified organized venous thrombus; commonly an incidental radiological finding in the pelvis.

Primary hemostasis

Formation of the platelet plug by adhesion (vWF–GpIb), activation and aggregation (fibrinogen–GpIIb/IIIa). Its failure causes **muco-cutaneous** bleeding.

Protein C / protein S

Vitamin K-dependent anticoagulants; activated protein C, with protein S as cofactor, inactivates factors Va and VIIIa.

Purpura

Haemorrhage 3–5 mm (up to ~1 cm) into skin or mucosa. *Palpable* purpura implies vasculitis; flat purpura implies a platelet defect.

Recanalization

Formation of new endothelial-lined channels through an organized thrombus, partly restoring flow but never to normal.

Red (haemorrhagic) infarct

Infarct that remains full of blood: caused by venous occlusion, or arterial occlusion in a loose, spongy, dual-supply organ (lung, bowel, testis, ovary), or after reperfusion.

Rhexis, haemorrhage per

Haemorrhage through a frank structural breach of the vessel wall, as opposed to diapedesis.

Saddle embolus

A large thromboembolus straddling the bifurcation of the main pulmonary artery; usually immediately fatal.

Schistocyte

A fragmented red cell, sheared as it passes through intravascular fibrin strands; the blood-film marker of microangiopathic haemolysis in DIC and TTP.

Secondary hemostasis

Deposition of fibrin by the coagulation cascade, reinforcing the platelet plug. Its failure causes **deep** and **delayed** bleeding.

Septic infarct

An infarct seeded by organisms — usually from an infected embolus — which is converted into an **abscess** because necrotic tissue has no blood supply to deliver defences.

Stasis

Slowing or cessation of blood flow; the dominant flow abnormality in **venous** thrombosis.

Starling forces

The four pressures governing fluid movement across a capillary: capillary hydrostatic (out), plasma oncotic (in), interstitial hydrostatic (in) and interstitial oncotic (out).

Thrombomodulin

Endothelial receptor that binds thrombin and converts it into an activator of protein C — turning the body's most powerful procoagulant into an anticoagulant.

Thrombosis

Formation of an adherent clotted mass of blood within the cardiovascular system of a **living** body.

Transudate

Protein-poor, low-specific-gravity, acellular fluid produced by a **Starling-force imbalance** across an intact vessel wall; does not clot.

Trousseau syndrome

Migratory thrombophlebitis caused by procoagulants released from a tumour; may be the first sign of an occult visceral cancer.

Vegetation

A thrombotic mass on a heart valve. Infective (destructive, contains organisms), marantic (sterile, malignancy) or Libman–Sacks (sterile, SLE, on both leaflet surfaces).

Virchow's triad

The three predisposing influences to thrombosis: **endothelial injury**, **abnormal blood flow** (stasis or turbulence), and **hypercoagulability**.

von Willebrand factor

Plasma multimer that bridges subendothelial collagen to platelet GpIb, enabling adhesion under high shear; also the carrier protein for factor VIII.

White (pale) infarct

Infarct that loses its blood: arterial occlusion in a solid, compact, end-arterial organ (kidney, spleen, heart). Sharply demarcated and yellow-white.

End of Session 3 — Disorders of Vascular Flow.